

MATH601: Abstract Algebra II

Notes by Yuval Amit

Instructor: Harry Tamvakis

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Modules

Definition

For an R -module M , where R is a commutative ring, we defined **tensor powers** $T^n M$ to be $M \otimes \cdots \otimes M$, and **symmetric powers** $S^n M$ to be $T^n M / I$. Moreover, we define **tensor algebras** to be $T^\bullet M = \bigoplus_{n \geq 0} T^n M$ and **symmetric algebras** to be $S^\bullet M = \bigoplus_{n \geq 0} S^n M$

1. Exterior Powers and Exterior Algebras

Definition

1.1

For $n \geq 1$, we define the n -th **exterior power** $\wedge^n M$ to be an R -module, together with an R -multilinear map $\zeta : M \times \cdots \times M \rightarrow \wedge^n M$ which is **alternating**. The map sends $(x_1, \dots, x_n) \mapsto x_1 \wedge \cdots \wedge x_n$.

Alternating means $\zeta(x_1, \dots, x_n) = 0$ whenever $x_i = x_j$ for some $i \neq j$.

Proposition

1.2

Alternating implies (but is not equivalent to) the condition that

$$\zeta(x_{\sigma(1)}, \dots, x_{\sigma(n)}) = \text{sgn}(\sigma) \zeta(x_1, \dots, x_n)$$

for all $\sigma \in S_n$

Proposition

1.3

Let $n = 2$. Given a bilinear map $\varphi : M \times M \rightarrow A$ for some R -module A , we have

$$\varphi(x + y, x + y) = \varphi(x, x) + \varphi(x, y) + \varphi(y, x) + \varphi(y, y).$$

If $\varphi(x, x) = 0$ for all $x \in M$, then this implies $\varphi(y, x) = -\varphi(x, y)$ for all $x, y \in R$.

The converse is true if $2\varphi(x, x) = 0 \implies \varphi(x, x) = 0$. This holds if multiplication by 2 is injective in the R -module A .

Theorem

1.4

The pair $(\wedge^n M, \zeta)$ satisfies the universal property that for any R -module A and alternating multilinear map $f : M \times \cdots \times M \rightarrow A$, there exists a unique R -linear map $\tilde{f} : \wedge^n M \rightarrow A$ such that $\tilde{f} \circ \zeta = f$. That is, the diagram below commutes:

$$\begin{array}{ccc}
 M \times \cdots \times M & \xrightarrow{\zeta} & \bigwedge^n M \\
 & \searrow f & \downarrow \exists! \tilde{f} \\
 & & A
 \end{array}$$

Proof. We start with existence. Let $\bigwedge^n M = T^n M / J$, where J is the R -submodule of $T^n M = M \otimes \cdots \otimes M$ generated by the tensors $x_1 \otimes \cdots \otimes x_n$, where $x_i = x_j$ for some $i \neq j$. **Exercise:** check this. $\square$

Remark

1.5

We have $\bigwedge^1 M = M$, and $\bigwedge^0 M = R$ by convention.

Similar to the case of symmetric powers, we have bilinear maps $\bigwedge^a M \otimes \bigwedge^b M \rightarrow \bigwedge^{a+b} M$, where

$$(x_1 \wedge \cdots \wedge x_a) \otimes (y_1 \wedge \cdots \wedge y_b) \mapsto x_1 \wedge \cdots \wedge x_a \wedge y_1 \wedge \cdots \wedge y_b$$

Definition

1.6

These make the R -module

$$\bigwedge^\bullet M := \bigoplus_{n=0}^{\infty} \bigwedge^n M$$

into an R -algebra called the **exterior algebra** of M . $\bigwedge^\bullet M$ is a graded, skew-commutative R -algebra.

Skew-commutative means that for $x \in \bigwedge^a M, y \in \bigwedge^b M$, we have

$$y \wedge x = (-1)^{ab} x \wedge y$$

(To see this, think about how x has a wedges, y has b wedges, and the sign flips every time they cross.)

Theorem

1.7

For any R -algebra A and map $f : M \rightarrow A$ of R -modules such that $f(x)^2 = 0$ in A for any $x \in M$, there exists a unique homomorphism of R -algebras $\tilde{f} : \bigwedge^\bullet M \rightarrow A$, making the below diagram commute:

$$\begin{array}{ccc}
 M & \xrightarrow{\quad} & \bigwedge^\bullet M \\
 & \searrow f & \swarrow \tilde{f} \\
 & & A
 \end{array}$$

Proof. Construction: Let

$$\bigwedge^\bullet M := T^\bullet M / \langle 2\text{-sided ideals generated by } x \otimes x \in T^2 M, \forall x \in M \rangle$$

 $\square$

Proposition**1.8**

$$\bigwedge^n (M \oplus N) \cong \bigoplus_{a+b=n} \bigwedge^a M \otimes \bigwedge^b N$$

Proof. We have maps $\bigwedge^a M \otimes \bigwedge^b N \rightarrow \bigwedge^{a+b} (M \oplus N)$ which map

$$(x_1 \wedge \cdots \wedge x_a) \otimes (y_1 \wedge \cdots \wedge y_b) \mapsto (x_1 \wedge \cdots \wedge x_a) \wedge (y_1 \wedge \cdots \wedge y_b).$$

Here, $x \in M$ is defined with $(x, 0) \in M \oplus N$, and $y \in N \equiv (0, y) \in M \oplus N$. That is, we have a natural inclusion $M \hookrightarrow M \oplus N, N \hookrightarrow M \oplus N$.

Adding these together produces a map

$$g : \bigoplus_{a+b=n} \bigwedge^a M \otimes \bigwedge^b N \rightarrow \bigwedge^n (M \oplus N).$$

The inverse of g is harder to define. For $a, b \geq 0$ such that $a + b = n$, we will consider partitions of the set $[1, n] := \{1, 2, \dots, n\}$ into (disjoint) subsets I and J , with $|I| = a, |J| = b$. We write

$$I = \{i_1 < \cdots < i_a\}, J = \{j_1 < \cdots < j_b\}, I \sqcup J = [1, n].$$

For each pair (I, J) , let $\text{sgn}(I, J)$ be the sign of the permutation

$$\begin{pmatrix} 1 & 2 & \cdots & n \\ i_1 & i_2 \cdots i_a & \cdots & j_1 \cdots j_b \end{pmatrix} \in S_n.$$

Consider the R -multilinear map

$$f : (M \oplus N) \times \cdots \times (M \oplus N) \rightarrow \bigoplus_{a+b=n} \bigwedge^a M \otimes \bigwedge^b N$$

given by

$$\begin{aligned} & f((x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)) \\ &= \bigoplus_{a+b=n} \left(\sum_{I \sqcup J = [1, n], |I|=a, |J|=b} \text{sgn}(I, J) (x_{i_1} \wedge \cdots \wedge x_{i_a}) \otimes (y_{j_1} \wedge \cdots \wedge y_{j_b}) \right) \end{aligned}$$

For example, when $n = 2$, we have

$$f((x_1, y_1), (x_2, y_2)) = x_1 \wedge x_2 + x_1 \otimes y_2 - x_2 \otimes y_1 + y_1 \wedge y_2.$$

To understand the sign, consider the identity:

$$(x_1 + y_1) \wedge (x_2 + y_2) = x_1 \wedge x_2 + x_1 \wedge y_2 + y_1 \wedge x_2 + y_1 \wedge y_2.$$

□

Example 1.9. When $n = 2$, we have

$$\bigwedge^2 (M \oplus N) \cong (\bigwedge^2 M) \oplus (M \otimes N) \oplus (\bigwedge^2 N)$$

Example 1.10. For $M = R$, we have $\bigwedge^n R = 0$ if $n > 1$. This is because if $a, b \in R$, we have $a \wedge b = ab(1 \wedge 1) = 0$.

Corollary**1.11**

If M is a free R -module with basis $e_1, \dots, e_n$, then $\bigwedge^k M$ is free with basis $e_{i_1} \wedge \dots \wedge e_{i_k}$ for $1 \leq i_1 < \dots < i_k \leq n$. So, $\bigwedge^k M$ has rank $\binom{n}{k}$.

Proof. We have $\bigwedge^0 R = \bigwedge^1 R = R$, and $\bigwedge^a R = 0$ if $a > 1$. Since M is free on $e_1, \dots, e_n$, we have

$$M \cong \bigoplus_{i=1}^n Re_i.$$

The previous proposition and induction give

$$\begin{aligned} \bigwedge^k M &= \bigwedge^k (Re_1 \oplus \dots \oplus Re_n) \cong \bigoplus_{a_1 + \dots + a_n = k, a_i \geq 0} \bigwedge^{a_1} Re_1 \otimes \dots \otimes \bigwedge^{a_n} Re_n \\ &\cong \bigoplus_{a_1 + \dots + a_n = k, a_i \in \{0,1\}} \bigwedge^{a_1} Re_1 \otimes \dots \otimes \bigwedge^{a_n} Re_n \end{aligned}$$

as R -modules. This is free of rank $\binom{n}{k}$, with basis $1 \otimes \dots \otimes e_{i_1} \otimes \dots \otimes 1 \otimes e_{i_k} \otimes 1 \otimes \dots \otimes 1$, and $\{i_1, \dots, i_k\} \subset [1, n]$. The inverse isomorphism maps these to the elements $e_{i_1} \wedge \dots \wedge e_{i_k}$ in $\bigwedge^k M$. $\square$

Remark**1.12**

The definition of tensor, symmetric, and exterior powers are **functorial**: given a map $f : M \rightarrow N$ of R -modules, we get maps $T^n f : T^n M \rightarrow T^n N$, $S^n f : S^n M \rightarrow S^n N$ and $\bigwedge^n f : \bigwedge^n M \rightarrow \bigwedge^n N$ for all $n \geq 0$.

For example, $\bigwedge^n f(x_1 \wedge \dots \wedge x_n) := f(x_1) \wedge \dots \wedge f(x_n)$

$$\begin{array}{ccc} (x_1, \dots, x_n) & \xrightarrow{\quad} & f(x_1) \wedge \dots \wedge f(x_n) \\ \downarrow \tilde{f}^{(n)} & \nearrow & \uparrow \\ T^n M & & \\ & \nearrow & \\ x_1 \otimes \dots \otimes x_n & \xrightarrow{\quad} & \bigwedge^n f \\ \downarrow & & \end{array}$$

(Note: I'm not sure if this is correct)

Corollary**1.13**

If the R -module M has n generators, then $\bigwedge^k M = 0$ for all $k > n$.

Proof. To say that M has n generators is equivalent to saying that there is a surjective R -linear map $\pi : F \rightarrow M$, where

$$F = \bigoplus_{i=1}^n Re_i \cong R^n$$

is a free R -module of rank n . We get an induced map $\bigwedge^k \pi : \bigwedge^k F \rightarrow \bigwedge^k M$ which is also surjective

[In fact, if $f : M \rightarrow N$ is surjective, then $T^n f, S^n f, \wedge^n f$ are all surjective]. If $k > n$, then the corollary gives $\wedge^k F = 0$, hence $\wedge^k M = 0$ also. $\square$

Challenge

If M and N are free R -modules, and $f : M \hookrightarrow N$ is injective, what about $T^n f, S^n f$, and $\wedge^n f$?

Example 1.14. For M a free R -module of rank n with basis $e_1, \dots, e_n$, $\wedge^\bullet M$ is free of rank 2^n with basis $e_{i_1} \wedge \dots \wedge e_{i_k}$ for $1 \leq i_1 < \dots < i_k \leq n$. As an algebra, $\wedge^\bullet M$ is generated by $e_1, \dots, e_n$ modulo the relations $e_i \wedge e_i = 0$ and $e_i \wedge e_j + e_j \wedge e_i = 0$ for $i, j \in [1, n]$. (Recall: $S^\bullet M \equiv R[e_1, \dots, e_n]$, $T^\bullet M \equiv R\langle e_1, \dots, e_n \rangle$).

2. Determinants

Definition

2.1

Let F be a free R -module on n generators, and $T : F \rightarrow F$ be an R -linear map. Then, we get an induced map $\wedge^n T : \wedge^n F \rightarrow \wedge^n F$. Here, $\wedge^n F$ is free on one generator (namely $e_1 \wedge \dots \wedge e_n$, where $e_1, \dots, e_n$ is an R -basis of F). This implies that $\wedge^n T(x) = \lambda x$ for all $x \in \wedge^n F$, and for some $\lambda \in R$. This λ is called the **determinant** of T , denoted $\det T$.

Note

The determinant $\det T$ is defined independently of the choice of basis. So, we have

$$\wedge^n T(x) = (\det T) \cdot x$$

for all $x \in \wedge^n F$.

Example 2.2 (Formula for $\det T$). Let $e_1, \dots, e_n$ be a basis for F . For any $\sigma \in S_n$, we can think of σ as a linear map $\sigma : F \rightarrow F$ given by $\sigma(e_i) := e_{\sigma(i)}$. Then, we have

$$\wedge^n \sigma(e_1 \wedge \dots \wedge e_n) = \sigma(e_1) \wedge \dots \wedge \sigma(e_n) = \operatorname{sgn}(\sigma) \cdot e_1 \wedge \dots \wedge e_n,$$

hence $\det(\sigma) = \operatorname{sgn}(\sigma)$.

Formula for $\det T$

2.3

In general, if $T : F \rightarrow F$ is an R -linear map, write $T(e_j) = \sum_{i=1}^n a_{ij} e_i$, $1 \leq j \leq n$. The matrix $A := \{a_{ij}\}_{1 \leq i, j \leq n}$ is $(T)_E$, i.e. the matrix representing T with respect to the ordered basis $E := (e_1, \dots, e_n)$. Then,

$$\det(T) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{\sigma(1),1} a_{\sigma(2),2} \cdots a_{\sigma(n),n}$$

Proof. We have

$$\begin{aligned} \wedge^n T(e_1 \wedge \dots \wedge e_n) &= T(e_1) \wedge \dots \wedge T(e_n) = \sum_{i_1} a_{i_1 1} e_{i_1} \wedge \sum_{i_2} a_{i_2 2} e_{i_2} \wedge \dots \wedge \sum_{i_n} a_{i_n n} e_{i_n} \\ &= \sum_{i_1, \dots, i_n} a_{i_1 1} a_{i_2 2} \cdots a_{i_n n} \cdot e_{i_1} \wedge \dots \wedge e_{i_n} = \sum_{\sigma \in S_n} a_{\sigma(1)1} a_{\sigma(2)2} \cdots a_{\sigma(n)n} \cdot e_{\sigma(1)} \wedge \dots \wedge e_{\sigma(n)} \end{aligned}$$

$$= \left(\sum_{\sigma \in S_n} \text{sgn}(\sigma) a_{\sigma(1)1} \cdots a_{\sigma(n)n} \right) e_1 \wedge \cdots \wedge e_n$$

□

Remark**2.4**

We denote $\det T$ by $\det A$ for any $A \in M_n(R)$ representing T .

Example 2.5.

$$\det \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} = a_{11}a_{22} - a_{21}a_{12}$$

where the first term corresponds to the identity permutation, and the second corresponds to the permutation $(1\ 2) \in S_2$.

Proposition**2.6**

If $S, T : F \rightarrow F$ are 2 R -linear maps, then

$$\det(T \circ S) = \det(T) \cdot \det(S)$$

Proof. $\wedge^n F \xrightarrow{\wedge^n S} \wedge^n F \xrightarrow{\wedge^n T} \wedge^n F$ commutes.

$$\searrow \quad \xrightarrow{\wedge^n(T \circ S)} \quad \nearrow$$

□

Proposition**2.7**

For any $A \in M_n(R)$, we have $\det(A^t) = \det(A)$.

Proof. If $A = \{a_{ij}\}$, then $A^t = \{b_{ij}\}$ where $b_{ij} = a_{ji}$. Therefore,

$$\begin{aligned} \det(A^t) &= \sum_{\sigma \in S_n} \text{sgn}(\sigma) a_{1\sigma(1)} a_{2\sigma(2)} \cdots a_{n\sigma(n)} = \sum_{\sigma \in S_n} \text{sgn}(\sigma) a_{\sigma^{-1}(1)1} \cdots a_{\sigma^{-1}(n)n} \\ &= \sum_{\sigma \in S_n} \text{sgn}(\sigma^{-1}) a_{\sigma^{-1}(1)1} \cdots a_{\sigma^{-1}(n)n} = \det(A) \end{aligned}$$

□

Cramer's Rule**2.8**

Let $A = \{a_{ij}\} \in M_n(R)$. If $A \cdot \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}$ with $a_{ij}, x_i, b_i \in R$, then we have

$$\det(A) \cdot x_i = \det \begin{pmatrix} a_{11} & \cdots & b_1 & \cdots & a_{1n} \\ \vdots & & \vdots & & \vdots \\ a_{1n} & \cdots & b_n & \cdots & a_{nn} \end{pmatrix}$$

where the b column is the i -th column.

Proof. Let $F = R^n$, and regard its elements as column vectors. Let $A = (\vec{a}_1 \cdots \vec{a}_n)$ where $\vec{a}_1, \dots, \vec{a}_n$

are the columns of A , and $\vec{x}, \vec{b} = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} \in F$. Then, $A\vec{x} = \vec{b}$ implies that $\vec{b} = x_1\vec{a}_1 \cdots x_n\vec{a}_n$.

Therefore,

$$\det(\vec{a}_1 \cdots \vec{b} \cdots \vec{a}_n) = \det(\vec{a}_1 \cdots \sum_{j=1}^n x_j \vec{a}_j \cdots \vec{a}_n) = \sum_{j=1}^n x_j \det(\vec{a}_1 \cdots \vec{a}_j \cdots \vec{a}_n) = \det(A)x_i.$$

□

Corollary

2.9

If $\det T$ is not a zero divisor in R , then T is injective.

Proof. By Cramer's Rule,

$$x = \sum_{i=1}^n x_i e_i \in \ker T \iff T(x) = 0$$

implies that $\det(T) \cdot x_i = 0$ for all $1 \leq i \leq n$. Since $\det(T)$ is not a zero divisor in R , we get $x_i = 0, 1 \leq i \leq n$. □

Exercises

1. Prove the converse to the above corollary.
2. Prove the usual Laplace expansion of $\det(A)$ along rows and columns. A generalization is on the homework.

Remark

2.10

1. Here is a different method of defining determinants:

Let $M := R^n$. Then, $\det : M \times \cdots \times M \rightarrow R$ is the unique alternating multilinear map such that $\det(I_n) = 1$. This defines $\det(A)$ for any $A \in M_n(R)$

2. Determinants arise naturally as signed volumes:

For $v_1, \dots, v_n \in \mathbb{R}^n$, let $V_S(v_1, \dots, v_n)$ denote the signed volume of the polytope

$$P := \left\{ \sum_{i=1}^n \lambda_i v_i \mid 0 \leq \lambda_i \leq 1 \right\}$$

with "sides" $v_1, \dots, v_n$. Precisely, we have

$$V_S(v_1, \dots, v_n) = \begin{cases} 0 & \text{if } v_1, \dots, v_n \text{ are linearly dependent} \\ \text{Vol}(P) & \text{if } v_1, \dots, v_n \text{ are a positively oriented basis} \\ -\text{Vol}(P) & \text{otherwise} \end{cases}$$

where positively oriented means $\det(v_1 \cdots v_n) > 0$. Then,

$$\text{Vol}(P) = |\det(\vec{v}_1 \cdots \vec{v}_n)|$$

3. Localization of Rings and Modules

Definition

3.1

Let R be a commutative ring. A subset S of R with $1 \in S$ is **multiplicatively closed** if $x, y \in S \implies xy \in S$. For such S , there exists a commutative ring $S^{-1}R$ with a map of rings $\pi : R \rightarrow S^{-1}R$ such that the following universal property holds:

For any ring homomorphism $f : R \rightarrow R'$ such that $f(s)$ is a unit in R' for each $s \in S$, there exists a unique map $\tilde{f} : S^{-1}R \rightarrow R'$ such that $\tilde{f} \circ \pi = f$. That is, the below diagram commutes:

$$\begin{array}{ccc} R & \xrightarrow{\pi} & S^{-1}R \\ & \searrow f & \swarrow \exists! \tilde{f} \\ & R' & \end{array}$$

Proof. Construction:

$$S^{-1}R := R \times S / \sim$$

where $(r, s) \sim (r', s') \iff u(rs' - sr') = 0$ for some $u \in S$. Denote the equivalence classes of (r, s) by $\frac{r}{s}$, and define the operations in $S^{-1}R$ by the usual addition and multiplication rules of fractions. Then, the map $\pi : R \rightarrow S^{-1}R$ sends r to $\frac{r}{1}$.

Remark

3.2

$$\ker \pi = \{r \in R \mid sr = 0 \text{ for some } s \in S\}$$

so π is injective

S contains neither zero nor any zero divisor in R . Also, $S^{-1}R = 0$ if $0 \in S$. □

Definition

3.3

This definition extends to modules M over R (with S still a subring of R). We define $S^{-1}M$ as a set of equivalence classes $[\frac{m}{s}]$, where $m \in M$ and $s \in S$, and where $\frac{m}{s} \sim \frac{m'}{s'} \iff s''(s'm - sm') = 0$ for some $s'' \in S$.

Remark

3.4

There is a map $\pi : M \rightarrow S^{-1}M$ with $\pi(m) = [\frac{m}{1}]$ for all $m \in M$. Also,

$$\ker \pi = \{m \in M \mid sm = 0 \text{ for some } s \in S\}.$$

Theorem

3.5

If M and N are R -modules, and $f : M \rightarrow N$ is a module homomorphism, then for $S \subset R$ multiplicatively closed, we get an induced $S^{-1}R$ -module map $S^{-1}M \rightarrow S^{-1}N$ sending $\frac{m}{s} \mapsto \frac{f(m)}{s}$.

Proposition

3.6

$$S^{-1}M \cong S^{-1}R \otimes_R M$$

as $S^{-1}R$ modules. That is, $S^{-1}M$ is the $S^{-1}R$ module obtained by extension of scalars from the R -module M .

Proof. We have a map $S^{-1}R \otimes M \rightarrow S^{-1}M$ sending $\frac{r}{s} \otimes m \mapsto \frac{rm}{s}$ with inverse $S^{-1}M \rightarrow S^{-1}R \otimes M$ sending $\frac{m}{s} \mapsto \frac{1}{s} \otimes m$. To see that the latter map is well defined, note that if $\frac{m}{s} = \frac{m'}{s'}$ in $S^{-1}M$, then $x(s'm - sm') = 0$ for some $x \in S$. So,

$$\frac{1}{s} \otimes m = \frac{1}{xs's} \otimes (s'xm) = \frac{1}{xs's} \otimes (xsm') = \frac{1}{s} \otimes m'$$

□

We'll use these constructions when $S^{-1}R$ is the fraction field K of an integral domain (so when $S = R \setminus \{0\}$).

Exercise

Show that

$$S^{-1}(M \oplus N) \cong S^{-1}M \oplus S^{-1}N$$

for any two R -modules M and N .

4. Finitely Generated Modules Over a PID

Goal

Classify all finitely generated modules over a PID, up to isomorphism.

Example 4.1. When $R = \mathbb{Z}$, this gives a classification of all finitely generated abelian groups.

Definition

4.2

Given any R -module M , the **torsion submodule** of M , denoted M_{tors} , is defined by

$$M_{\text{tors}} = \{x \in M \mid rx = 0 \text{ for some non-zero } r \in R\}$$

Definition

4.3

We say that M is **torsion free** if $M_{\text{tors}} = 0$.

Example 4.4. 1. Any free R -module R^n is torsion free, provided R is a domain.

2. The converse of (1) is false, e.g. $\mathbb{Q}$ is a torsion free $\mathbb{Z}$ -module, but $\mathbb{Q}$ is not free as a $\mathbb{Z}$ -module.

3. If $M = \mathbb{Z}^{10} \oplus \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/8\mathbb{Z}$, then $M_{\text{tors}} = \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/8\mathbb{Z}$.

Proposition

4.5

We have an exact sequence

$$0 \rightarrow M_{\text{tors}} \rightarrow M \rightarrow M/M_{\text{tors}} \rightarrow 0$$

for any module M . The quotient M/M_{tors} is torsion free.

Suppose that R is an integral domain with quotient field K . That is, let $K = S^{-1}R$, where $S = R \setminus \{0\}$. Moreover, for any R -module M , we set

$$M_K := S^{-1}M,$$

which is a K -vector space. The natural map $\alpha : M \rightarrow M_K$ defined by $m \mapsto \left[\frac{m}{1}\right]$ has

$$\ker \alpha = \left\{m \in M \mid \frac{m}{1} \sim \frac{0}{1}\right\} = \{m \in M \mid \exists r \in R \setminus \{0\}, rm = 0\} = M_{\text{tors}}.$$

Hence, M is torsion free $\iff M \xrightarrow{\alpha} M_K$ as R -modules.

Example 4.6. 1. If $M = R$, then $M_K = K$. More generally,, if $M = R^n$ is free of rank n , then

$$M_K = K^n$$

2. Suppose $M = \mathbb{Z}/n\mathbb{Z}$ as a $\mathbb{Z}$ -module. Then, $M_{\mathbb{Q}} = \{0\}$. Indeed, for each $k \in \mathbb{Z}^\times$, we have

$$\frac{x}{k} = \frac{nx}{nk} = \frac{0}{nk} = 0 \text{ for all } x \in M. \text{ More generally, if } M = \mathbb{Z}^n \oplus \mathbb{Z}/k_z\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/k_r\mathbb{Z}, \text{ then } M_{\mathbb{Q}} \cong \mathbb{Q}^n.$$

Definition

4.7

Let R be a domain, and M and R -module. The **rank** of M , denote $\text{rank}(M)$, is the dimension of M_K as a K -vector space.

Remark

4.8

If M is finitely generated, then $\text{rank}(M)$ is finite.

Theorem

4.9

Suppose R is a PID, and M is a finitely generated torsion-free R -module. Then, M is free.

Proof. We induct on $\text{rank}(M)$. If $\text{rank}(M) = 0$, then $M = 0$, since M is torsion free. Now, suppose $\text{rank}(M) > 0$. Then, M_K is a finite dimensional non-zero K -vector space, so there exists a surjective map $M_K \rightarrow K$. We then have $M \xrightarrow{\alpha} M_K \rightarrow K$, and $\text{Im}(\alpha)$ generates M_K . It follows that there exists a non-zero map $\beta : M \rightarrow K$. Let $M = \langle m_1, \dots, m_r \rangle$, so then multiplying β by a suitable element in R to clear denominators, we obtain a non-zero map $\varphi : M \rightarrow R$. Now, $\text{Im} \varphi = \langle b \rangle$ is an ideal of R (which is the same as a submodule of R). Choose $a \in M$ with $\varphi(a) = b$.

Now, we claim $M \cong \ker \varphi \oplus Ra \cong \ker \varphi \oplus R$ as R -modules: Let $N := \ker \varphi$. Then, given $m \in M$, we write $\varphi(m) = xb = x\varphi(a) = \varphi(xa)$, so $m - xa \in M$. Therefore, $m \in N + Ra$, so $M = N + Ra$. If $ra \in N$ for some $r \in R$, then $0 = \varphi(ra) = r\varphi(a) = rb$, hence $r = 0$, so $ra = 0$. Hence $N \cap Ra = \{0\}$, proving the claim.

The claim implies that $M \cong N \oplus R$. Hence, $M_K \cong N_K \oplus K$, while $\text{rank}(N_K) < \text{rank}(M)$. We hence

conclude by induction. $\square$

Proposition

4.10

If F is finitely generated and free over a PID R , and $N \subset F$ is a submodule. Then, there exists a basis $e_1, \dots, e_n$ of F , and $d_1 \mid d_2 \mid \dots \mid d_k \in R$ for $k \leq n$ (so $\langle d_1 \rangle \supset \dots \supset \langle d_k \rangle$) such that N has a basis $d_1 e_1, \dots, d_k e_k$. In particular, N is free. Moreover, the integer k , and the ideals $\langle d_1 \rangle \supset \dots \supset \langle d_k \rangle$ are uniquely determined by F and N .

Proof. We may assume $N \neq 0$. We use induction on $\text{rank}(F)$. If $\text{rank}(F) = 1$, then $F = R$, so $N = (d) = Rd$ for some $d \in R$.

In general, suppose $N \subset F$ is any submodule. The idea is that a general element of N is $\sum_{i=1}^k r_i d_i e_i = d_1 y$ for some y . Hence, for all $\varphi \in F^* := \text{Hom}(F, R)$, $\varphi(N) \subset (d_1)$. Moreover, there exists $e_1^* \in F^*$ such that $e_1^*(N) = (d_1)$. So, consider the set

$$S := \{\varphi(N) \mid \varphi \in F^*\}$$

of ideals in R . Since R is Noetherian (it's a PID), we know S has a maximal element $=: I$. Write $I = \langle d_1 \rangle$, and let $x \in N$ and $\varphi \in F^*$ be such that $\varphi(x) = d_1$.

Now, let $x = \alpha_1 e_1 + \dots + \alpha_n e_n$. Then, consider $\langle d_1, \alpha_i \rangle = \langle s d_1 + t \alpha_i \rangle$ for some $s, t \in R$. Then,

$$(s\varphi + t e_i^*)(x) = s d_1 + t \alpha_i,$$

so $\langle d_1 \rangle \subset \langle d_1, \alpha_i \rangle \subset (s\varphi + t e_i^*)(N)$. Now, since $\langle d_1 \rangle$ is maximal in S , we deduce that $\langle d_1 \rangle = \langle d_1, \alpha_i \rangle$, hence $d \mid \alpha_i$. Therefore, $x = d_1 y$ for some $y \in F$. Note that $\varphi(y) = 1$. Let $F' := \ker \varphi$. Arguing as in the previous proposition, we get $F \cong F' \oplus Ry$. Moreover, $N = N' \oplus Rx = N' \oplus R d_1 y$, where $N' = N \cap F'$.

Since F' is torsion free, we know by the previous proposition that F' is free, and $\text{rank}(F') = n - 1$, while $N' \subset F'$. By induction on $\text{rank}(F)$, we have a basis $e_2, \dots, e_n$ of F' , and $d_2 \mid d_3 \mid \dots \mid d_k$ such that $N' = R d_2 e_2 \oplus \dots \oplus R d_k e_k$. Now, set $e_1 := y$. Then, $N = R d_1 e_1 \oplus \dots \oplus R d_k e_k$. Moreover, we have $\langle d_1 \rangle \supset \langle d_2 \rangle$, because if not, we can produce $\psi \in F^*$ with $\psi(N) \supset \langle d_1, d_2 \rangle \supsetneq \langle d_1 \rangle$ as before, which gives a contradiction. We have thus proven existence.

For uniqueness, consider

$$M := F/N \cong R^{n-k} \oplus R/\langle d_1 \rangle \oplus \dots \oplus R/\langle d_k \rangle.$$

Then, $n - k = \text{rank}(M) = \text{rank}(F/N)$, so k is uniquely determined. Moreover,

$$M_{\text{tors}} = R/\langle d_1 \rangle \oplus \dots \oplus R/\langle d_k \rangle.$$

Definition

4.11

Given any R -module T , the **annihilator** of T is the ideal of R given by

$$\text{Ann}(T) := \{r \in R \mid rt = 0 \ \forall t \in T\}.$$

Now,

$$\langle d_k \rangle = \text{Ann}(M_{\text{tors}}), \langle d_{k-1} \rangle = \text{Ann}(\bigwedge^2 M_{\text{tors}}), \dots, \langle d_{k+1-m} \rangle = \text{Ann}(\bigwedge^m M_{\text{tors}}), \dots$$

To see this, we use

$$\bigwedge^2(R/\langle d \rangle) = 0 \iff (r_1 + \langle d \rangle) \wedge (r_2 + \langle d \rangle) = r_1 r_2 (1 + \langle d \rangle) \wedge (1 + \langle d \rangle) = 0.$$

Now, recall from last year that

$$R/I \otimes_R R/J \cong R/(I + J)$$

by the map $(r_1 + I) \otimes (r_2 + I) \mapsto r_1 r_2 + (I + J)$. So,

$$\begin{aligned} \bigwedge^m M_{\text{tors}} &= \bigwedge^m \left(\bigoplus_{j=1}^k R/\langle d_j \rangle \right) = \bigoplus_{j_1 < \dots < j_m} R/\langle d_{j_1} \rangle \otimes \dots \otimes R/\langle d_{j_m} \rangle \\ &= \bigoplus_{j_1 < \dots < j_m} R/[\langle d_{j_1} \rangle + \dots + \langle d_{j_m} \rangle] = \bigoplus_{j_1 < \dots < j_m} R/\langle d_{j_1} \rangle \\ &\implies \text{Ann}(\bigwedge^m M_{\text{tors}}) = \langle d_{k+1-m} \rangle \end{aligned}$$

□

Structure Theorem (Invariant Factor Form)

4.12

Let R be a PID, and M a finitely generated R -module. Then, we have

$$M \cong R^m \oplus R/\langle d_1 \rangle \oplus \dots \oplus R/\langle d_k \rangle,$$

where $\langle d_1 \rangle \supset \dots \supset \langle d_k \rangle \neq 0$. Moreover, m, k , and the ideals $\langle d_1 \rangle, \dots, \langle d_k \rangle$ are uniquely determined by M .

Proof. We can produce a surjection $\varphi : F \rightarrow M$, where F is a free module. Let $N := \ker \varphi$, and apply the previous theorem to prove existence. To prove uniqueness, we see that $m = \text{rank}(M)$ is uniquely determined. The rest follows as in the last proposition. □

Definition

4.13

The integer m is the **rank** of M (and equals $\dim_K(M_K)$). The elements $d_1, \dots, d_k$ of R , defined up to units, are called the **invariant factors** of M .

Remark

4.14

If

$$d = \prod_{i=1}^r p_i^{k_i}$$

is the factorization of $d \in R$ into primes, then by the Chinese Remainder Theorem

$$R/\langle d \rangle \cong R/\langle p_1^{k_1} \rangle \times \dots \times R/\langle p_r^{k_r} \rangle$$

as rings. Using this in the last result gives the following theorem.

Structure Theorem (Elementary Divisor Form)

4.15

Given M , a finitely generated module over a PID R , we have

$$M \cong R^m \oplus \bigoplus_{p \text{ prime}} \left(R/\langle p^{n_1(p)} \rangle \oplus \dots \oplus R/\langle p^{n_k(p)} \rangle \right)$$

with $1 \leq n_1(p) \leq \cdots \leq n_k(p)$, where all are uniquely determined. The prime powers $p^{n_j(p)}$ that occur are called the **elementary divisors** of M . (Note: the $\oplus$ above sums over some finite set of primes in R .)

Fundamental Theorem of Finitely Generated Abelian Groups

4.16

If G is any finitely generated abelian group, then

$$G \cong \mathbb{Z}^m \oplus \mathbb{Z}/d_1\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/d_k\mathbb{Z},$$

with $d_1 \mid \cdots \mid d_k$. Moreover, $m, k, d_1, \dots, d_k$ are uniquely determined positive integers.

Example 4.17. 1. Find all abelian groups of order 100: we have $100 = 2 \cdot 50 = 5 \cdot 20 = 10 \cdot 10$, giving 4 groups in invariant factor form:

$$\mathbb{Z}_{100}, \mathbb{Z}_2 \oplus \mathbb{Z}_{50}, \mathbb{Z}_5 \oplus \mathbb{Z}_{20}, \mathbb{Z}_{10} \oplus \mathbb{Z}_{10}$$

and in elementary divisor form:

$$\mathbb{Z}_{2^2} \oplus \mathbb{Z}_{5^2}, \mathbb{Z}_2 \oplus \mathbb{Z}_2 \oplus \mathbb{Z}_{5^2}, \mathbb{Z}_5 \oplus \mathbb{Z}_5 \oplus \mathbb{Z}_{2^2}, \mathbb{Z}_2 \oplus \mathbb{Z}_2 \oplus \mathbb{Z}_5 \oplus \mathbb{Z}_5.$$

2. Write $\mathbb{Z}_2 \oplus \mathbb{Z}_4 \oplus \mathbb{Z}_3 \oplus \mathbb{Z}_5 \oplus \mathbb{Z}_{25}$ in invariant form:

$$\mathbb{Z}_{10} \oplus \mathbb{Z}_{300}$$

3. For any fixed prime p , and $n \geq 1$, the number of isomorphism classes of abelian groups of order p^n is $\pi(n)$, the number of partitions of n . For example, for $n = 4$, we have $\pi(4) = 5$, given by $4 = 3 + 1 = 2 + 2 = 2 + 1 + 1 = 1 + 1 + 1 + 1$.

Definition

4.18

An R -module M is **cyclic** if it is generated by a single element $m \in M$. Equivalently, there exists a surjective linear map $\varphi : R \rightarrow M$, mapping $r \mapsto rm$. If R is a PID, we deduce that $M \cong R/\ker \varphi \cong R/\langle a \rangle$ for some $a \in R$. So, the theorem we proved shows that a finitely generated module M over a PID R is a direct sum of cyclic modules.

5. Canonical Forms for Linear Operators

Let F be a field, V an F -vector space, and $T : V \rightarrow V$ an F -linear operator on V . Then, V becomes an $F[x]$ -module, by the action $f(x) \cdot v := f(T)(v)$ for any $f \in F[x], v \in V$.

Conversely, if M is any $F[x]$ -module, then M is an F -module, i.e. an F -vector space. Also, the action of $X : M \rightarrow M, m \mapsto x \cdot m$ is an F -linear operator on M .

Remark

5.1

An $F[x]$ -submodule of M is just a T -invariant subspace of M .

Example 5.2. 1. Consider $V = F[x]$, the free $F[x]$ module of rank 1. Then, V is infinite dimensional as an F -vector space: $V \cong \bigoplus_{i=0}^{\infty} Fx^i$. The linear map $x : V \rightarrow V$ is the **shift operator**.

2. Suppose $\dim_F V < \infty$, and $T \in \text{Hom}(V, V)$, inducing an $F[x]$ -module structure on V . Since V is finite dimensional, it is a finitely generated $F[x]$ -module, and $F[x]$ is a PID, so the structure theorem applies to V . Since $F[x]^k$ is infinite dimensional, V must be a torsion $F[x]$ -module. (This also follows from the fact that there exists a non-zero polynomial $p \in F[x]$ such that $p(T) = 0$. Therefore, $p(x) \cdot v = 0$ for all $v \in V$.)

Consider $\text{Ann}(V) \subset F[x]$. We must have $\text{Ann}(V) = \langle m(x) \rangle$ for a unique monic polynomial $m(x) \in F[x]$. In particular, $m(x)$ is the monic polynomial of least degree such that $m(T) = 0$. Also, if $q(T) = 0$ for some $q(x) \in F[x]$, then $m(x) \mid q(x)$. We call $m(x)$ the **minimal polynomial** of T .

Definition

5.3

Let $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0$ be a monic polynomial. The **companion matrix** of f is the $n \times n$ matrix

$$C(f) = \begin{pmatrix} -a_{n-1} & 1 & 0 & \cdots & 0 \\ -a_{n-2} & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & 1 \\ -a_0 & 0 & \cdots & 0 & 0 \end{pmatrix}$$

Proof.

$$\begin{aligned} p_{C(f)}(x) &= \det(xI_n - C(f)) = \det \begin{pmatrix} x + a_{n-1} & -1 & & & \\ & a_{n-2} & x - 1 & & \\ & \vdots & & \ddots & \\ & & & & -1 \\ a_0 & & & & x \end{pmatrix} = \cdots \\ &= a_0 + x(a_1 + x(a_2 + \cdots + (a_{n-1} + x) \cdots)). \end{aligned}$$

□

Remark

5.4

Suppose V is a finite dimensional F -vector space, and $T \in \text{Hom}(V, V)$ giving an $F[x]$ -module structure to V . Then, the structure theorem implies $V = W_1 \oplus \cdots \oplus W_K$ splits as a direct sum of T -invariant subspaces, with $W_i \cong F[x] / \langle d_i(x) \rangle$, where $d_i(x) \in F[x]$ are unique monic polynomials in $F[x]$ such that $d_1 \mid d_2 \mid \cdots \mid d_K$.

Corollary

5.5

- (i) V has an ordered basis $\mathcal{C}$ such that $(T)_{\mathcal{C}}$ has block diagonal form, where the blocks have size $\dim W_1, \dots, \dim W_K$
- (ii) The action of T on W_i corresponds to multiplication by x in $F[x] / \langle d_i \rangle$. This allows us to choose a basis of W_i so that $T|_{W_i}$ is represented by a simple matrix.
- (iii) $\text{Ann}(V) = \langle d_K \rangle$. This implies the next proposition.

Proposition

5.6

The minimal polynomial m_T of T equals the largest invariant factor of V . Moreover, all the invariant factors divide $m_T(x)$.

Consider a single invariant subspace $W \cong F[x]/\langle d \rangle$, with

$$d(x) = x^r + a_{r-1}x^{r-1} + \cdots + a_1x + a_0,$$

and $a_i \in F$. Choose an ordered basis

$$(e_1, \dots, e_r) := (\bar{x}^{r-1}, \bar{x}^{r-2}, \dots, \bar{x}, \bar{1})$$

of $F[x]/\langle d \rangle$. Let $\varphi : F[x]/\langle d \rangle \rightarrow W$ be the respective isomorphism. Then, the operator X on $F[x]/\langle d \rangle$ has the form:

$$xe_1 = \bar{x}^r = -a_{r-1}e_1 - \cdots - a_0e_r$$

$$xe_2 = e_1$$

$$\vdots$$

$$xe_r = e_{r-1}.$$

If $w_i = \varphi(e_i) \in W$ for $1 \leq i \leq r$, then W has an ordered basis $\mathcal{B} := (w_1, \dots, w_r)$ with $T(w_1) = -a_{r-1}w_1 - \cdots - a_0w_r$, and $T(w_i) = w_{i-1}$ for $i \geq 2$. Hence we have

$$(T)_{\mathcal{B}} = \begin{pmatrix} -a_{r-1} & 1 & & \\ -a_{r-2} & 0 & 1 & \\ \vdots & & 0 & \ddots \\ & & & \ddots & 1 \\ -a_0 & & & & 0 \end{pmatrix} = C(d).$$

The lemma implies that the characteristic polynomial of $T|_W = \text{char polyn}(C(d)) = d(x)$. Applying this analysis to all the invariant subspaces W_i of T , we obtain a basis $\mathcal{C}$ of V such that

$$(T)_{\mathcal{C}} = \begin{pmatrix} C(d_1) & & \\ & C(d_2) & 0 \\ & 0 & \ddots \\ & & & C(d_k) \end{pmatrix} \quad (1)$$

Definition

5.7

A matrix in the form (1) is said to be in **rational canonical form**. The polynomials $d_1 \mid \cdots \mid d_k$ are the invariant factors of the matrix.

Theorem

5.8

- (i) There is a basis $\mathcal{C}$ for V with respect to which $(T)_{\mathcal{C}}$ is in rational canonical form.
- (ii) The rational canonical form for T is unique. In particular, two linear transformations $V \rightarrow V$ are similar if and only if they have the same rational canonical form.

Proof. (Proving ii) Note that the process of construction the rational canonical form (R.C.F) of T is invertible: if $\mathcal{C}$ is a basis of V so that $(T)_{\mathcal{C}}$ is in rational canonical form, then $V \cong \bigoplus_{i=1}^k W_i$, and it is easy to check that each W_i is a cyclic $F[x]$ -module, with annihilator $d_i(x)$. Uniqueness now follows from the corresponding uniqueness statement in the structure theorem. $\square$

Corollary**5.9**

Let $A \in M_n(F)$ be an $n \times n$ matrix.

- (i) The characteristic polynomial of A is the product $d_1 \cdots d_k$ of all invariant factors of A .
- (ii) (Cayley-Hamilton Theorem) $m_A(x)$ divides the characteristic polynomial $p_A(x)$.
- (iii) The characteristic polynomial $p_A(x)$ divides some power of $m_A(x)$ (namely, $m_A^k(x)$). In particular, these two polynomials have the same roots, not counting multiplicities.

Corollary**5.10**

Let $A \in M_n(F)$ be a nilpotent matrix. Then, A is similar to an upper triangular matrix, whose non-zero entries are 1's adjacent to the main diagonal.

6. Jordan Canonical Form

Assume that F is algebraically closed, e.g. $F = \mathbb{C}$. Then, the elementary divisors of V over $F[x]$ have the form $(x - \lambda)^r$ for some $\lambda \in F, r \geq 1$, while $V \cong \bigoplus_i F[x]/(x - \lambda_i)^{n_i}$ as an $F[x]$ -module.

Suppose now that $W \cong F[x]/(x - \lambda)^r$ is a T -invariant subspace. Then, we choose a basis $(e_1, \dots, e_r)$ for this space with

$$e_1 := (\bar{x} - \lambda)^{r-1}, \dots, e_{r-1} := (\bar{x} - \lambda), e_r := 1.$$

Then,

$$\bar{x}e_1 = \lambda e_1 + (\bar{x} - \lambda)^r = \lambda e_1$$

$$\bar{x}e_2 = e_1 + \lambda e_2$$

$$\bar{x}e_3 = e_2 + \lambda e_3$$

$$\vdots$$

$$\bar{x}e_r = e_{r-1} + \lambda e_r.$$

So, the action of T on W is given by the matrix

$$J_r(\lambda) := \begin{pmatrix} \lambda & 1 & & \\ & \lambda & 1 & 0 \\ & & \ddots & \ddots \\ 0 & & & \ddots & 1 \\ & & & & \lambda \end{pmatrix}$$

Definition

6.1

$J_r(\lambda)$ is called a **Jordan block** of size r with eigenvalue λ . A matrix is said to be in **Jordan canonical form** if it is a block diagonal matrix with Jordan blocks along the main diagonal.

Theorem

6.2

Let $T \in \text{Hom}(V, V)$, and assume F contains all eigenvalues of T . Then, there exists a basis $\mathcal{C}$ of V such that $(T)_{\mathcal{C}}$ is in Jordan canonical form. Moreover, the Jordan canonical form is unique, up to a permutation of the Jordan blocks along the diagonal.

7. Converting a Matrix Into Jordan Canonical Form

Remark

7.1

Every Jordan block $J_k(\lambda)$ has a unique eigenvector (up to scale), namely $e_1 = (1, 0, \dots, 0)$, as $Te_1 = \lambda e_1$. Indeed, the λ -eigenspace $W_\lambda = \ker(T|_W - \lambda I) = \langle e_1 \rangle$ is 1-dimensional. This allows us to equate the dimension of the eigenspace $V_\lambda = \ker(T - \lambda I)$ corresponding to λ with the number of Jordan blocks with eigenvalue λ .

Case 1: The roots of the characteristic polynomial $p_T(x)$ of T are distinct.

Then, the Jordan form is diagonal.

Case 2: The root α of $p_T(x)$ has multiplicity r .

Here, there are various possibilities for the part of the Jordan matrix with α on the diagonal.

$$r = 2: \begin{pmatrix} \alpha & 1 \\ & \alpha \end{pmatrix}, \begin{pmatrix} \alpha & \\ & \alpha \end{pmatrix}$$

$$r = 3: \begin{pmatrix} \alpha & 1 & \\ & \alpha & 1 \\ & & \alpha \end{pmatrix}, \begin{pmatrix} \alpha & 1 & \\ & \alpha & \\ & & \alpha \end{pmatrix}, \begin{pmatrix} \alpha & & \\ & \alpha & \\ & & \alpha \end{pmatrix}$$

$$r = 4: \begin{pmatrix} \alpha & 1 & & \\ & \alpha & 1 & \\ & & \alpha & 1 \\ & & & \alpha \end{pmatrix}, \begin{pmatrix} \alpha & 1 & & \\ & \alpha & 1 & \\ & & \alpha & \\ & & & \alpha \end{pmatrix}, \begin{pmatrix} \alpha & 1 & & \\ & \alpha & & \\ & & \alpha & 1 \\ & & & \alpha \end{pmatrix}, \begin{pmatrix} \alpha & 1 & & \\ & \alpha & & \\ & & \alpha & \\ & & & \alpha \end{pmatrix}, \begin{pmatrix} \alpha & & & \\ & \alpha & & \\ & & \alpha & \\ & & & \alpha \end{pmatrix}$$

The cases $r = 2$ and $r = 3$ are distinguished by computing $\dim V_\alpha$. When $r = 4$, we see that $\dim V_\alpha = \dim \ker(A - \alpha I)$ is given by 1, 2, 2, 3, 4 respectively. The remaining 2 cases are distinguished by computing $\dim(A - \alpha I)^2$ (the matrix $(A - \alpha I)^2$ is non-zero, or zero, respectively).

In general, one can show that the dimensions of the null spaces of the matrices $(A - \alpha I)^k$ for $k = 1, 2, \dots, \frac{r}{2}$ distinguish the Jordan forms in all cases. The spaces $\ker(A - \alpha I)^k$ are called **generalized eigenspaces**.

Remark

7.2

If $M \cong R / \langle a \rangle$ is cyclic, for R a PID, then $\langle a \rangle = \text{Ann}(M)$.

Moreover, any submodule $N \subset M$ is cyclic, with $\text{Ann}(N) = \langle b \rangle$, where $b \mid a$. Conversely, for any divisor $b \mid a$, M has a *unique* submodule N , with $\text{Ann}(N) = \langle b \rangle$

Proof. Submodules of M are given by ideals of $R / \langle a \rangle$, so they're of the form J / I , where $I \subset J \subset R$. Therefore, they're of the form $\langle b' \rangle / \langle a \rangle$, where $\langle a \rangle \subset \langle b' \rangle$, i.e. $b' \mid a$. If we then write $a = bb'$, then the map

$$\begin{aligned} Rb' / Ra &\rightarrow R / Rb = R / \langle b \rangle \\ rb' + Rbb' &\mapsto r + Rb \end{aligned}$$

is an R -module isomorphism, so $\text{Ann}(N) = \langle b \rangle$, and the converse also follows. $\square$

8. Normal Linear Operators, and The Spectral Theorem

Let V be a finite dimensional complex vector space, equipped with a hermitian inner product $\langle \cdot, \cdot \rangle$. Last semester (in the homework), we showed that for all $T \in \text{Hom}(V, V)$, there exists a unique $T^* \in \text{Hom}(V, V)$, called the **adjoint** of T , characterized by the property

$$\langle Tx, y \rangle = \langle x, T^*y \rangle \quad \forall x, y \in V.$$

Moreover, if $A \in M_n(\mathbb{C})$, the **adjoint matrix** is $A^* = \overline{A}^t$.

Remark

8.1

Our constructions/results have analogues for real inner product spaces V , with almost the same proofs.

Lemma

8.2

Let $T \in \text{Hom}(V, V)$ and $\mathcal{B} = (b_1, \dots, b_n)$ be an ordered orthonormal basis of V . Moreover, suppose $A := (T)_{\mathcal{B}} = \{a_{ij}\}$. Then, $(T^*)_{\mathcal{B}} = A^*$.

Proof. Since $T(b_j) = \sum_{i=1}^n a_{ij}b_i$, and $\mathcal{B}$ is orthonormal, we have

$$\begin{aligned} a_{ij} &= \langle T(b_j), b_i \rangle = \langle b_j, T^*(b_i) \rangle = \overline{\langle T^*b_i, b_j \rangle} \\ &\implies \langle T^*(b_j), b_i \rangle = \overline{a_{ji}}. \end{aligned}$$

 $\square$ **Remark**

8.3

The operation $*$ on $M_n(\mathbb{C})$ or on $\text{Hom}(V, V)$ is an analogue of complex conjugation on $\mathbb{C}$. One can check that

- (i) $(S + T)^* = S^* + T^*$
- (ii) $(\lambda T)^* = \overline{\lambda} T^*$

$$(iii) (ST)^* = T^*S^*$$

$$(iv) (T^*)^* = T$$

Motivating Question

Given $T \in \text{Hom}(V, V)$, when is there an orthonormal basis of eigenvectors for T ?

Necessary condition: If $\mathcal{B}$ is orthonormal, and

$$(T)_{\mathcal{B}} = D = \begin{pmatrix} c_1 & & \\ & \ddots & \\ & & c_n \end{pmatrix}$$

is diagonal, then

$$(T^*)_{\mathcal{B}} = \overline{D} = \begin{pmatrix} \bar{c}_1 & & \\ & \ddots & \\ & & \bar{c}_n \end{pmatrix}.$$

If V is a real inner product space, we have $c_i \in \mathbb{R}$, hence $T = T^*$. If V is a $\mathbb{C}$ -inner product space, we may have $T \neq T^*$, however $TT^* = T^*T$.

Definition

8.4

$T \in \text{Hom}(V, V)$ is **self-adjoint** if $T = T^*$, or **normal** if $TT^* = T^*T$

Proposition

8.5

Let V be a hermitian inner product space, $\dim_{\mathbb{C}} V < \infty$, and $T : V \rightarrow V$ a normal linear operator on V .

1. If $f \in \mathbb{C}[x]$ is any polynomial, then $f(T)$ is normal.
2. $\|Tv\| = \|T^*v\|$ for all $v \in V$.
3. $\ker T = (\text{Img } T)^{\perp}$.
4. $T^2v = 0 \implies Tv = 0$ for all $v \in V$.
5. $v \in V$ is an eigenvector for T with eigenvalue $\lambda \in \mathbb{C} \iff v$ is an eigenvector for T^* with eigenvalue $\bar{\lambda} \in \mathbb{C}$.
6. The eigenspaces of distinct eigenvalues of T are orthogonal.

Proof.

(i) Since $(\lambda T^n)^* = \bar{\lambda}(T^*)^n$, this claim follows directly from the definition of normality.

(ii)

$$\|Tv\|^2 = \langle Tv, Tv \rangle = \langle v, T^*Tv \rangle = \langle v, TT^*v \rangle = \langle T^*v, T^*v \rangle = \|T^*v\|^2$$

(iii)

$$u \in (\text{Img } T)^{\perp} \iff \langle u, Tv \rangle = 0 \ \forall v \in V \iff \langle T^*u, v \rangle = 0$$

$$\iff T^*u = 0 \stackrel{(ii)}{\iff} Tu = 0 \iff u \in \ker T$$

(iv)

$$T^2v = 0 \implies Tv \in \operatorname{Im} T \cap \ker T \stackrel{(iii)}{=} \{0\}$$

(v) By (i), $T - \lambda I$ is normal for all $\lambda \in \mathbb{C}$. Then, by (ii),

$$\|(T - \lambda I)(v)\| = 0 \iff \|(T^* - \bar{\lambda}I)(v)\| = 0,$$

hence v is an eigenvector for T with eigenvalue $\lambda \in \mathbb{C} \iff v$ is an eigenvector for T^* with eigenvalue $\bar{\lambda} \in \mathbb{C}$.

(vi) Suppose $Tv = \lambda v$ and $Tv' = \mu v'$, where $v, v' \neq 0$, and $\lambda \neq \mu$. Then, we compute

$$\langle v, v' \rangle = \left\langle \frac{1}{\lambda - \mu} (T - \mu I)(v), v' \right\rangle = \frac{1}{\lambda - \mu} \langle v, (T - \mu I)^*(v') \rangle = \frac{1}{\lambda - \mu} \langle v, 0 \rangle = 0$$

because v' is an eigenvector for T^* with eigenvalue $\bar{\mu}$, by (v).

□

Spectral Theorem (Complex Case)

8.6

Suppose V is a finite dimensional Hermitian space, and $T \in \operatorname{Hom}(V, V)$ is normal. Then, V has an orthonormal basis of eigenvectors for T .

Proof. We claim that the minimal polynomial $m_T(x)$ is a product of distinct irreducible factors. Indeed, if $d(x)$ is an irreducible factor of $m_T(x)$, then we show that $d^2(x)$ does not divide $m_T(x)$. Suppose in fact that $d^2(x) \mid m_T(x)$. Recall that, as a $\mathbb{C}[x]$ -module,

$$V \cong \mathbb{C}[x] / \langle m_T \rangle \oplus \cdots$$

By [Remark 7.2](#), we therefore have that $\mathbb{C}[x] / \langle d^2 \rangle$ is a cyclic submodule of V . Hence, there is a vector $v \in V$ with $d^2(T)(v) = 0$, but $d(T)(v) \neq 0$. However by (i) from [Proposition 8.5](#), we know $d(T)$ is normal, but this contradicts (iv). Thus, as every irreducible polynomial in $\mathbb{C}[x]$ is linear, it follows that $m_T(x)$ is a product of distinct linear factors. Hence, T is diagonalizable (see HW), and the eigenspaces for T are orthogonal by (vi). So, we choose an orthonormal basis for each eigenspace. □

Spectral Theorem (Real Case)

8.7

Suppose V is a finite dimensional real inner product space, and $T : V \rightarrow V$ is a self-adjoint operator. Then, V has an orthonormal basis of eigenvectors for T .

Proof. Let $V_{\mathbb{C}} := V \oplus V$, and we can make $V_{\mathbb{C}}$ into a $\mathbb{C}$ -vector space by setting

$$(a + bi) \cdot (u, v) = (au - bv, bu + av)$$

for all $a, b \in \mathbb{R}$ and $u, v \in V$. Then, we define an inner product on $V_{\mathbb{C}}$ by

$$\langle (u, v), (u', v') \rangle = \langle u, u' \rangle + \langle v, v' \rangle + i(\langle v, u' \rangle - \langle u, v' \rangle)$$

(one can check this). Next, extend T to a linear map $T_{\mathbb{C}} : V_{\mathbb{C}} \rightarrow V_{\mathbb{C}}$ by $T_{\mathbb{C}}(u, v) := (T(u), T(v))$. One can check that $T_{\mathbb{C}}$ is $\mathbb{C}$ -linear on $V_{\mathbb{C}}$, and also self-adjoint. By (v) from [Proposition 8.5](#), all

the eigenvalues of $T_{\mathbb{C}}$ are real, and $m_{T_{\mathbb{C}}}(x)$ is a product of distinct (real) linear factors, hence $m_{T_{\mathbb{C}}} \in \mathbb{R}[x]$. If $f \in \mathbb{R}[x]$, then $f(T_{\mathbb{C}})(u, v) = (f(T)(u), f(T)(v))$. Hence, $m_T(x) = m_{T_{\mathbb{C}}}(x)$, and we conclude that T is diagonalizable. The rest of the proof is the same as in the complex case. $\square$

Remark

8.8

We have $V_{\mathbb{C}} = \mathbb{C} \otimes_{\mathbb{R}} V$, and $T_{\mathbb{C}} = 1_{\mathbb{C}} \otimes_{\mathbb{R}} T$

Proof. Indeed, note that

$$\mathbb{C} \otimes_{\mathbb{R}} V = (R \oplus i\mathbb{R}) \otimes_{\mathbb{R}} V \cong (1 \otimes V) \oplus (i \otimes V) = V \oplus iV,$$

and

$$(a + bi)(1 \otimes u + i \otimes v) = (a + bi) \otimes u + (ai - b) \otimes v = 1 \otimes (au - bv) + i \otimes (bu + av)$$

 $\square$ **Spectral Theorem Matrix Form**

8.9

- (a) If $A \in M_n(\mathbb{C})$ is a normal matrix, then there exists a unitary matrix U such that $A = UDU^{-1} = UDU^*$, where D is diagonal.
- (b) If $A \in M_n(\mathbb{R})$ is symmetric, then there exists an orthogonal matrix P such that $A = PDP^{-1} = PDP^t$, where D is diagonal.

Example 8.10. $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ has eigenvalues 3 and -1 , with corresponding eigenvectors $(1, 1)$ and $(-1, 1)$, respectively. We deduce that

$$A = P \begin{pmatrix} 3 & 0 \\ 0 & -1 \end{pmatrix} P^t,$$

where

$$P = \begin{pmatrix} \sqrt{2}/2 & -\sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{pmatrix}$$

9. Bilinear Forms**Definition**

9.1

Let V be a finite dimensional vector space over a field F , with basis $e_1, \dots, e_n$. Write

$$x = \sum_{i=1}^n x_i e_i, y = \sum_{j=1}^n y_j e_j$$

with $x_i, y_j \in F$. A **bilinear form** on V is a bilinear map $B : V \times V \rightarrow F$, such that

$$B(x, y) = \sum_{i,j=1}^n x_i y_j B(e_i, e_j).$$

The matrix $A := \{B(e_i, e_j)\}_{1 \leq i, j \leq n}$ is called the matrix of B relative to the basis $E = (e_1, \dots, e_n)$.

Conversely, if $A = \{a_{ij}\}$ is any $n \times n$ matrix with entries in F , we define the function

$$B_A(x, y) := x^t A y = \sum_{i,j=1}^n a_{ij} x_i y_j.$$

The determinant $\det\{B(e_i, e_j)\}$ is called the **discriminant** of B .

Suppose we change the basis E to another basis $E' = (e'_1, \dots, e'_n)$, and let $P := (\text{id})_{E, E'}$ be the change of basis matrix, so that if $P = \{p_{ij}\}$, then $e'_i = \sum_{j=1}^n p_{ij} e_j$ for all $i \in \{1, \dots, n\}$. Then,

$$B(e'_i, e'_j) = B\left(\sum_k p_{ik} e_k, \sum_l p_{jl} e_l\right) = \sum_{k,l} p_{ik} B(e_k, e_l) p_{jl}.$$

Hence, if $A = \{B(e_i, e_j)\}$ and $A' = \{B(e'_i, e'_j)\}$, then,

$$A' = P A P^t.$$

We therefore see that $\det(A') = (\det P)^2 \cdot \det(A)$, so the discriminant $\det A$ of B is changed to $(\det P)^2 \cdot \det A$ upon changing the basis. That is, the discriminant does depend on choice of basis.

Definition

9.2

We define $V^L := \{x \in V \mid B(x, v) = 0 \text{ for all } v \in V\}$. We call this the **left radical** of B . Similarly, we have the **right radical** $V^R := \{y \in V \mid B(v, y) = 0 \text{ for all } v \in V\}$

Theorem

9.3

The following are equivalent:

- (i) $V^L = 0$
- (ii) $V^R = 0$
- (iii) The matrix A of B relative to any basis is invertible.

Proof. Let $e_1, \dots, e_n$ be any basis of V , and $A = \{B(e_i, e_j)\}$. Then, clearly

$$x \in V^L \iff B(x, e_i) = 0 \quad \forall i = 1, 2, \dots, n$$

$$x \in V^R \iff B(e_i, x) = 0 \quad \forall i = 1, 2, \dots, n.$$

So, if we write $x = \sum x_j e_j$, then $V^L = 0$ (resp. $V^R = 0$) is equivalent to saying that the linear system $A^t x = 0$ (resp. $Ax = 0$) has only the zero solution. These are equivalent to the condition $\det A \neq 0$. $\square$

Definition

9.4

A bilinear form B is called **non-degenerate** if its matrix with respect to any basis is invertible.

Fix $y \in V$. Then, y gives rise to a linear functional on V , namely $x \mapsto B(x, y)$. This functional only depends on the equivalence class of y modulo V^R . Hence, we get a homomorphism $V \rightarrow V^*$ whose kernel is V^R by definition. Hence, we obtain an injection $V/V^R \hookrightarrow V^*$. If B is non-degenerate, we obtain an isomorphism $V \cong V^*$ by $y \mapsto (x \mapsto B(x, y))$. A similar analysis applies

to the second variable and V^L .

Proposition

9.5

Suppose B is non-degenerate and $\dim_F V < \infty$. Then, any linear functional on V has the form $x \mapsto B(x, y)$ (and also $x \mapsto B(y, x)$) for some $y \in V$.

Definition

9.6

We say that x is **orthogonal** to y if $B(x, y) = 0$. We denote this $x \perp y$.

Theorem

9.7

The relation $\perp$ is symmetric $\iff$

1. B is **alternating**: $B(x, x) = 0$ OR
2. B is **symmetric**: $B(x, y) = B(y, x)$.

Proof. On homework. □

10. Symmetric Bilinear Forms and Quadratic Forms

Suppose $B : V \times V \rightarrow F$ is a symmetric bilinear form, and $\text{char } F \neq 2$.

Definition

10.1

The corresponding **quadratic form** to B is $Q(x) := B(x, x)$, which is a function $Q : V \rightarrow F$.

Remark

10.2

We can recover B from Q by

$$B(x, y) = \frac{1}{2}(Q(x+y) - Q(x) - Q(y))$$

By choosing a basis E of V , we can represent B by a symmetric matrix A so that $B(x, y) = x^t A y$, and $Q(x) = x^t A x$.

Example 10.3.

$$Q(x, y, z) = x^2 + 2y^2 - z^2 + 10xy + 6xz - 4yz = \begin{pmatrix} x & y & z \end{pmatrix} \begin{pmatrix} 1 & 5 & 3 \\ 5 & 2 & -2 \\ 3 & -2 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

Definition

10.4

Suppose $Q = Q_A(x) = x^t A x$ for $A \in M_n(\mathbb{R})$ symmetric. We say that Q is

- (i) **positive definite** if $Q(x) > 0$ for all $x \neq 0$.
- (ii) **negative definite** if $Q(x) < 0$ for all $x \neq 0$.
- (iii) **indefinite** if $Q(x) > 0$ and $Q(x') < 0$ for some x, x' .

To determine which property Q satisfies, we can use the spectral theorem.

Write $A = PDP^t$, where P is orthogonal and $D = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}$ is diagonal. Then, we introduce new coordinates $y := (x)_P = P^{-1}x \iff x = Py$. So,

$$Q = x^t Ax = (Py)^t A(Py) = y^t Dy$$

Example 10.5. Let $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$, so $Q_A(x_1, x_2) = x_1^2 + x_2^2 + 4x_1x_2$. We have $A = PDP^t$ with

$$D = \begin{pmatrix} 3 & 0 \\ 0 & -1 \end{pmatrix}, P = \frac{\sqrt{2}}{2} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}.$$

Then,

$$y = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = P^t \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \frac{\sqrt{2}}{2} \begin{pmatrix} x_1 + x_2 \\ x_1 - x_2 \end{pmatrix},$$

so

$$\begin{aligned} Q(y) &= \begin{pmatrix} y_1 & y_2 \end{pmatrix} \begin{pmatrix} 3 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = 3 \left(\frac{\sqrt{2}}{2}(x_1 + x_2) \right)^2 - \left(\frac{\sqrt{2}}{2}(x_1 - x_2) \right)^2 \\ &= \frac{1}{2}(3(x_1 + x_2)^2 - (x_1 - x_2)^2). \end{aligned}$$

Hence, Q is indefinite.

Theorem

10.6

For any real quadratic form $Q = x^t Ax$ in n variables, there is an orthogonal change of basis in which that form becomes

$$Q(y_1, \dots, y_n) = \lambda_1 y_1^2 + \dots + \lambda_n y_n^2, \quad \lambda_i \in \mathbb{R}.$$

Q is positive (resp. negative) definite if $\lambda_i > 0$ (resp. $\lambda_i < 0$) for all $i = 1, \dots, n$. Q is indefinite if $\lambda_i > 0$ and $\lambda_j < 0$ for some i, j .

Definition

10.7

We say that Q is **positive semi-definite** (resp. **negative semi-definite**) if $Q(x) \geq 0$ (resp. $Q(x) \leq 0$) for all $x \in \mathbb{R}^n$.

Over any field F , one can show:

Theorem

10.8

Let B be a symmetric bilinear form on a vector space V over a field F with $\text{char } F \neq 2$. Then,

V has a basis $E = (e_1, \dots, e_n)$ such that the matrix of B relative to E has the form

$$\begin{pmatrix} \lambda_1 & & & & \\ & \ddots & & & \\ & & \lambda_r & & \\ & & & 0 & \\ & & & & \ddots \\ & & & & & 0 \end{pmatrix}$$

where $\lambda_i \in F \setminus \{0\}$ for all $i \in \{1, \dots, r\}$. Note that $B(e_i, e_j) = 0$ for all $i \neq j$, so E is an **orthogonal** basis of V .

11. Alternating Bilinear Forms

Let V be a vector space over F , and $B : V \times V \rightarrow F$ an alternating ($B(x, x) = 0$) bilinear form on V . Then, $B(x, y) = -B(y, x)$ for all $x, y \in V$.

If $E = (e_1, \dots, e_n)$ is any basis of V , we see that $B(e_i, e_j) = -B(e_j, e_i)$ for $i \neq j$, and $B(e_i, e_i) = 0$ for all i . Hence, if $A = \{B(e_i, e_j)\}_{i,j}$, then $A + A^t = 0$, i.e. A is **skew-symmetric**.

Conversely, if A is a skew-symmetric matrix in $M_n(F)$, then $B(x, y) := x^t A y$ is an alternating form, since

$$B(x, x) = \sum_{i,j} a_{ij} x_i x_j = \sum_{i < j} (a_{ij} + a_{ji}) x_i x_j = 0.$$

Theorem

11.1

If B is an alternating bilinear form, then there exists a basis of V such that the matrix of B relative to this basis has the form

$$A' := \begin{pmatrix} S & & & \\ & \ddots & & \\ & & S & \\ & & & 0 & \\ & & & & \ddots \\ & & & & & 0 \end{pmatrix},$$

where $S = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$.

Proof. If $B = 0$, the result is trivial, so assume $B \neq 0$. Then, we can find $u, v \in V$ such that $B(u, v) \neq 0$. Since $B(u, \lambda u) = 0$ for all $\lambda \in F$, we know u and v are linearly independent. Say $B(u, v) = \lambda \neq 0$, and define $u_1 := u, v_1 := \lambda^{-1}v$. Then, $B(u_1, v_1) = 1 = -B(v_1, u_1)$.

Inductively, suppose we found linearly independent vectors $E_k := (u_1, v_1, \dots, u_k, v_k)$ such that $B(u_i, v_i) = 1 = -B(v_i, u_i)$ for all $i \in \{1, \dots, k\}$, and $B(z, w) = 0$ for every other pair $z, w \in E_k$.

Let $V_k := \text{Span}\{u_1, v_1, \dots, u_k, v_k\}$. We claim $V = V_k \oplus V_k^\perp$. Indeed, since the matrix $B|_{V_k}$ is the diagonal matrix with k S 's, which is invertible, we know that $B|_{V_k}$ is non-degenerate. So, $V_k \cap V_k^\perp = 0$. Moreover, if $x \in V$, we may define $y := x - \sum_{i=1}^k B(x, v_i)u_i + \sum_{i=1}^k B(x, u_i)v_i$. Then, check that $B(y, u_j) = B(y, v_j) = 0$ for all $j \in \{1, \dots, k\}$. Hence $y \in V_k^\perp$, so it follows that $x \in V_k \oplus V_k^\perp$.

Now, if $B|_{V_k^\perp} = 0$, then choose a basis $w_1, \dots, w_{n-2k}$ for $V_k^\perp$, and append it to E_k to obtain the required basis. Otherwise, we can choose $u_{k+1}, v_{k+1} \in V_k^\perp$ such that $B(u_{k+1}, v_{k+1}) = 1 = -B(v_{k+1}, u_{k+1})$, and then proceed inductively. $\square$

Corollary**11.2**

The rank of a skew-symmetric matrix with entries in F is even, and its determinant is a square.

Proof. If $A + A^t = 0$, and $B_A(x, y) = x^t A y$, then the previous theorem gives $A = P A' P^t$, where A' is the matrix from the theorem, and P is a change of basis matrix. So, P is invertible, hence A and A' have the same rank, so A has even rank. Also, $\det(A) = (\det P)^2 \cdot \det(A') = (\det P)^2$ (or 0). $\square$

12. The Pfaffian

If $A = \{a_{ij}\}_{1 \leq i < j \leq n}$ is skew-symmetric, then $\det A$ is a polynomial in the variables a_{ij} for $i < j$, i.e. in $R := \mathbb{Z}[a_{ij}, i < j]$.

Example 12.1. If n is odd, then $\det A = 0$. We can also see this as

$$\det A = \det(A^t) = \det(-A) = (-1)^n \det A$$

Example 12.2.

$$\det \begin{pmatrix} 0 & a_{12} & a_{13} & a_{14} \\ & 0 & a_{23} & a_{24} \\ & & 0 & a_{34} \\ & & & 0 \end{pmatrix} = (a_{12}a_{34} - a_{13}a_{24} + a_{14}a_{23})^2.$$

We'll call the polynomial inside the square the *Pfaffian*.

The quotient field of R is $\mathbb{Q}(a_{ij}, i < j)$. The corollary implies $\det A = \left(\frac{f}{g}\right)^2$ for some $f, g \in R$. Therefore, $f^2 = \det(A) \cdot g^2$, hence $g^2 \mid f^2$ in R , so $g \mid f$ in R . That is, $\frac{f}{g} = h$ for some $h \in R$, hence $\det A = h^2$, and h is determined up to a sign ± 1 by this relation. When $A = A' = \text{diag}(S, S, \dots, S)$, we obtain $1 = \det A' = h(a'_{12} = 1, \dots)^2$. We fix the sign of h so that $h(a'_{12} = 1, \dots) = 1$.

Definition**12.3**

The resulting polynomial from the discussion above is denoted $\text{Pf}(A)$, the **Pfaffian** of A .

In general,

$$\text{Pf}(A) = \sum \text{sgn}(\sigma) a_{\sigma(1)\sigma(2)} \cdots a_{\sigma(2n-1)\sigma(2n)}$$

summed over all $\sigma \in S_{2n}$ such that $\sigma(2r-1) < \sigma(2r)$ for all r , and $\sigma(1) < \sigma(3) < \cdots < \sigma(2n-1)$.

There are $(2n-1)!! = (2n-1)(2n-3) \cdots 3 \cdot 1$ terms in this sum.

There also exists a Laplace type expansion

$$\text{Pf } A = a_{12} \text{Pf } A_{12} - a_{13} \text{Pf } A_{13} + \cdots + (-1)^n a_{1n} \text{Pf } A_{1n}$$

Categories

13. Categories, Morphisms, and Functors

Definition

13.1

A **category** $\mathcal{C}$ consists of a class $\text{Ob } \mathcal{C}$ of **objects**, and for every pairs (A, B) of objects, a set $\text{Hom}_{\mathcal{C}}(A, B)$ (or $\text{Hom}(A, B)$) whose elements are called **morphisms** with **domain** A and **codomain** B (or ‘from A to B ’). Moreover, for each triple (A, B, C) of objects, we are given a map (**composition law**) $\text{Hom}(A, B) \times \text{Hom}(B, C) \rightarrow \text{Hom}(A, C)$ sending $(f, g) \mapsto gf$ such that the following are true:

- (i) If $(A, B) \neq (C, D)$, then $\text{Hom}(A, B) \cap \text{Hom}(C, D) = \emptyset$
- (ii) If $f \in \text{Hom}(A, B)$ (denoted $f : A \rightarrow B$), $g : B \rightarrow C$, and $h : C \rightarrow D$, then $(hg)f = h(gf)$.
- (iii) For every $A \in \text{Ob } \mathcal{C}$, there exists an element $1_A \in \text{Hom}(A, A)$ such that $f1_A = f$ for all $f \in \text{Hom}(A, B)$, and $1_A g = g$ for all $g \in \text{Hom}(B, A)$ [for all $B \in \text{Ob } \mathcal{C}$].

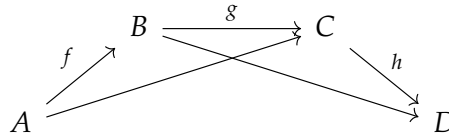
The morphisms in $\mathcal{C}$ are called **arrows**.

Category theory is the theory of the arrows.

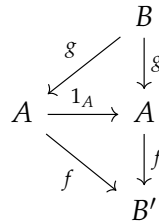
Remark

13.2

Property (ii) is equivalent to saying that the diagram



commutes. Property (iii) is equivalent to saying that the diagram



commutes.

Definition

13.3

$\mathcal{C}$ is a **subcategory** of $\mathcal{D}$ if $\text{Ob } \mathcal{C} \subset \text{Ob } \mathcal{D}$, and for all $A, B \in \text{Ob } \mathcal{C}$, we have $\text{Hom}_{\mathcal{C}}(A, B) \subset \text{Hom}_{\mathcal{D}}(A, B)$.

Example 13.4. 1. *Sets*, the category of sets, with the usual notion $\text{Hom}(A, B) =: B^A :=$ all functions

$$A \rightarrow B.$$

2. The category **Grp** of groups, with group homomorphisms. It has a subcategory **Ab** of abelian groups.
3. The category **Ring** of rings and ring homomorphisms.
4. The category **Top** of topological spaces and continuous maps.
5. For any ring R , we have the category $R\text{-mod}$ of (left) R -modules.
6. Given a category $\mathcal{C}$, the **opposite category** is defined by $\mathcal{C}^{\text{op}}$ has $\text{Ob } \mathcal{C}^{\text{op}} = \text{Ob } \mathcal{C}$ and $\text{Hom}_{\mathcal{C}^{\text{op}}}(A, B) := \text{Hom}_{\mathcal{C}}(B, A)$. So, $\mathcal{C}^{\text{op}}$ is $\mathcal{C}$ with the arrows reversed.
7. Let G be any group. Then, G defines a category $\mathbf{G}$ with a single object $\cdot$, so $\text{Ob } \mathbf{G} = \{\cdot\}$, and $\text{Hom}(\cdot, \cdot) := G$, with $1 := e \in G$ and $fg \in \text{Hom}(\cdot, \cdot)$ the product in G .
8. We can define **isomorphisms** in a category in the obvious way. However, there are several ways to define monomorphisms (injective) and epimorphisms (surjective).

Definition

13.5

Let $\mathcal{C}$ and $\mathcal{D}$ be two categories. A **(covariant) functor** $\mathcal{F} : \mathcal{C} \rightarrow \mathcal{D}$ consists of

- (i) A map $\text{Ob } \mathcal{C} \rightarrow \text{Ob } \mathcal{D}$ with $A \mapsto \mathcal{F}(A)$
- (ii) For every pair (A, B) of objects of $\mathcal{C}$, a map $\text{Hom}_{\mathcal{C}}(A, B) \rightarrow \text{Hom}_{\mathcal{D}}(\mathcal{F}(A), \mathcal{F}(B))$ with $f \mapsto \mathcal{F}f$ such that $\mathcal{F}(gf) = \mathcal{F}g\mathcal{F}f$ and $\mathcal{F}1_A = 1_{\mathcal{F}(A)}$.

A **contravariant functor** $\mathcal{F} : \mathcal{C} \rightarrow \mathcal{D}$ is a covariant functor from $\mathcal{C}^{\text{op}} \rightarrow \mathcal{D}$. So, for each $f \in \text{Hom}_{\mathcal{C}}(A, B)$, we have $\mathcal{F}f \in \text{Hom}_{\mathcal{D}}(\mathcal{F}(B), \mathcal{F}(A))$ and $\mathcal{F}(gf) = \mathcal{F}f\mathcal{F}g$. So, contravariant functors reverse arrows.

Example 13.6. 1. $\mathcal{F} : \mathbf{Grp} \rightarrow \mathbf{Sets}$ maps any group G to the underlying set G , and $\varphi : G_1 \rightarrow G_2$ to the underlying set map. The functor $\mathcal{F}$ is called the **forgetful functor**.

2. The abelianization functor $\mathbf{Grp} \rightarrow \mathbf{Ab}$ sends G to $G^{\text{ab}} := G/[G, G]$, where $[G, G]$ is the **commutator subgroup**. Each group homomorphism $f : G_1 \rightarrow G_2$ maps $[G_1, G_1]$ to $[G_2, G_2]$, so induces a map $f^{\text{ab}} : G_1^{\text{ab}} \rightarrow G_2^{\text{ab}}$
3. The contravariant **power set functor** $\mathcal{P} : \mathbf{Sets} \rightarrow \mathbf{Sets}$ sends any set A to its power set $\mathcal{P}(A)$. Given any set map $f : A \rightarrow B$, we have $\mathcal{P}f : \mathcal{P}(B) \rightarrow \mathcal{P}(A)$ mapping $U \subset B$ to $f^{-1}(U) \subset A$.
4. For any R -module M , we get a covariant functor $\text{Hom}_R(M, -) : R\text{-mod} \rightarrow \mathbf{Ab}$ sending N to $\text{Hom}_R(M, N)$ and $\varphi : N \rightarrow N'$ to $\text{Hom}(M, \varphi : \text{Hom}(M, N) \rightarrow \text{Hom}(M, N')$, sending $f \mapsto \varphi \circ f$. We also have a contravariant functor $\text{Hom}_R(-, M) : R\text{-mod} \rightarrow \mathbf{Ab}$. These are the **Hom functors**.
5. For any R -module N , we get a covariant functor $- \otimes_R N : R\text{-mod} \rightarrow R\text{-mod}$ defined by $M \mapsto M \otimes_R N$, sending $f : M \rightarrow M'$ to $f \otimes 1 : M \otimes N \rightarrow M' \otimes N$.

Definition

13.7

Let $\mathcal{C}$ and $\mathcal{D}$ be categories, and $\mathcal{F}, \mathcal{G} : \mathcal{C} \rightarrow \mathcal{D}$ be covariant functors. A **natural transformation** from $\mathcal{F}$ to $\mathcal{G}$ is a map η that assigns to each $A \in \text{Ob } \mathcal{C}$ a morphism $\eta_A : \mathcal{F}(A) \rightarrow \mathcal{G}(A)$, so that for all $A, B \in \text{Ob } \mathcal{C}$, and for all $f \in \text{Hom}_{\mathcal{C}}(A, B)$, we have $(\mathcal{G}f)\eta_A = \eta_B(\mathcal{F}f)$. That is, the

following diagram commutes:

$$\begin{array}{ccc}
 \mathcal{F}(A) & \xrightarrow{\eta_A} & \mathcal{G}(A) \\
 \mathcal{F}f \downarrow & \circlearrowleft & \downarrow \mathcal{G}f \\
 \mathcal{F}(B) & \xrightarrow{\eta_B} & \mathcal{G}(B)
 \end{array}$$

Example 13.8. Consider the functor $GL_n : \mathbf{CommRings} \rightarrow \mathbf{Grp}$ defined by $R \mapsto GL_n(R)$, and $\mathcal{G} : \mathbf{CommRings} \rightarrow \mathbf{Grp}$ defined by $R \mapsto R^\times$, the group of unit functors.

The determinant $\det$ is a natural transformation $GL_n \rightarrow \mathcal{G}$ since it is defined by the same polynomial for all rings, so that for any ring homomorphisms $f : R \rightarrow S$, the following diagram commutes:

$$\begin{array}{ccc}
 GL_n(R) & \xrightarrow{\det} & R^\times \\
 GL_n(f) \downarrow & & \downarrow f \\
 GL_n(S) & \xrightarrow{\det} & S^\times
 \end{array}$$

Definition

13.9

If $\mathcal{F}, \mathcal{G} : \mathcal{C} \rightarrow \mathcal{D}$ are two functors, we say that a natural transformation $\eta : \mathcal{F} \rightarrow \mathcal{G}$ is an **isomorphism** if each $\eta_A : \mathcal{F}(A) \rightarrow \mathcal{G}(A)$ is an isomorphism, for all $A \in \text{Ob } \mathcal{C}$.

Example 13.10. Consider the identity functor $1_{R\text{-mod}}$ and the double dual functor D^2 on $R\text{-mod}$. For each R -module M , there is a natural map $\eta_M : M \rightarrow M^{**}$ sending $x \mapsto [\hat{x}(f) := f(x)]$. Then, for any morphisms $f : M \rightarrow N$ in $R\text{-mod}$, the diagram

$$\begin{array}{ccc}
 M & \xrightarrow{\eta_M} & M^{**} \\
 f \downarrow & & \downarrow f^{**} \\
 N & \xrightarrow{\eta_N} & N^{**}
 \end{array}$$

commutes. Hence, $\eta : 1_{R\text{-mod}} \rightarrow D^2$ is a natural transformation. η becomes an isomorphism if we define it on the category of finite dimensional vector spaces over a fixed field.

Definition

13.11

Two categories $\mathcal{C}$ and $\mathcal{D}$ are called **isomorphic** if there are functors $\mathcal{F} : \mathcal{C} \rightarrow \mathcal{D}$ and $\mathcal{G} : \mathcal{D} \rightarrow \mathcal{C}$ such that $\mathcal{G}\mathcal{F} = 1_{\mathcal{C}}$ and $\mathcal{F}\mathcal{G} = 1_{\mathcal{D}}$. This notion is often too strong, so we say that $\mathcal{C}$ and $\mathcal{D}$ are **equivalent categories** if there exist functors $\mathcal{F} : \mathcal{C} \rightarrow \mathcal{D}$ and $\mathcal{G} : \mathcal{D} \rightarrow \mathcal{C}$ such that $\mathcal{G}\mathcal{F} \cong 1_{\mathcal{C}}$ and $\mathcal{F}\mathcal{G} \cong 1_{\mathcal{D}}$, where $\cong$ denotes a natural isomorphism of functors. This defines an equivalence relation on categories.

Example 13.12 (Known Theorem). The categories $R\text{-mod}$ and $M_n(R)\text{-mod}$ are equivalent for all $n \geq 1$.

14. Yoneda's Lemma

In many categories whose objects are sets with structure, the underlying set $|X|$ of an object X can be described as the set of morphisms from a **universal object** ω to X .

Example 14.1. For any $G \in \text{Ob}(\mathbf{Grp})$, we have $|G| = \text{Hom}(\mathbb{Z}, G)$. For any $R \in \mathbf{Ring}$, we have $|R| = \text{Hom}(\mathbb{Z}[x], R)$. In general, such a universal ω may not exist.

Enter Grothendieck

Grothendieck's Idea

14.2

Consider not one ω , but *all* objects ω .

We attach to $X \in \text{Ob } \mathcal{C}$ the whole set $\bigcup_{\omega} \text{Hom}(\omega, X)$. More precisely, we consider the Hom functor $\text{Hom}_{\mathcal{C}}(-, X)$, the “functor of points of X ”. The idea is that we can recover X from the functor.

For any category $\mathcal{C}$, and for any $X \in \mathcal{C}$, we have the contravariant **hom functor** $h_X : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Sets}$ defined by $Z \mapsto \text{Hom}(Z, X)$. Now, suppose $f : X \rightarrow Y$ is a morphism in $\mathcal{C}$. Then, for all $Z \in \text{Ob } \mathcal{C}$, the composition $h_{f,Z} : h_X(Z) \rightarrow h_Y(Z)$ given by $g \mapsto fg$ defines a morphism (i.e. a natural transformation) $h_f : h_X \rightarrow h_Y$ of functors. We obtain a covariant functor $h : \mathcal{C} \rightarrow \widehat{\mathcal{C}} := \mathbf{Funct}(\mathcal{C}^{\text{op}}, \mathbf{Sets})$. We claim h is an embedding.

Let $\mathcal{F} : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Sets}$ be a functor and $X \in \text{Ob } \mathcal{C}$. Let $\alpha : h_X \rightarrow \mathcal{F}$ be a morphism of functors. Then, for all objects Y , we are given a map $\alpha_Y : h_X(Y) \rightarrow \mathcal{F}(Y)$. We can consider $\alpha_X(1_X) \in \mathcal{F}(X)$.

Yoneda's Lemma

14.3

The map $\text{Hom}_{\widehat{\mathcal{C}}}(h_X, \mathcal{F}) \rightarrow \mathcal{F}(X)$ sending $\alpha \mapsto \alpha_X(1_X)$ is a bijection, and functorial in X .

Proof. For any $\omega \in \mathcal{F}(X)$, we define a map $(\alpha_{\omega})_Y : h_X(Y) \rightarrow \mathcal{F}(Y)$ given by $f \mapsto F(f)(\omega)$, where $h_X(Y) = \text{Hom}(Y, X)$. The rest is left as an exercise. $\square$

We apply Yoneda's lemma to the special case when $\mathcal{F} = h_Y$ for an object Y of $\mathcal{C}$. We therefore see that the functor $X \mapsto h_X$ induces a bijection $\text{Hom}_{\mathcal{C}}(X, Y) \xrightarrow{\sim} \text{Hom}_{\widehat{\mathcal{C}}}(h_X, h_Y)$. This means that $X \mapsto h_X$ is a **fully faithful** functor $\mathcal{C} \rightarrow \widehat{\mathcal{C}}$, so the category $\mathcal{C}$ embeds into a category of functors from $\mathcal{C}^{\text{op}} \rightarrow \mathbf{Sets}$.

15. The Category $R\text{-mod}$

In $R\text{-mod}$, the Hom sets have a natural abelian group structure, and the product of morphisms (i.e. composition) of morphisms in $R\text{-mod}$ is distributive. That is,

$$h(f + g) = hf + hg$$

$$(f + g)h = fh + gh.$$

When considering functors between categories of modules, we focus on the additive ones, i.e. such that $f \mapsto \mathcal{F}f$ from $\text{Hom}(M, N)$ to $\text{Hom}(\mathcal{F}M, \mathcal{F}N)$ is a map of abelian groups.

In $R\text{-mod}$, we have notions of kernels, cokernels, short exact sequences, etc.

Definition**15.1**

A functor $\mathcal{F} : R\text{-mod} \rightarrow R'\text{-mod}$ is called **exact** if it maps short exact sequences of modules to short exact sequence. $\mathcal{F}$ is called **left exact** whenever

$$0 \rightarrow M' \xrightarrow{f} M \xrightarrow{g} M''$$

being exact implies

$$0 \rightarrow \mathcal{F}(M') \xrightarrow{\mathcal{F}f} \mathcal{F}(M) \xrightarrow{\mathcal{F}g} \mathcal{F}(M'')$$

is exact. We similarly define **right exact** functors.

Analogous definition apply to contravariant functors: a **left exact contravariant functor** $\mathcal{C} \rightarrow \mathcal{D}$ is a left exact functor $\mathcal{C}^{\text{op}} \rightarrow \mathcal{D}$.

Theorem**15.2**

The hom functor $\text{Hom}(M, -) : R\text{-mod} \rightarrow \mathbb{Z}\text{-mod}$ is left exact.

Proof. We will show that if

$$0 \rightarrow N' \xrightarrow{f} N \xrightarrow{g} N''$$

is exact, then

$$0 \rightarrow \text{Hom}(M, N') \xrightarrow{\alpha} \text{Hom}(M, N) \xrightarrow{\beta} \text{Hom}(M, N'')$$

is exact, where $\alpha(\varphi) = f\varphi$ and $\beta(\psi) = g\psi$.

We first claim that $\ker \beta = \text{Img } \alpha$.

Proof. Suppose $\psi \in \ker \beta$. Then,

$$g\psi = 0 \iff \forall m \in M, g\psi(m) = 0 \iff \psi(m) \in \ker g = \text{Img } f,$$

i.e. there exists a unique (by injectivity) $n' \in N'$ such that $\psi(m) = f(n')$. We thus obtain a well-defined map $\varphi : M \rightarrow N'$ sending m to n' . One checks from the definition that $f\varphi = \psi$, and $\varphi \in \text{Hom}(M, N')$. This proves that $\ker \beta \subset \text{Img } \alpha$. For the reverse inclusion, if $\varphi \in \text{Hom}(M, N')$, then

$$gf = 0 \implies gf\varphi = 0 \implies \forall m \in M, gf\varphi(m) = 0 \implies \beta\alpha = 0,$$

hence $\text{Img } \alpha \subset \ker \beta$. □

The rest of the proof is left as an exercise. □

Exercise

Show that the contravariant functor $\text{Hom}(-, M)$ is left exact.

16. Projective Modules

While free modules are nice, it's impossible to neatly describe them from a categorical perspective within the category of modules. Projective modules generalize free modules to fix this.

Definition

16.1

A module P is **projective** if given any surjective map $\pi : M \twoheadrightarrow N$, and any map $f : P \rightarrow N$, there exists a $\tilde{f} : P \rightarrow M$ such that $\pi\tilde{f} = f$. That is, the diagram

$$\begin{array}{ccccc} & & P & & \\ & \swarrow \exists \tilde{f} & \downarrow f & & \\ M & \xrightarrow{\pi} & N & \longrightarrow & 0 \end{array}$$

commutes.

Example 16.2 (Free modules are projective.). Let F be any free module, i.e. $F = \bigoplus_{i \in I} Re_i$. Then, $f : F \rightarrow N$ is defined by $f(e_i)$ for all $i \in I$. By surjectivity, there exists $m_i \in M$ such that $\pi(m_i) = f(e_i)$. So, we define $\tilde{f} : F \rightarrow M$ by $\tilde{f}(e_i) = m_i$ for all $i \in I$.

The definition of projective modules directly implies:

Proposition

16.3

A module P is projective $\iff \text{Hom}(P, -)$ is an exact functor.

Question

How close are projective modules to being free?

Definition

16.4

A short exact sequence $0 \rightarrow M' \xrightarrow{f} M \xrightarrow{g} M'' \rightarrow 0$ **splits** if there exists $\varepsilon : M'' \rightarrow M$ such that $g\varepsilon = 1_{M''}$.

Equivalently, the sequence splits if there exists an $\eta : M \rightarrow M'$ such that $\eta f = 1_{M'}$.

We then have a canonical isomorphism $M \cong M' \oplus M''$ sending $m \mapsto (f\eta(m), \varepsilon g(m))$

The details of these claims are left as an exercise.

Theorem

16.5

The following are equivalent:

- (a) P is projective.
- (b) Any short exact sequence $0 \rightarrow M \rightarrow N \rightarrow P \rightarrow 0$ splits.
- (c) There exists a free module F such that $F \cong P \oplus P'$, for some P' .

Since (a) and (c) are equivalent, it follows that P' is also projective.

Proof.

(a) $\implies$ (b): Given $0 \rightarrow M \rightarrow N \xrightarrow{g} P \rightarrow 0$ exact, we may consider the identity map $1_P : P \rightarrow P$. Then, since P is projective, there exists $\varepsilon : P \rightarrow N$ such that $g\varepsilon = 1_P$.

(b) $\implies$ (c): Since any module is a homomorphic image of a free module F , there is an exact sequence $0 \rightarrow \ker g \rightarrow F \xrightarrow{g} P \rightarrow 0$. By (b), this sequence splits, so $F \cong P \oplus \ker g$.

(c) $\implies$ (a): We are given a split exact sequence $0 \rightarrow P' \rightarrow F \rightarrow P \rightarrow 0$, with F free. Consider the diagram:

$$\begin{array}{ccc} & P & \\ & \downarrow f & \\ M & \xrightarrow{\alpha} N & \longrightarrow 0 \end{array}$$

Then, combining the diagrams, we get

$$\begin{array}{ccccccc} 0 & \longrightarrow & P' & \xrightarrow{i} & F & \xrightarrow{\pi} & P \longrightarrow 0 \\ & & & & \downarrow g & \searrow f\pi & \downarrow f \\ & & & & M & \xrightarrow{\alpha} N & \longrightarrow 0 \end{array}$$

Since F is free, hence projective, we get $g : F \rightarrow M$ such that $f\pi = \alpha g$. Moreover, we have a splitting map $\varepsilon : P \rightarrow F$ such that $\pi\varepsilon = 1_P$. Then,

$$f = f1_P = f\pi\varepsilon = \alpha g\varepsilon,$$

so $g\varepsilon : P \rightarrow M$ satisfies the definition of projective.

$$\begin{array}{ccccccc} 0 & \longrightarrow & P' & \xrightarrow{i} & F & \xleftarrow{\varepsilon} P & \longrightarrow 0 \\ & & & & \downarrow g & \swarrow \pi & \downarrow f \\ & & & & M & \xleftarrow{g\varepsilon} N & \longrightarrow 0 \end{array}$$

□

Example 16.6. Let $R = \mathbb{Z}/6\mathbb{Z}$ and $M = \mathbb{Z}/6\mathbb{Z}$, so clearly M is a free R -module of rank 1. Moreover, $M \cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z}$, so $\mathbb{Z}/2\mathbb{Z}$ and $\mathbb{Z}/3\mathbb{Z}$ are projective R -modules, but not free.

17. Significance of Projective Modules

Definition

17.1

Let F be $\mathbb{R}$ or $\mathbb{C}$, and X be a compact Hausdorff space. Let $C(X)$ be the ring of continuous F -valued functions on X . An **F -vector bundle** over X consists of a topological space E , and a continuous surjective map $\pi : E \rightarrow X$ which is **locally trivial** in the sense that for any $x \in X$, there exists a neighborhood U of x and a homeomorphism $\varphi : \pi^{-1}(U) \rightarrow U \times F^n$ for some fixed

$n \geq 0$ such that the diagram

$$\begin{array}{ccc} \pi^{-1}(U) & \xrightarrow{\varphi} & U \times F^n \\ \downarrow & & \downarrow pr_1 \\ U & \xrightarrow{1_U} & U \end{array}$$

commutes. A **continuous section** of E is a continuous map $s : X \rightarrow E$ such that $\pi \circ s = 1_X$. The space $\Gamma(X, E)$ of all continuous sections of E is a $C(X)$ -module in an obvious way: for $f \in C(X)$ and $s \in \Gamma(X, E)$, we have $(fs)(x) := f(x)s(x)$.

Swan's Theorem

17.2

The module $\Gamma(X, E)$ is finitely generated and projective over $C(X)$.

Conversely, every finitely generated projective $C(X)$ -module arises in this way.

Remark

17.3

The projectivity of $\Gamma(X, E)$ is equivalent to the existence of a vector bundle E' on X such that $E \oplus E' \cong X \times F^n$, the trivial vector bundle for some n .

The category $\mathbf{Vect}_X$ of (finite rank) vector bundles on X admits direct sums $E \oplus E'$. We obtain a semigroup $\mathcal{V}$ of isomorphism classes of vector bundles in $\mathbf{Vect}_X$. The abelian group associated to $\mathcal{V}$ is definition the **topological K-theory** of X , denoted $K^0(X)$.

Similarly, for any commutative ring R , the semigroup $\mathcal{S}$ of isomorphism classes of finitely generated projective R -modules (with the operation of direct sums) has an associated abelian group $K_0(R)$.

Definition

17.4

$K_0(R)$ is defined as equivalence classes of pairs (M, N) with $M, N \in \mathcal{S}$, where $(M, N) \sim (M', N') \iff$ there exists $P \in \mathcal{S}$ with $M \oplus N' \oplus P \cong M' \oplus N \oplus P$. Addition is given by

$$[(M, N)] + [(M', N')] := [(M + M', N + N')]$$

Remark

17.5

If R is a PID, then any finitely generated projective R -module is isomorphic to R^n for some n , and the rank of modules induces an isomorphism $K_0(R) \cong \mathbb{Z}$.

If R is a Dedekind domain (e.g. $R = \mathfrak{O}_K$), then $K_0(R) \cong \mathbb{Z} \oplus C(R)$, where $C(R)$ is the class group of R .

18. Injective Modules

Injective Modules

18.1

A module Q is **injective** if, for modules N, M and injective $i : N \hookrightarrow M$, and any $f : N \rightarrow Q$, there exists $g : M \rightarrow Q$ such that $f = g \circ i$. That is, the diagram commutes:

$$\begin{array}{ccccc} 0 & \longrightarrow & N & \xhookrightarrow{i} & M \\ & & \downarrow f & \swarrow g & \\ & & Q & & \end{array}$$

Proposition

18.2

The following are equivalent:

1. Q is injective
2. If N is a submodule of M , then any homomorphism $N \rightarrow Q$ can be extended to a homomorphism $M \rightarrow Q$
3. The contravariant functor $\text{Hom}(*, Q)$ is exact. This just requires that for any exact sequence $0 \rightarrow N' \xrightarrow{i} N$, the sequence $\text{Hom}(N, Q) \xrightarrow{\text{Hom}(i, Q)} \text{Hom}(N', Q) \rightarrow 0$ is also exact.
4. Any short exact sequence $0 \rightarrow Q \rightarrow M \rightarrow N \rightarrow 0$ splits.

Our goal is to prove $1 \iff 4$. We begin by focusing on $\mathbb{Z}$ -module.

Definition

18.3

An abelian group T is **divisible** if for any integer $m \neq 0$, the map $x \mapsto mx$ is surjective onto T .

Proposition

18.4

If T is a divisible abelian group, then T is injective as a $\mathbb{Z}$ -module.

Proof. Suppose $M' \subset M$ is a subgroup of the abelian group M , and $f : M' \rightarrow T$ is a group homomorphism. Pick $x \in M$, and we'll extend f to the submodule $\langle M', x \rangle$.

If x is free over M' (i.e. $mx \notin M'$ for all $m \in \mathbb{Z} \setminus \{0\}$), then we can select any $t \in T$, and extend f to $\langle M', x \rangle$ by setting $f(x) := t$. Suppose now that $dx \in M'$ for some minimal $d \in \mathbb{Z}^+$. Since T is divisible, there exists $u \in T$ such that $du = f(dx)$. Hence, for any $m' \in M'$ and $k \in \mathbb{Z}$, define

$$f(m' + kx) = f(m') + ku. \quad (2)$$

Note that (2) is well defined: $m'_1 + kx = m'_2 + lx$ implies

$$(k - l)x = m'_2 - m'_1 \in M' \implies k - l = dr$$

for some r , hence

$$f(m'_2) - f(m'_1) = f(drx) = rdu = ku - lu$$

Consider the set S of pairs (N, g) where N is a submodule with $M' \subset N \subset M$, and g is an extension of f to N . We say that $(N, g) \leq (N', g')$ if $N \subset N'$ and $g'|_N = g$. The hypotheses of Zorn's lemma hold for $(S, \leq)$, so let (N, g) be a maximal element of S . Then, if $N \neq M$, there exists $x \in M \setminus N$, and we can apply the first part to extend g to $\langle N, x \rangle$, contradicting the maximality of (N, g) . $\square$

Corollary**18.5**

Any $\mathbb{Z}$ -module can be embedded in an injective $\mathbb{Z}$ -module.

Proof. If $F \cong \bigoplus_{i \in I} \mathbb{Z}e_i$ is any free abelian group, then $F' := \bigoplus_{i \in I} \mathbb{Q}e_i$ is divisible and $F \hookrightarrow F'$. More generally, if M is any abelian group, then $M \cong F/K$ for some free $\mathbb{Z}$ -module F . Then, F'/K is divisible, hence injective, and $F/K \cong M$ is a subgroup of F'/K . $\square$

Now, what about beyond $\mathbb{Z}$ -modules?

Let A be a commutative ring, and T be any abelian group. Then, $\text{Hom}_{\mathbb{Z}}(A, T)$ becomes an A -module: for any $f \in \text{Hom}_{\mathbb{Z}}(A, T)$, and $a \in A$, set $(af)(x) := f(ax)$.

Proposition**18.6**

For any A -module M , there is a natural isomorphism of abelian groups

$$\text{Hom}_{\mathbb{Z}}(M, T) \cong \text{Hom}_A(M, \text{Hom}_{\mathbb{Z}}(A, T))$$

Proof. Suppose $\psi : M \rightarrow T$ is a $\mathbb{Z}$ -homomorphism. We map ψ to $f : M \rightarrow \text{Hom}_{\mathbb{Z}}(A, T)$ with $f(m)(a) := \psi(am)$. Observe that $m \mapsto f(m)$ is an A -module homomorphism, giving a map

$$\text{Hom}_{\mathbb{Z}}(M, T) \rightarrow \text{Hom}_A(M, \text{Hom}_{\mathbb{Z}}(A, T)).$$

Conversely, given an A -homomorphism $f : M \rightarrow \text{Hom}_{\mathbb{Z}}(A, T)$, let $\psi : M \rightarrow T$ be given by $\psi(m) := f(m)(1)$. One checks that this ψ is the inverse of the above map. $\square$

Lemma**18.7**

If T is a divisible abelian group, then $\text{Hom}_{\mathbb{Z}}(A, T)$ is injective in the category of A -modules.

Proof. Suppose $0 \rightarrow N \rightarrow M$ is exact in $A\text{-mod}$. Then, $\text{Hom}_{\mathbb{Z}}(M, T) \rightarrow \text{Hom}_{\mathbb{Z}}(N, T) \rightarrow 0$ is exact since T is an injective $\mathbb{Z}$ -module. Now, we have a commutative diagram

$$\begin{array}{ccccc} \text{Hom}_{\mathbb{Z}}(M, T) & \xrightarrow{\quad\quad\quad} & \text{Hom}_{\mathbb{Z}}(N, T) & \xrightarrow{\quad\quad\quad} & 0 \\ \downarrow \cong & & \downarrow \cong & & \\ \text{Hom}_A(M, \text{Hom}_{\mathbb{Z}}(A, T)) & \xrightarrow{\quad\quad\quad} & \text{Hom}_A(N, \text{Hom}_{\mathbb{Z}}(A, T)) & \xrightarrow{\quad\quad\quad} & 0 \end{array}$$

Thus, the bottom row is exact, proving the lemma. $\square$

Theorem**18.8**

Every A -module embeds in an injective A -module.

Proof. Let M be any A -module. Embed $(M, +)$ into a divisible abelian group T . We then get an A -module homomorphism $\varphi : M \rightarrow \text{Hom}_{\mathbb{Z}}(A, T)$, with $m \mapsto f_m$, where $f_m(a) := am$ for all $a \in A$. Since $f_m(1) = m$, φ embeds M into $\text{Hom}_{\mathbb{Z}}(A, T)$, which is injective. $\square$

We're now equipped to prove the earlier $1 \iff 4$ equivalence.

Proposition**18.9**

A module Q is injective if and only if every short exact sequence $0 \rightarrow Q \rightarrow M \rightarrow N \rightarrow 0$ splits (if and only if Q is a direct summand of every module which contains it).

Proof. Suppose Q is injective, and $0 \rightarrow Q \xrightarrow{i} M \rightarrow N \rightarrow 0$ is exact. Then, with $1_Q : Q \rightarrow Q$, we induce a map $\eta : M \rightarrow Q$ such that $\eta \circ i = 1$, hence the sequence splits.

$$\begin{array}{ccccc} 0 & \longrightarrow & Q & \xrightarrow{i} & M \\ & & \downarrow 1_Q & \swarrow \eta & \\ & & Q & & \end{array}$$

Conversely, suppose Q has the splitting property. By the previous theorem, we have an exact sequence $0 \rightarrow Q \xrightarrow{i} M$ where M is injective. We can extend this to a short exact sequence

$$0 \rightarrow Q \xrightarrow{i} M \xrightarrow{\pi} M/Q \rightarrow 0,$$

and by our assumption there exists a map $\eta : M \rightarrow Q$ such that $\eta \circ i = 1_Q$. Now, suppose that we have a diagram:

$$\begin{array}{ccccc} 0 & \longrightarrow & N' & \xrightarrow{j} & N \\ & & \downarrow f & & \\ & & Q & & \end{array}$$

Since M is injective, we can enlarge this to a commutative diagram

$$\begin{array}{ccccc} 0 & \longrightarrow & N' & \xrightarrow{j} & N \\ & & \downarrow f & & \uparrow g \\ & & Q & & \\ & & \uparrow i & \swarrow \eta & \\ & & M & & \end{array}$$

where there exists $g : N \rightarrow M$ such that $g \circ j = i \circ f$. Hence, we have $\eta \circ g : N \rightarrow Q$, and

$$f = (\eta \circ g) \circ j$$

$$\begin{array}{ccccc}
 0 & \longrightarrow & N' & \xrightarrow{j} & N \\
 & & \downarrow f & \swarrow \eta \circ g & \\
 & & Q & & \\
 & & \uparrow i & \nwarrow g & \\
 & & M & &
 \end{array}$$

□

19. Flat Modules

Definition

19.1

Let R be a commutative ring, and M an R -module. We define a functor

$$M \otimes_R * : R\text{-mod} \rightarrow R\text{-mod}$$

sending any R -module N to $M \otimes_R N$, and any morphism $f : N' \rightarrow N$ to

$$1 \otimes f : M \otimes_R N' \rightarrow M \otimes_R N.$$

Theorem

19.2

The functor $M \otimes_R *$ is right exact. That is, given any exact sequence of R -modules

$$N' \xrightarrow{f} N \xrightarrow{g} N'' \rightarrow 0,$$

the sequence

$$M \otimes N' \xrightarrow{1 \otimes f} M \otimes N \xrightarrow{1 \otimes g} M \otimes N'' \rightarrow 0$$

is exact.

Proof. Surjectivity of $1 \otimes g$ follows from surjectivity of g . We check that $\ker(1 \otimes g) = \text{Img}(1 \otimes f)$. Since $gf = 0$, we have

$$(1 \otimes g) = \left(\sum_i m_i \otimes f(n'_i) \right) = \sum_i m_i \otimes gf(n'_i) = 0$$

hence $\text{Img}(1 \otimes f) \subset \ker(1 \otimes g)$. In particular, we have a map

$$\pi : (M \otimes_R N) / \text{Img}(1 \otimes f) \rightarrow (M \otimes_R N) / \ker(1 \otimes g) \cong M \otimes N''.$$

We claim that π is an isomorphism. Indeed, define $\pi' : M \times N'' \rightarrow (M \otimes N) / \text{Img}(1 \otimes f)$ with $(m, n'') \mapsto [m \otimes n]$ for any n such that $g(n) = n''$. If n_1, n_2 are such that $g(n_1) = g(n_2) = n''$, then $n_1 - n_2 \in \ker g = \text{Img } f$, so $n_1 - n_2 = f(n')$, thus $m \otimes n_1 = m \otimes n_2$ modulo $\text{Img}(1 \otimes f)$. Hence, π' is well defined. Now, π' is R -bilinear, inducing a map

$$\pi' : M \otimes_R N' \rightarrow (M \otimes_R N) / \text{Img}(1 \otimes f).$$

It follows that π and π' are inverses. $\square$

Example 19.3. The functor $M \otimes_R *$ is not always left exact: For example, the injection $i : \mathbb{Z} \hookrightarrow \mathbb{Q}$ of $\mathbb{Z}$ -modules induces the zero map from $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Q}$

We always know that the tensor functor $M \otimes_R *$ is right exact. When it's also left exact, we say M is flat.

Definition

19.4

We say that an R -module M is **flat** if the functor $M \otimes_R *$ is exact. Concretely, M is flat if for any injection $N' \xhookrightarrow{i} N$ of R -modules, the map $1 \otimes i : M \otimes_R N' \rightarrow M \otimes_R N$ is injective.

Lemma

19.5

$M := \bigoplus_{i \in I} M_i$ is flat $\iff M_i$ is flat.

Proof. Suppose $f : N' \hookrightarrow N$ is an injection of R -modules. We must show that

$$1_M \otimes f : M \otimes_R N' \rightarrow M \otimes_R N$$

is injective if and only if

$$1_{M_i} \otimes f : M_i \otimes_R N' \rightarrow M_i \otimes_R N$$

is injective. We have the commutative diagram

$$\begin{array}{ccc} M \otimes_R N' & \xrightarrow{\sim} & \bigoplus_i (M_i \otimes_R N') \\ \downarrow 1 \otimes f & \circlearrowleft & \downarrow \bigoplus_i (1 \otimes f) \\ M \otimes_R N & \xrightarrow{\sim} & \bigoplus_i (M_i \otimes_R N) \end{array}$$

so the result is clear. $\square$

Corollary

19.6

Projective modules are flat.

Proof. It's easy to see that R is a flat R -module: $R \otimes_R N \cong N$ for any R -module N . So, an injection $N' \hookrightarrow N$ induces an injection $R \otimes_R N' \hookrightarrow R \otimes_R N$. The lemma we just proved implies that any free R -module F is flat. Since projective modules are direct summands of free modules, another application of the lemma shows that they are flat. $\square$

Homological Algebra

20. Cochain Complexes and Cohomology

Definition

20.1

Let R be a commutative ring. A **cochain complex** $C^\bullet$ for R is a sequence

$$0 \rightarrow C^0 \xrightarrow{d_1} C^1 \xrightarrow{d_2} C^2 \rightarrow \cdots \rightarrow C^{n-1} \xrightarrow{d_n} C^n \rightarrow \cdots$$

of R -modules and R -linear maps such that $d_{i+1} \circ d_i = 0$, for all $i \geq 1$.

If $(C^\bullet, d)$ and $(\tilde{C}^\bullet, \tilde{d})$ are two cochain complexes, a **homomorphism** $\alpha : C^\bullet \rightarrow \tilde{C}^\bullet$ is a sequence of homomorphisms $\alpha_i : C^i \rightarrow \tilde{C}^i$ such that for every n , the diagram

$$\begin{array}{ccccccc} \cdots & \longrightarrow & C^n & \longrightarrow & C^{n+1} & \longrightarrow & \cdots \\ & & \downarrow \alpha_n & \circlearrowleft & \downarrow \alpha_{n+1} & & \\ \cdots & \longrightarrow & \tilde{C}^n & \longrightarrow & \tilde{C}^{n+1} & \longrightarrow & \cdots \end{array}$$

commutes.

We let $R\text{-comp}$ denote the category of cochain complexes for R .

Definition

20.2

If $(C^\bullet, d)$ is a cochain complex, its n th **cohomology** group is the quotient group

$$H^n(C^\bullet) := \ker d_{n+1} / \text{Img } d_n.$$

We use the notation $Z^n := \ker d_{n+1}$ (n -**cocycles**) and $B^n := \text{Img } d_n$ (n -**coboundaries**), so

$$H_n := Z^n / B^n$$

Check

If $\alpha : C^\bullet \rightarrow \tilde{C}^\bullet$ is a morphism of cochain complexes, then α maps cocycles to cocycles and coboundaries to coboundaries. Hence, for each n , α induces a map on cohomology groups

$$\tilde{\alpha}_n : H^n(C^\bullet) \rightarrow H^n(\tilde{C}^\bullet)$$

sending $z + B^n$ to $\alpha_n(z) + \tilde{B}^n$. This gives an n -th **cohomology functor**

$$H^n : R\text{-comp} \rightarrow R\text{-mod}$$

Remark

20.3

1. There is a dual notion of **chain complexes** $C_\bullet$, which has the arrows reversed:

$$\cdots \xrightarrow{d_{n+1}} C_n \xrightarrow{d_n} C_{n-1} \rightarrow \cdots \xrightarrow{d_1} C_0 \rightarrow 0$$

where $d_i \circ d_{i-1} = 0$ for all $i \geq 2$. Then, the **homology groups** of C_0 are

$$H_n(C_\bullet) := Z_n / B_n := \ker d_n / \text{Im } d_{n+1}$$

2. Observe that H^n (resp. H_n) measures the failure of exactness of $C^\bullet$ (resp $C_\bullet$) at position n .

The Long Exact Cohomology Sequence

20.4

Let

$$0 \rightarrow \mathcal{A} \xrightarrow{\alpha} \mathcal{B} \xrightarrow{\beta} \mathcal{C} \rightarrow 0$$

be a short exact sequence of cochain complexes. That is, the sequences

$$0 \rightarrow A^n \xrightarrow{\alpha_n} B^n \xrightarrow{\beta_n} C^n \rightarrow 0$$

are exact for all $n \geq 1$. Then, there is an induced long exact sequence of cohomology groups

$$0 \rightarrow H^0(\mathcal{A}) \rightarrow H^0(\mathcal{B}) \rightarrow H^0(\mathcal{C}) \xrightarrow{\delta_0} H^1(\mathcal{A}) \rightarrow H^1(\mathcal{B}) \rightarrow H^1(\mathcal{C}) \xrightarrow{\delta_1} \cdots$$

The maps δ_n are called **connecting homomorphisms**.

Proof. We give the definition of $\delta_n : H^n(\mathcal{C}) \rightarrow H^{n+1}(\mathcal{A})$.

$$\begin{array}{ccccccc} & & & B^{n-1} & \xrightarrow{\beta_{n-1}} & C^{n-1} & \longrightarrow 0 \\ & & & \downarrow & & \downarrow & \\ 0 & \longrightarrow & A^n & \xrightarrow{\alpha_n} & B^n & \xrightarrow{\beta_n} & C^n \longrightarrow 0 \\ & & \downarrow d_{n+1} & & \downarrow d_{n+1} & & \downarrow d_{n+1} \\ 0 & \longrightarrow & A^{n+1} & \xrightarrow{\alpha_{n+1}} & B^{n+1} & \xrightarrow{\beta_{n+1}} & C^{n+1} \longrightarrow 0 \\ & & \downarrow d_{n+2} & & \downarrow d_{n+2} & & \downarrow d_{n+2} \\ 0 & \longrightarrow & A^{n+2} & \xrightarrow{\alpha_{n+2}} & B^{n+2} & \xrightarrow{\beta_{n+2}} & C^{n+2} \longrightarrow 0 \end{array}$$

Pick $x \in H^n(\mathcal{C})$ and let $c \in C^n$ represent the class of x in $H^n(\mathcal{C})$. Then, there exists $b \in B^n$ such that $\beta_n(b) = c$. Now,

$$\beta_{n+1}d_{n+1}(b) = d_{n+1}\beta_n(b) = d_{n+1}(c) = 0$$

since $c \in Z^n(\mathcal{C})$. Since $\ker \beta_{n+1} = \text{Im } \alpha_{n+1}$, and α_{n+1} is injective, there exists a unique $a \in A^{n+1}$ such that $\alpha_{n+1}(a) = d_{n+1}(b)$. Note that,

$$\alpha_{n+2}d_{n+2}(a) = d_{n+2}\alpha_{n+1}(a) = d_{n+2}d_{n+1}(b) = 0$$

hence α_{n+2} being injective implies that $d_{n+2}(a) = 0$. So, a is a cocycle, and so represents a class

$[a] \in H^{n+1}(\mathcal{A})$. Define

$$\delta_n(x) := [a].$$

To show that δ_n is well defined, we must show that $[a]$ is independent of the choices of c and b . Then, we'll have that δ_n is a homomorphism $H^n(\mathcal{C}) \rightarrow H^{n+1}(\mathcal{A})$.

Independence of c : If $c = d_n c'$ for some $c' \in C^{n-1}$, there exists some $b' \in B^{n-1}$ such that $\beta_{n-1}(b') = c'$ by surjectivity. Then,

$$\beta_n(d_n b') = d_n \beta_{n-1} b' = d_n c' = c,$$

so we can take $b := d_n b'$, and this leads to $a = 0$.

Independence of b : Exercise.

The check of exactness of the long exact sequence is also routine. For example, we'll show that $\ker \tilde{\alpha} = \text{Im} \delta_n$, where $\tilde{\alpha} : H^{n+1}(\mathcal{A}) \rightarrow H^{n+1}(\mathcal{B})$. Indeed, for $[a] \in \text{Im} \delta_n$, we have $\alpha_{n+1}(a) = d_{n+1}(b)$, so it is clear that $\text{Im} \delta_n \subset \ker \tilde{\alpha}$, because $d_{n+1}(b)$ is zero in $H^{n+1}(\mathcal{A})$. Conversely, if $a \in A^{n+1}$ is such that $\alpha_{n+1}(a) = d_{n+1}(b)$ (i.e. $a \in \ker \tilde{\alpha}$), then clearly $a = \delta_n(x)$, where x denotes the class of $c := \beta_{n+1}(b)$. $\square$

21. Homotopy

Definition

21.1

Let f, g be homomorphisms of cochain complexes $(\mathcal{A}, d)$ into $(\mathcal{B}, d')$. Then, f is said to be **homotopic** to g if there exists module homomorphisms $\{s_n\}_{n \geq 1}$ with $s_n : A^n \rightarrow B^{n-1}$ for all $n \geq 1$ such that

$$f_n - g_n = d'_n s_n + s_{n+1} d_{n+1}$$

for all $n \geq 1$.

$$\begin{array}{ccccccc} \cdots & \longrightarrow & A^{n-1} & \xrightarrow{d_n} & A^n & \xrightarrow{d_{n+1}} & A^{n+1} & \longrightarrow & \cdots \\ & & \Downarrow & \swarrow s_n & \Downarrow & \swarrow s_{n+1} & \Downarrow & & \\ \cdots & \longrightarrow & B^{n-1} & \xrightarrow{d'_n} & B^n & \xrightarrow{d'_{n+1}} & B^{n+1} & \longrightarrow & \cdots \end{array}$$

We write $f \sim g$.

Check

This is an equivalence relation on cochain complexes $\mathcal{A} \rightarrow \mathcal{B}$.

Proposition

21.2

If $f \sim g$, then $\tilde{f}_n = \tilde{g}_n$, where $\tilde{f}_n$ and $\tilde{g}_n$ are the corresponding maps of cohomology modules $H^n(\mathcal{A}) \rightarrow H^n(\mathcal{B})$.

Proof. If $z_n \in Z^n(\mathcal{A})$, then we have

$$\begin{aligned}\tilde{f}_n(z_n) &= [f_n(z_n)] = [(g_n + d'_n s_n + s_{n+1} d_{n+1})(z_n)] \stackrel{d_{n+1} z_n = 0}{=} [g_n(z_n) + d'_n s_n(z_n)] \\ &= [g_n(z_n)] = \tilde{g}_n(z_n)\end{aligned}$$

□

22. The Ext Groups

Definition

22.1

Let M be an R -module. A **projective resolution** of M is an exact sequence

$$\cdots \rightarrow P_n \xrightarrow{d_n} P_{n-1} \rightarrow \cdots \xrightarrow{d_1} P_0 \xrightarrow{\varepsilon} M \rightarrow 0$$

such that each P_n is a projective R -module.

Remark

22.2

Every $M \in R\text{-mod}$ admits a projective resolution by *free* modules P_n .

Proof. Let P_0 be a free module surjecting onto M via $\varepsilon : P_0 \rightarrow M$. Let P_1 be any free module surjecting onto $\ker \varepsilon$ by a map $d_1 : P_1 \rightarrow P_0$, and continue in this way. □

Example 22.3. Any projective module P has a projective resolution

$$\cdots \rightarrow 0 \rightarrow 0 \rightarrow P \xrightarrow{\text{id}} P \rightarrow 0.$$

Given any R -module N , we can apply the contravariant functor $\text{Hom}(*, N)$ to a projective resolution of M to obtain a (not necessarily exact) sequence

$$0 \rightarrow \text{Hom}_R(M, N) \xrightarrow{\varepsilon} \text{Hom}_R(P_0, N) \xrightarrow{d_1} \text{Hom}_R(P_1, N) \xrightarrow{d_2} \cdots \quad (3)$$

(we're denoting $\text{Hom}_R(d_n, N)$ again by d_n for simplicity).

Definition

22.4

We define the **Ext groups** $\text{Ext}_R^\bullet(M, N)$ by

$$\text{Ext}_R^0(M, N) := \ker d_1$$

$$\text{Ext}_R^n(M, N) := \ker d_{n+1} / \text{Im } d_n \quad \forall n \geq 1.$$

The module $\text{Ext}_R^n(M, N)$ is the **n -th cohomology group derived from the functor $\text{Hom}_R(*, N)$** . Ext stands for extension.

Remark

22.5

The groups $\text{Ext}_R^n(M, N)$ are the cohomology groups of the cochain complex

$$0 \rightarrow \text{Hom}_R(P_0, N) \rightarrow \text{Hom}_R(P_1, N) \rightarrow \cdots$$

(where we replace $\text{Hom}_R(M, N)$ in (3) with zero).

Proposition**22.6**

$$\text{Ext}_R^0(M, N) = \text{Hom}_R(M, N).$$

Proof. Since $P_1 \xrightarrow{d_1} P_0 \xrightarrow{\varepsilon} M \rightarrow 0$ is exact, we deduce that

$$0 \rightarrow \text{Hom}(M, N) \xrightarrow{\varepsilon} \text{Hom}(P_0, N) \xrightarrow{d_1} \text{Hom}(P, N)$$

is exact. Then,

$$\text{Ext}_R^0(M, N) := \ker d_1 = \text{Im } \varepsilon \cong \text{Hom}(M, N)$$

□

Example 22.7. Suppose $a \in R$ is a not a zero divisor, and take $M := R/aR$. Then,

$$\text{Ext}_R^0(R/aR, N) = \text{Hom}(R/aR, N) = N[a] := \{x \in N \mid ax = 0\}.$$

We call this is a ***a-torsion submodule*** of N .

We have a projective resolution

$$0 \rightarrow R \xrightarrow{a} R \rightarrow R/aR \rightarrow 0$$

inducing the exact sequence

$$\begin{aligned} 0 \rightarrow \text{Hom}_R(R/aR, N) &\rightarrow \text{Hom}(R, N) \xrightarrow{a} \text{Hom}(R, N) \rightarrow \text{Hom}(0, N) \\ &\cong 0 \rightarrow \text{Hom}_R(R/aR, N) \rightarrow N \rightarrow N \rightarrow 0. \end{aligned}$$

We deduce that

$$\text{Ext}_R^n(R/aR, N) = \begin{cases} N[a] & n = 0 \\ N/aN & n = 1 \\ 0 & n \geq 2 \end{cases}$$

Our goal is to prove that Ext is well defined independent of choice of projective resolution.

Proposition**22.8**

Let $f : M \rightarrow M'$ be any R -linear map, and take projective resolutions M and M' . Then, for each $n \geq 0$, there is a lift f_n of f such that the following diagram commutes:

$$\begin{array}{ccccccc} \cdots & \xrightarrow{d_2} & P_1 & \xrightarrow{d_1} & P_0 & \xrightarrow{\varepsilon} & M \longrightarrow 0 \\ & & \downarrow f_1 & & \downarrow f_0 & & \downarrow f \\ \cdots & \xrightarrow{d'_2} & P'_1 & \xrightarrow{d'_1} & P'_0 & \xrightarrow{\varepsilon} & M' \longrightarrow 0 \end{array}$$

Proof. Since P_0 is projective, there exists $f_0 : P_0 \rightarrow P'_0$ such that $f\varepsilon = \varepsilon'f_0$. Now, observe that

$\varepsilon' f_0 d_1 = f \varepsilon d_1 = 0$, hence $f_0 d_1(P_1) \subset \ker \varepsilon' = \text{Im } d'_1$.

$$\begin{array}{ccccc} P_1 & \xrightarrow{d_1} & P_0 & & \\ & \searrow & \downarrow f_0 & & \\ P'_1 & \xrightarrow{d'_1} & d'_1(P'_1) & \longrightarrow & 0 \end{array}$$

So, as P_1 is projective, there exists $f_1 : P_1 \rightarrow P'_1$ such that $d'_1 f_1 = d_1 f_0$. This process continues in general for f_n ...

If we apply $\text{Hom}_R(*, N)$ to the previous diagram, we get

$$\begin{array}{ccccccc} 0 & \longrightarrow & \text{Hom}(M', N) & \longrightarrow & \text{Hom}(P'_0, N) & \longrightarrow & \text{Hom}(P'_1, N) \longrightarrow \dots \\ & & \downarrow f & & \downarrow f_0 & & \downarrow f_1 \\ 0 & \longrightarrow & \text{Hom}(M, N) & \longrightarrow & \text{Hom}(P_0, N) & \longrightarrow & \text{Hom}(P_1, N) \longrightarrow \dots \end{array}$$

We then get induced maps on cohomology groups.

Theorem

22.9

For every n , the induced maps $\varphi_n : \text{Ext}_R^n(M', N) \rightarrow \text{Ext}_R^n(M, N)$ depend only on f , and not on the choice of lifts f_n .

Proof. We may assume that $f = 0$, and show that the induced maps on cohomology are also zero. Using the projectivity of the modules P_n , we recursively construct a homotopy as follows: There exists maps $s_n : P_n \rightarrow P'_{n+1}$ such that for all n , we have $f_n = d'_{n+1} s_n + s_{n-1} d_n$. That is, we're trying to make the following diagram:

$$\begin{array}{ccccccccccc} \dots & \longrightarrow & P_2 & \xrightarrow{d_2} & P_1 & \xrightarrow{d_1} & P_0 & \xrightarrow{\varepsilon} & M & \longrightarrow & 0 \\ & & \downarrow f_2 & & \downarrow f_1 & & \downarrow f_0 & & \downarrow f=0 & & \\ \dots & \longrightarrow & P'_2 & \xrightarrow{d'_2} & P'_1 & \xrightarrow{d'_1} & P'_0 & \xrightarrow{\varepsilon'} & M' & \longrightarrow & 0 \end{array}$$

(Dashed arrows s_2, s_1, s_0 point from P_2, P_1, P_0 to P'_3, P'_2, P'_1 respectively.)

Since $d'_1 : P'_1 \rightarrow \ker \varepsilon'$ is surjective and $f = 0$ implies $f_0 P_0 \subset \ker \varepsilon'$, we have a map $s_0 : P_0 \rightarrow P'_1$ such that $d'_1 s_0 = f_0$.

Next, to obtain $s_1 : P_1 \rightarrow P'_2$ such that $f_1 = d'_2 s_1 + s_0 d_1$, consider $f - s_0 d_1$ and note that

$$d'_1(f_1 - s_0 d_1) = f_0 d_1 - (d'_1 s_0) d_1 = 0.$$

As before, we can construct $s_1 : P_1 \rightarrow P'_2$, and so on.

The above maps give a **chain homotopy** between the given resolution and 0. Taking homomorphisms into N gives a **cochain homotopy**, and the discussion from last time finishes the proof. $\square$

Theorem

22.10

The groups $\text{Ext}_R^n(M, N)$ depend only on M and N , i.e. are independent of the choice of projective resolution $P_\bullet$ of M .

Proof. Consider $1_M : M \rightarrow M$, set $M' := M$, and take two projective resolutions

$$\begin{array}{ccccccc} \cdots & \longrightarrow & P_1 & \longrightarrow & P_0 & \longrightarrow & M \longrightarrow 0 \\ & & \downarrow f_1 & & \downarrow f_0 & & \downarrow \\ \cdots & \longrightarrow & P'_1 & \longrightarrow & P'_0 & \longrightarrow & M' \longrightarrow 0 \end{array}$$

where the f_i 's are the choice of lifts of 1_M . Then, we can copy the top row to the bottom and provide lifts g_n for that:

$$\begin{array}{ccccccc} \cdots & \longrightarrow & P_1 & \longrightarrow & P_0 & \longrightarrow & M \longrightarrow 0 \\ & & \downarrow f_1 & & \downarrow f_0 & & \uparrow 1_M \\ \cdots & \longrightarrow & P'_1 & \longrightarrow & P'_0 & \longrightarrow & M' \longrightarrow 0 \\ & & \downarrow g_1 & & \downarrow g_0 & & \uparrow 1_M \\ \cdots & \longrightarrow & P_1 & \longrightarrow & P_0 & \longrightarrow & M \longrightarrow 0 \end{array}$$

The top two rows induce maps φ_n on Ext_R^n , and similarly the bottom two rows induce ψ_n . Now, the composites $\{g_n \circ f_n\}_{n \geq 0}$ are a lift of the identity map as well, and (by the previous theory) induce the same map on cohomology as the identity map on the top complex. Therefore, $\varphi_n \circ \psi_n$ is the identity map on $\text{Ext}_R^n(M, N)$. By a symmetric argument, $\psi_n \circ \varphi_n$ is the identity map on $\text{Ext}_R^n(M', N)$. Thus, both maps are isomorphisms.

We deduce that $\text{Ext}_R^n(*, N)$ is a contravariant functor from $R\text{-mod} \rightarrow R\text{-mod}$. $\square$

The next result shows that Ext_R^1 is the first measure of the failure of right exactness of the Hom functor $\text{Hom}_R(*, N)$.

Theorem

22.11

Let $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ be a short exact sequence of R -modules. Then, there is an induced long exact sequence

$$\begin{aligned} 0 \rightarrow \text{Hom}_R(M'', N) \rightarrow \text{Hom}_R(M, N) \rightarrow \text{Hom}_R(M', N) &\xrightarrow{\delta_0} \text{Ext}_R^1(M'', N) \\ &\rightarrow \text{Ext}_R^1(M, N) \rightarrow \text{Ext}_R^1(M', N) \xrightarrow{\delta_1} \text{Ext}_R^2(M'', N) \rightarrow \cdots \end{aligned}$$

Proof. Given projective resolutions $P'_\bullet$ and $P''_\bullet$ of M' and M'' , we construct another projective resolution $P_\bullet := P'_\bullet \oplus P''_\bullet$ of M such that the following diagram commutes:

$$\begin{array}{ccccccc} 0 & \longrightarrow & P'_1 & \longrightarrow & P'_1 \oplus P''_1 & \longrightarrow & P''_1 \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & P'_0 & \longrightarrow & P'_0 \oplus P''_0 & \longrightarrow & P''_0 \longrightarrow 0 \\ & & \downarrow \varepsilon' & & \downarrow \varepsilon & & \downarrow \varepsilon'' \\ 0 & \longrightarrow & M' & \xrightarrow{a} & M & \xrightarrow{b} & M'' \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ & & 0 & & 0 & & 0 \end{array}$$

where the rows are the obvious split exact sequence. To construct ε from ε' and ε'' , let $\eta : P_0'' \rightarrow M$ be a lifting of ε'' so that $b\eta = \varepsilon''$. Then, we set $\varepsilon(x, y) := a\varepsilon'(x) + \eta(y)$. The remaining vertical maps are constructed similarly.

Now, take $\text{Hom}(*, N)$ to get the commutative diagram

$$\begin{array}{ccccccc}
 & & 0 & & 0 & & 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & \text{Hom}_R(P_0'', N) & \longrightarrow & \text{Hom}_R(P_0' \oplus P_0'', N) & \longrightarrow & \text{Hom}_R(P_0', N) \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & \text{Hom}_R(P_1'', N) & \longrightarrow & \text{Hom}_R(P_1' \oplus P_1'', N) & \longrightarrow & \text{Hom}_R(P_1', N) \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 & & \vdots & & \vdots & & \vdots
 \end{array}$$

Then, the columns are cochain complexes and the rows are split exact, since

$$\text{Hom}_R(M \oplus N, D) \cong \text{Hom}_R(M, D) \oplus \text{Hom}_R(N, D)$$

for any three modules M, N, D . Hence, we get a short exact sequence of complexes. The theorem now follows from the induced long exact sequence of cohomology groups. $\square$

Proposition

22.12

For an R -module Q , the following are equivalent:

- (a) Q is injective
- (b) $\text{Ext}_R^1(M, Q) = 0$ for all R -modules M
- (c) $\text{Ext}_R^n(M, Q) = 0$ for all R -modules M and all $n \geq 1$.

Proof.

(b) $\implies$ (a): Since $\text{Hom}_R(*, Q)$ is exact if and only if Q is injective, the result follows from the last theorem.

(c) $\implies$ (b): Obvious

(a) $\implies$ (c): Suppose Q is injective, and take a projective resolution

$$\cdots \rightarrow P_n \rightarrow P_{n-1} \rightarrow \cdots \rightarrow P_0 \rightarrow M \rightarrow 0$$

for any R -module M . Since Q is injective, the sequence

$$0 \rightarrow \text{Hom}_R(M, Q) \rightarrow \text{Hom}_R(P_0, Q) \rightarrow \text{Hom}_R(P_1, Q) \rightarrow \cdots$$

is still exact (check this). So, all its cohomology groups vanish, hence $\text{Ext}_R^n(M, Q) = 0$ for all $n \geq 1$. $\square$

Definition

22.13

The functors $\text{Ext}_R^n(*, N)$ are called the **right derived functors** of the functor $\text{Hom}_R(*, N)$ (which is left exact). We can similarly construct (covariant) right derived functors for

$\text{Hom}_R(M, *)$. These two constructions produce the same Ext groups.

Theorem**22.14**

Let $0 \rightarrow N' \rightarrow N \rightarrow N'' \rightarrow 0$ be a short exact sequence of R -modules. Then, there is an induced long exact sequence

$$0 \rightarrow \text{Hom}_R(M, N') \rightarrow \text{Hom}_R(M, N) \rightarrow \text{Hom}_R(M, N'') \xrightarrow{\delta'_0} \text{Ext}_R^1(M, N') \\ \rightarrow \text{Ext}_R^1(M, N) \rightarrow \cdots$$

Proof. Let $P_\bullet \twoheadrightarrow M \rightarrow 0$ be a projective resolution of M . Apply the Hom functors to get an exact sequence of complexes

$$0 \rightarrow \text{Hom}(P_\bullet, N') \rightarrow \text{Hom}(P_\bullet, N) \rightarrow \text{Hom}(P_\bullet, N'') \rightarrow 0$$

which induces the result [omitted: a proof that the Ext groups in the theorem are the same as the ones defined earlier. Not obvious but true.] $\square$

Proposition**22.15**

For any R -module P , the following are equivalent

- (a) P is projective
- (b) $\text{Ext}_R^1(P, N) = 0$ for any R -module N
- (c) $\text{Ext}_R^n(P, N) = 0$ for any R -module N and $n \geq 1$.

Proof.

(c) $\implies$ (b) $\implies$ (a): As before.

(a) $\implies$ (c): If P is projective, then $0 \rightarrow P \xrightarrow{1_P} P \rightarrow 0$ is a projective resolution of P , which induces a cochain complex

$$0 \rightarrow \text{Hom}_R(P, N) \xrightarrow{1} \text{Hom}_R(P, N) \rightarrow 0 \rightarrow \cdots$$

This implies that $\text{Ext}_R^n(P, N) = 0$ for all R -modules N , and $n \geq 1$. $\square$

Remark**22.16**

1. One can dually define $\text{Ext}_R^n(M, N)$ using an **injective resolution** of N , instead of a projective resolution of M :

$$0 \rightarrow N \rightarrow Q_0 \rightarrow Q_1 \rightarrow \cdots$$

so that $\text{Ext}_R(M, N)$ is the n -th cohomology group of the cochain complex obtained by applying $\text{Hom}_R(M, *)$ to this resolution of N .

2. For any M, N , there is a bijection between $\text{Ext}_R^1(M, N)$ and the set of equivalence classes of **extensions** E of M by N : i.e. short exact sequences

$$0 \rightarrow N \rightarrow E \rightarrow M \rightarrow 0.$$

This explains the terminology “Ext”.

23. The Tor Groups

Let R be a commutative ring. We've seen that $* \otimes_R M$ is a right exact functor. The failure of left exactness is measured by the Tor group.

Definition

23.1

Given an R -module N , we let $P_\bullet \rightarrow N \rightarrow 0$ be a projective resolution, and then tensor with M to get a chain complex:

$$\cdots \rightarrow M \otimes P_n \xrightarrow{1 \otimes d_n} M \otimes P_{n-1} \rightarrow \cdots \xrightarrow{1 \otimes d_1} M \otimes P_0 \xrightarrow{1 \otimes \epsilon} M \otimes N \rightarrow 0. \quad (4)$$

We thus define the **Tor group** to be

$$\mathrm{Tor}_0^R(M, N) := \mathrm{coker}(1 \otimes d_1) := M \otimes P_0 / \mathrm{Im}(1 \otimes d_1),$$

$$\mathrm{Tor}_n^R(M, N) := \ker(1 \otimes d_n) / \mathrm{Im}(1 \otimes d_{n+1}).$$

Equivalently, $\mathrm{Tor}_n^R(M, N)$ is the n -th homology group obtained from (4) by removing the 1st term $M \otimes N$.

We now have analogues of all the previous results for Ext groups, with basically identical proofs.

Example 23.2. $\mathrm{Tor}_0^R(M, N) = M \otimes_R N$.

Proposition

23.3

The homology groups $\mathrm{Tor}_n^R(M, N)$ are independent of choice of projective resolution $P_\bullet$. They are functorial in the sense that $N \rightarrow N'$ induces $\psi_n : \mathrm{Tor}_n^R(M, N) \rightarrow \mathrm{Tor}_n^R(M, N')$ for all $n \geq 0$.

Theorem

23.4

Suppose $0 \rightarrow N' \rightarrow N \rightarrow N'' \rightarrow 0$ is a short exact sequence of R -modules. Then, there is an induced long exact sequence of abelian groups

$$\cdots \rightarrow \mathrm{Tor}_1^R(M, N) \rightarrow \mathrm{Tor}_1^R(M, N'') \rightarrow M \otimes N' \rightarrow M \otimes N \rightarrow M \otimes N'' \rightarrow 0$$

Proposition

23.5

For any module M , the following are equivalent.

- (a) M is a flat R -module
- (b) $\mathrm{Tor}_1^R(M, N) = 0$ for any R -module N
- (c) $\mathrm{Tor}_n^R(M, N) = 0$ for any R -module N , for all $n \geq 1$.

Remark

23.6

Since $M \otimes_R N \cong N \otimes_R M$ for all R -modules M, N , it follows that

$$\mathrm{Tor}_n^R(M, N) \cong \mathrm{Tor}_n^R(N, M)$$

Remark

23.7

One can show (see textbook) that if A is an abelian group, then $\operatorname{Tor}_1(A, B) = 0$ for every abelian group $B \iff A$ is torsion free $\iff A$ is flat as a $\mathbb{Z}$ -module. This explains the term “Tor”.

Fields

24. Field Extensions

Definition

24.1

An inclusion $E \supset F$ of fields is called a **field extension**. The **degree** $[E : F]$ is the dimension $\dim_F E$. If the degree is finite, we say that $E \supset F$ is a **finite extension**.

Example 24.2. Let F be any field, and consider the characteristic homomorphism $i : \mathbb{Z} \rightarrow F$. We have two possibilities:

- (a) i is injective, so $\mathbb{Z} \hookrightarrow F$, hence $\mathbb{Q} \subset F$.
- (b) $\ker i = p\mathbb{Z}$ for some prime p , so $\mathbb{F}_p := \mathbb{Z}/p\mathbb{Z} \subset F$.

We call $\mathbb{Q}$ or $\mathbb{F}_p$ the **prime subfield** of F . This is the smallest field contained in F .

Theorem

24.3

Suppose $F \subset K \subset L$ is a tower of three fields. Then, $[L : F]$ is finite $\iff [K : F]$ and $[L : K]$ are both finite. If this is the case, then

$$[L : F] = [L : K] \cdot [K : F]$$

Proof. First, assume $[L : F] < \infty$. Then, since K is an F -subspace of L , we have $[K : F] < \infty$. Moreover, any finite set which spans L over F must also span L over K , hence $[L : K] < \infty$.

Conversely, suppose $[L : K] = n$ and $[K : F] = m$. In particular, let $u_1, \dots, u_m$ be a basis of K over F , and $v_1, \dots, v_n$ is a basis of L over K . We claim that $\{u_i v_j : i \in [1, m], j \in [1, n]\}$ is a basis of L over F . Indeed, if $v \in L$, we have $v = \sum_{j=1}^n a_j v_j$ for some $a_j \in K$. Moreover, we may write $a_j = \sum_{i=1}^m a_{ij} u_i$ for some $a_{ij} \in F$. Hence, $v = \sum_{i,j} a_{ij} u_i v_j$. For linear independence,

$$\sum_{i,j} a_{ij} u_i v_j = 0 \implies \sum_{j=1}^n \left(\sum_{i=1}^m a_{ij} u_i \right) v_j = 0 \implies \sum_{i=1}^m a_{ij} u_i = 0 \text{ for all } j \implies a_{ij} = 0 \text{ for all } i.$$

□

Corollary

24.4

If $F \subset K \subset L$ and $[L : F] < \infty$, then $[K : F] \mid [L : F]$. So, if $[L : F] = p$ a prime, then there are no subfields K with $F \subsetneq K \subsetneq L$.

Definition

24.5

Let K be a field, and $S \subset K$ be any subset. Then, there is a smallest subfield $\langle S \rangle$ of K containing

the set S . In particular,

$$\langle S \rangle := \bigcap_{\substack{S \subset F \subset K \\ F \text{ a field}}} F$$

We are particular interested in the case when $F \subset K$ is an extension, and for $\alpha \in K \setminus F$, we take $S := F \cup \{\alpha\}$. We denote $\langle S \rangle := F(\alpha)$. In particular,

$$F(\alpha) = \left\{ \frac{p(\alpha)}{q(\alpha)} : p, q \in F[x], q(\alpha) \neq 0 \right\}$$

Definition

24.6

Consider the evaluation map $\varphi : F[x] \rightarrow F(\alpha)$ sending $f(x) \mapsto f(\alpha)$. We have two cases:

1. φ is injective, so $f(\alpha) \neq 0$ for all non-zero polynomials $f \in F[x]$. In this case, we say α is **transcendental** over F . We also see that $F(\alpha) \cong F(x)$, the quotient field.
2. There exists a non-zero polynomial $f \in F[x]$ with $f(\alpha) = 0$. In this case, we say that α is **algebraic** over F .

If $F = \mathbb{Q}$ and $\alpha \in \mathbb{C}$, we just say α is transcendental or algebraic (i.e. we omit “over F ”). Otherwise, these notions are relative to the base field F .

Example 24.7. Euler’s number e is transcendental over $\mathbb{Q}$, but algebraic over $\mathbb{Q}(e^2)$, since e is a root of the polynomial $x^2 - e^2 \in \mathbb{Q}(e^2)[x]$.

Proposition

24.8

Let $F \subset K$ be an extension, and let $\alpha \in K$ be algebraic over F . Let $f(x) \in F[x]$ be a monic polynomial of least degree n such that $f(\alpha) = 0$. Then,

- (a) f is unique; we call it the minimal polynomial of α over F .
- (b) f is irreducible in $F[x]$
- (c) A polynomial $g \in F[x]$ satisfies $g(\alpha) = 0$ if and only if $f \mid g$.
- (d) $[F(\alpha) : F] = n$, and $1, \alpha, \dots, \alpha^{n-1}$ form a basis of $F(\alpha)$ as an F -vector space.

Proof. We have the evaluate map $\varphi : F[x] \rightarrow F(\alpha)$ sending $f \mapsto f(\alpha)$. The image of φ is denoted $F[\alpha] = \{f(\alpha) \mid f \in F[x]\}$, a subring of $F(\alpha)$. Hence, $F[\alpha]$ is a domain. Now, $F[x] / \ker \varphi \cong F[\alpha]$. Since $F[x]$ is a PID, we have $\ker \varphi = \langle f \rangle$ for a unique $f \in F[x]$ up to units, which we can make completely unique by requiring that f is monic. Now, $F[x] / \langle f \rangle$ is a domain, so $f(x)$ is irreducible. Hence, $\langle f \rangle$ is also a maximal ideal, so $F[\alpha]$ is a field. By definition, we therefore must have $F[\alpha] = F(\alpha)$. This proves (a)-(c).

Recall that $F[x] / \langle f \rangle$ has a basis $\bar{1}, \bar{x}, \dots, \bar{x}^{n-1}$. Since $F[x] / \langle f \rangle \cong F(\alpha)$, we have the identification $\bar{x} \leftrightarrow \alpha$, proving (d). $\square$

Example 24.9. Consider $\mathbb{Q}(\sqrt[3]{5})$. The minimal polynomial of $\sqrt[3]{5}$ over $\mathbb{Q}$ is $x^3 - 5$ (irreducible by Eisenstein’s criterion), so

$$\mathbb{Q}(\sqrt[3]{5}) = \mathbb{Q}[\sqrt[3]{5}] = \mathbb{Q} \oplus \mathbb{Q} \cdot \sqrt[3]{5} \oplus \mathbb{Q} \cdot \sqrt[3]{25}$$

All Degree 2 Extensions**24.10**

Assume $\text{char } F \neq 2$. Then, any extension $K \supset F$ of degree 2 can be obtained by adjoining a square root to F , i.e. $K = F(\delta)$, where $\delta^2 \in F$.

Conversely, if $\delta \notin F$, but $\delta^2 \in F$, then $[F(\delta) : F] = 2$.

Proof. Assume $[K : F] = 2$. Choose any $\alpha \in K \setminus F$. Then, $\{1, \alpha\}$ forms a basis of K over F (they are linearly independent), hence there exists $b, c \in F$ such that $\alpha^2 + b\alpha + c = 0$. By the quadratic formula, we want

$$\alpha = \frac{-b \pm \sqrt{b^2 - 4c}}{2}.$$

Let $\delta := 2\alpha + b$. Then, since $2 \neq 0$, we see that $F(\alpha) = F(\delta)$, and $\delta \in K \setminus F$. Moreover, $\delta^2 = b^2 - 4c \in F$, so δ is a square root.

Conversely, if $\delta \notin F$ and $\delta^2 \in F$, then δ has degree 2 over F (its minimal polynomial is $x^2 - \delta^2 \in F[x]$). Hence, $[F(\delta) : F] = 2$. $\square$

Corollary**24.11**

Suppose that α, β are algebraic over the field F . Then, the following are equivalent

1. There exists an isomorphism of fields $\sigma : F[\alpha] \rightarrow F[\beta]$ which is the identity map on F (i.e. $\sigma(x) = x$ for all $x \in F$) and $\sigma(\alpha) = \beta$
2. α and β have the same minimal polynomial over F .

Proof. Assume that α and β have the same minimal polynomial $f(x)$. Then, we have isomorphisms $F[\alpha] \xrightarrow{\varphi} F[x]/\langle f \rangle \xrightarrow{\psi} F[\beta]$ defined by $\alpha \mapsto \bar{x} \mapsto \beta$. We thus define $\sigma = \psi \circ \varphi$, and it satisfies the desired properties.

Conversely, suppose $\sigma : F[\alpha] \rightarrow F[\beta]$ is an isomorphism such that $\sigma(x) = x$ for all $x \in F$, and $\sigma(\alpha) = \beta$. Then, if $f(x) = c_0 + c_1x + \cdots + c_nx^n \in F[x]$ and $f(\alpha) = 0$, we have

$$\begin{aligned} f(\beta) &= f(\sigma(\alpha)) = c_0 + c_1\sigma(\alpha) + \cdots + c_n\sigma(\alpha)^n = \sigma(c_0 + c_1\alpha + \cdots + c_n\alpha^n) \\ &= \sigma(f(\alpha)) = \sigma(0) = 0. \end{aligned}$$

Similarly, $f(\beta) = 0 \implies f(\alpha) = 0$, so α and β are roots of the same set of polynomials in $F[x]$, so they must have the same minimal polynomial. $\square$

25. Algebraic Field Extensions**Definition****25.1**

A field extension $K \supset F$ is **algebraic over F** if each element of K is algebraic over F . Otherwise, K is **transcendental over F** .

If α is algebraic over F , the **degree** of α over F is $[F(\alpha) : F]$.

Note

If $[K : F] = n$ is finite, then K is algebraic over F .

Proof. For any $\alpha \in K$, the elements $1, \alpha, \dots, \alpha^n$ are linearly dependent. $\square$

Proposition

25.2

If $K \supset F$ is a field extension, the elements of K which are algebraic over F form a subfield of K .

Proof. Suppose α, β are algebraic over F , where α has degree n and β has degree m . Then,

$$[F(\alpha, \beta) : F] = [F(\alpha)(\beta) : F(\alpha)] \cdot [F(\alpha) : F] < \infty.$$

But, $\alpha \pm \beta, \alpha \cdot \beta, \frac{\alpha}{\beta} \in F(\alpha, \beta)$ which we've shown has finite degree, so they are all algebraic over F . $\square$

Definition

25.3

A complex number α is **algebraic** if it's algebraic over $\mathbb{Q}$. Otherwise, α is **transcendental**. The subfield $\overline{\mathbb{Q}} \subset \mathbb{C}$ of all algebraic numbers is countable.

26. Construction with Straight Edge and Compass

Constructable Numbers

26.1

Start with a point $(a, b) \in \mathbb{Z}^2$ as known. A point $(x, y) \in \mathbb{R}^2$ is **constructable** if you can construct it by straight edge and compass. A real number x is **constructable** if $(x, 0)$ is constructable.

Remark

26.2

Using analytic geometry, one sees that if $\alpha \in \mathbb{R}$ is constructable, then α lies in a field K such that

$$\mathbb{Q} = K_0 \subset K_1 \subset K_2 \subset \dots \subset K_n = K$$

where $K_i = K_{i-1}(\sqrt{\alpha_i})$ for some $\alpha_i \in K_{i-1}$. Therefore $[K : \mathbb{Q}] = 2^n$. Therefore $\deg(\alpha) = 2^r$.

Proof. The intersection of two lines, the intersection of a line and a circle, and the intersection of two circles can always be described with square roots. $\square$

Proposition

26.3

$\sqrt[3]{2}$ is not constructable, since it has degree 3 over $\mathbb{Q}$. Similarly, π is transcendental, so it's not constructable.

27. Splitting Fields

Example 27.1. Let $f(x) = x^2 + 1 \in \mathbb{R}[x]$. We can then define the field $\mathbb{C} := \mathbb{R}[x] / \langle x^2 + 1 \rangle$, which is a field where f has a root (namely $\bar{x}$).

Proposition

27.2

Given a field F and $f \in F[x]$, there exists a field K in which f has a root.

Proof. If f is irreducible over F , then set $K := F[x] / \langle f \rangle$, a field. Then, $\bar{x} := x + \langle f \rangle$ is a root of f in K , since $f(\bar{x}) = \overline{f(x)} = 0 \in K$.

In general, if p is an irreducible factor of f , then set $K := F[x] / \langle p \rangle$, so that p has a root in K , and therefore f does as well. $\square$

Corollary

27.3

Suppose $f \in F[x]$ has degree n . Then, there exists an extension K of F with $[K : F] \leq n!$ in which f has n roots. In this case, we say f **splits completely** in K .

Proof. Follows from the proof of Proposition 27.2 by induction on n . $\square$

Definition

27.4

The extension $K \supset F$ is a **splitting field** for $f \in F[x]$ if

1. f splits completely in $K[x]$
2. f does not split completely over any proper subfield of K containing F .

In this case, $K = F(\alpha_1, \dots, \alpha_n)$, where $\alpha_1, \dots, \alpha_n$ are the roots of f in K .

Example 27.5. 1. The splitting field of $x^3 - 5$ over $\mathbb{Q}$ is $K = \mathbb{Q}(\sqrt[3]{5}, \omega\sqrt[3]{5}, \omega^2\sqrt[3]{5})$ where $\omega = e^{2\pi i/3} = \frac{-1+i\sqrt{3}}{2}$. It's easy to see that $K = \mathbb{Q}(\sqrt[3]{5}, \omega)$. Observe that $1 + \omega + \omega^2 = 0$, so ω has degree 2. Therefore, $2, 3 \mid [K : \mathbb{Q}] \leq 3! = 6$, hence $[K : \mathbb{Q}] = 6$.

2. $[\mathbb{Q}(\sqrt[4]{5}), \mathbb{Q}] = 4$ since $x^4 - 5$ is irreducible over $\mathbb{Q}$, and $[\mathbb{Q}(i) : \mathbb{Q}] = 2$. Moreover

$$\mathbb{Q} \stackrel{4}{\subset} \mathbb{Q}(\sqrt[4]{5}) \stackrel{2}{\subset} \mathbb{Q}(\sqrt[4]{5})(i) = K,$$

which we know because $i \notin \mathbb{Q}(\sqrt[4]{5})$ (it's totally real).

Theorem

27.6

Let $\sigma : F \rightarrow F'$ be an isomorphism of fields, let $f \in F[x]$ be irreducible, and let $f' := \sigma f \in F'[x]$ be the (irreducible) polynomial obtained by applying σ to the coefficients of $f(x)$. Then,

1. Let a be a root of f in some extension of F , and b a root of σf in some extension of F' . Then, there is an isomorphism $\tilde{\sigma} : F(a) \rightarrow F'(b)$ with $\tilde{\sigma}(a) = b$, and $\tilde{\sigma}|_F = \sigma$.
2. Let K be a splitting field for f over F and K' be a splitting field for σf over F' . Then, σ extends to an isomorphism $\tilde{\sigma} : K \rightarrow K'$ such that $\tilde{\sigma}|_F = \sigma$.

Proof.

1. $\tilde{\sigma}$ is defined as the composite map

$$F(a) \xrightarrow{\sim} F[x] / \langle f \rangle \xrightarrow{\sigma} F'[x] / \langle \sigma f \rangle \xrightarrow{\sim} F'(b)$$

2. We induct on $n = \deg f$. Observe that if f splits completely in $F[x]$, then σf splits completely in $F'[x]$, so $K = F, K' = F'$, hence $\tilde{\sigma} = \sigma$. This completes the base case.

Now, assume $n > 1$, and suppose $p(x)$ is an irreducible factor of $f(x)$ of degree ≥ 2 . Then, σp is an irreducible factor of σf with the same degree as p . So, let $\alpha \in K$ be a root of p and $\beta \in K'$ a root of σp . Then, by (1), σ extends to an isomorphism $\sigma' : F(\alpha) \rightarrow F'(\beta)$ such that $\sigma'(\alpha) = \beta$.

Let $F_1 := F(\alpha), F'_1 := F'(\beta)$. Then, $f(x) = (x - \alpha)f_1(x)$, and $\sigma f(x) = (x - \beta)\sigma' f_1(x)$, so $\deg f_1 = \deg \sigma f_1 = n - 1$. Moreover, K is a splitting field of f_1 over F_1 , and similarly K' is a splitting field of $\sigma' f_1$ over F'_1 . By induction, there exists an isomorphism $\tilde{\sigma} : K \rightarrow K'$ extending σ' , which induces an extension of σ .

$$\begin{array}{ccc} K & \xrightarrow{\tilde{\sigma}} & K' \\ | & & | \\ F(\alpha) & \xrightarrow{\sigma'} & F'(\beta) \\ | & & | \\ F & \xrightarrow{\sigma} & F' \end{array}$$

□

Definition

27.7

Let $K, L \supset F$ be two field extensions. A homomorphism $\sigma : K \rightarrow L$ is said to be an **F -homomorphism** if $\sigma(x) = x$ for all $x \in F$. If σ is also an isomorphism, we say that K and L are **F -isomorphic**.

Corollary

27.8

Any two splitting fields for a polynomial $f \in F[x]$ are F -isomorphic.

Proof. Take $\sigma : F \rightarrow F$ to be the identity map id_F in Theorem 27.6. □

Definition

27.9

An extension $K \supset F$ is a **normal extension** if K is algebraic over F , and if every irreducible polynomial $f \in F[x]$ with one root in K splits completely in $K[x]$. In this case, we say **K is normal over F** .

Theorem

27.10

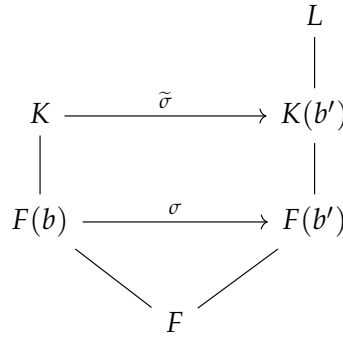
Let $f \in F[x]$ and K be a splitting field of f over F . Then, K is a normal extension of F .

Proof. Let $a_1, \dots, a_n$ be the roots of f in K , so $K = F(a_1, \dots, a_n)$. Let $g \in F[x]$ be irreducible with a root $b \in K$. Let L be a splitting field of g over K , and b' be any root of g in L .

Claim: $b' \in K$, so g splits in $K[x]$.

Let $\sigma : F(b) \xrightarrow{\sim} F(b')$ be an isomorphism with $\sigma(b) = b'$. Note that K is a splitting field for f over $F(b)$, and $K(b') = F(a_1, \dots, a_n, b')$ is a splitting field of $\sigma f = f$ over $F(b')$. Therefore, by the

previous theorem, we obtain an isomorphism $\tilde{\sigma} : K \rightarrow K(b')$ extending σ .



In particular $\sigma(b) = b'$. Now, b lies in $F(a_1, \dots, a_n) = F[a_1, \dots, a_n]$, so $b = h(a_1, \dots, a_n)$ for some polynomial $h \in F[x_1, \dots, x_n]$. Furthermore, since $\tilde{\sigma}f = f$, we know $\tilde{\sigma}(a_1), \dots, \tilde{\sigma}(a_n)$ are roots of f in $K(b')$, so they are just $a_1, \dots, a_n$ again in some different order. Therefore,

$$b' = \tilde{\sigma}(b) = \tilde{\sigma}(h(a_1, \dots, a_n)) = h(\tilde{\sigma}(a_1), \dots, \tilde{\sigma}(a_n)) \in K,$$

proving the claim. $\square$

Definition

27.11

We say that two elements $a, b \in K$ are **conjugate** over F if they have the same minimal polynomial over F .

Corollary

27.12

Let $K \supset F$ be a finite extension. Then, TFAE:

- (i) K is normal
- (ii) K is the splitting field of an $f \in F[x]$
- (iii) Every conjugate of an element of K lies in K .

Proof. Exercise. $\square$

Definition

27.13

A field $\bar{F}$ is called an **algebraic closure** of F if $\bar{F}$ is algebraic over F , and if every polynomial $f \in F[x]$ splits completely over $\bar{F}$. A field K is **algebraically closed** if every non-constant polynomial $f \in K[x]$ has a root in K (hence splits completely in K).

Remark

27.14

If $\bar{F}$ is an algebraic closure of F , then $\bar{F}$ is algebraically closed.

Proof. If $f \in \bar{F}[x]$, and α is a root of f , then $\bar{F}(\alpha) \supset \bar{F}$ is an algebraic extension. Since $\bar{F} \supset F$ is algebraic, we deduce that $\bar{F}(\alpha) \supset F$ is algebraic, hence α is algebraic over F . Thus, $\alpha \in \bar{F}$. $\square$

Example 27.15. $\mathbb{C}$ is the algebraic closure of $\mathbb{R}$, and algebraically closed.

27.1. Construction of Algebraic Closure of a Field (E. Artin)

Proposition

27.16

For any field F , there exists an algebraically closed field K containing F .

Proof. To each polynomial $f \in F[x]$ of degree ≥ 1 , we associate a letter x_f . Let S be the set of all such letters, and form the polynomial ring $F[S]$. Let I be the ideal of $F[S]$ generated by all polynomials $f(x_f)$. We claim $I \neq F[S]$: Indeed, otherwise, we have a finite combination

$$g_1 f_1(x_{f_1}) + \cdots + g_n f_n(x_{f_n}) = 1 \quad (5)$$

with $g_i \in F[S]$. For simplicity, write $x_i := x_{f_i}$. The polynomials g_i involve a finite number of variables, says $x_1, \dots, x_N$ with $N \geq n$. Then, the relation (5) now reads as

$$\sum_{i=1}^n g_i(x_1, \dots, x_N) f_i(x_i) = 1. \quad (6)$$

Now, let $F' \supset F$ be a finite extension in which each f_i has a root α_i . Let $\alpha_i := 0$ for $i > n$. Then, substituting into (6) yields $0 = 1$.

Now, knowing $I \neq F[S]$, we have $I \subset M$ for some maximal ideal M . Now, $K_1 := F[S]/M$ is a field containing (an isomorphic copy of) F . Moreover, each of the polynomials $f \in F[x]$ has a root in K_1 by construction, namely the image of x_f , since $f(x_f) \in I \subset M$.

Now, performing the same construction with K_1 in place of F , we obtain a field $K_2 \supset K_1$, where all non-constant polynomials in $K_1[x]$ have a root. Continuing this way, we obtain a sequence of fields $F := K_0 \subset K_1 \subset K_2 \subset \cdots$, where each non-constant polynomial in $K_n[x]$ has a root in K_{n+1} . Let $K = \bigcup_{n=0}^{\infty} K_n \supset F$, and we can guarantee K is algebraically closed, since any polynomial $h \in K[x]$ lies in $K_n[x]$ for some sufficiently large n . Then, h has a root in $K_{n+1} \subset K$. Therefore, K is algebraically closed. $\square$

Theorem

27.17

Every field F has an algebraic closure. Moreover, the algebraic closure is unique up to isomorphism.

Proof. By the previous proposition, there exists an algebraically closed field $K \supset F$. Let $\bar{F}$ be the collection of elements of K which are algebraic over F . By definition, $\bar{F}$ is an algebraic extension of F . If $f \in F[x]$, then f splits completely over K into linear factors $x - \alpha$, but each α is a root of f , so algebraic over F , so $\alpha \in \bar{F}$. Thus, each $f \in F[x]$ splits completely in $\bar{F}[x]$.

Uniqueness is proved along the same lines as the uniqueness of splitting fields, using Zorn's Lemma, but working with a family $\mathcal{F}$ of polynomials in $F[x]$ to be split, instead of a single $f(x)$. $\square$

28. Separable Extensions

Definition

28.1

Let $f \in F[x]$ with leading coefficient $a \in F \setminus \{0\}$. Over a splitting field for f , we can factor

$$f(x) = a(x - \alpha_1)^{n_1} \cdots (x - \alpha_k)^{n_k},$$

where the α_i are distinct, and $n_i \geq 1$. We say that α_i is a **simple root** if $n_i = 1$, and a **multiple root** if $n_i > 1$. This n_i is called the **multiplicity** of α_i .

Definition

28.2

A polynomial $f \in F[x]$ is called **separable** if it has no multiple roots. Otherwise, f is called **inseparable**.

Notice that if $f(x)$ has distinct roots in one splitting field, then $f(x)$ has distinct roots in any splitting field, since we have an F -isomorphism between any two splitting fields for f that is bijective on the set of roots.

Our focus is on the separability of irreducible polynomials. We'll see that in fields of characteristic > 0 , there are many irreducible polynomials which are inseparable.

Example 28.3. Let $F := \mathbb{F}_2(t)$, where $\mathbb{F}_2 = \mathbb{Z}/2\mathbb{Z}$. Consider $f(x) := x^2 - t \in F[x]$. It's easy to check that f has no roots in F , hence irreducible over F . However, if α is a root of f in an extension field of F , then $\alpha^2 = t$, and since $\text{char } F = 2$, we have $(x - \alpha)^2 = x^2 - \alpha^2 = x^2 - t$. Therefore, f is inseparable over F .

Definition

28.4

Given $f(x) = a_n x^n + \cdots + a_1 x + a_0 \in F[x]$, the **derivative** of f is the polynomial $f'(x) = na_n x^{n-1} + \cdots + 2a_2 x + a_1$.

Proposition

28.5

A polynomial $f(x)$ has a multiple root $\alpha \iff \alpha$ is also a root of $f'(x)$, i.e. f and $f'(x)$ are both divisible by the minimal polynomial of α over F .

In particular, $f(x)$ is separable $\iff \gcd(f, f') = 1$.

Proof. If $f(x) = (x - \alpha)^n g(x)$ for some $n \geq 2$, then

$$f'(x) = n(x - \alpha)^{n-1} g(x) + (x - \alpha)^n g'(x)$$

has α as a root. Conversely, if α is a root of both f and f' , then $f(x) = (x - \alpha)h(x)$ and $f'(x) = h(x) + (x - \alpha)h'(x)$, so $h(\alpha) = 0$, hence $h(x) = (x - \alpha)h_1(x)$, so $f(x) = (x - \alpha)^2 h_1(x)$. $\square$

Corollary

28.6

In characteristic 0, every irreducible polynomial f is separable.

Proof. Since $\deg f'(x) = \deg f(x) - 1$ in characteristic 0, we must have $\gcd(f, f') = 1$ since f is

irreducible. □

Corollary

28.7

In characteristic p , an irreducible polynomial f is separable, unless it has the form $f(x) = b_0 + b_1x^p + \cdots + b_mx^{mp} = g(x^p)$ for some $g \in F[x]$.

Definition

28.8

Let F be a field of characteristic $p > 0$. The p -th power map $\varphi(x) := x^p$ is the **Frobenius endomorphism** of F .

Remark

28.9

φ is an injective field homomorphism $F \rightarrow F$. In particular, if F is a finite field, then φ is an isomorphism, so every element of F is a p -th power.

Proof. Follows from $(x + y)^p = x^p + y^p$ and $(xy)^p = x^p y^p$ in characteristic p . □

Definition

28.10

A field F of characteristic $p > 0$ is called **perfect** if every element in F is a p -th power, i.e. $F = F^p$, i.e. φ is an isomorphism. Any field of characteristic 0 is also called perfect.

Proposition

28.11

Every irreducible polynomial over a perfect field F is separable. A polynomial in $F[x]$ is separable if and only if it is the product of distinct irreducible polynomials in $F[x]$.

Proof. Suppose $\text{char } F = p > 0$, and $f(x) \in F[x]$ is irreducible. If f is inseparable, then $f(x) = q(x^p)$ for some $q(x) = a_mx^m + \cdots + a_1x + a_0 \in F[x]$. Since F is perfect, let $a_i = b_i^p$ for $0 \leq i \leq m$. Then,

$$f(x) = b_m^p x^{pm} + \cdots + b_1^p x^p + b_0^p = (b_m x^m + \cdots + b_1 x + b_0)^p.$$

This contradicts the irreducibility of $f(x)$. □

Remark

28.12

One can show that if $F \neq F^p$, there exists irreducible inseparable polynomials in $F[x]$.

Definition

28.13

A field extension $K \supset F$ is **separable** if its algebraic, and if every element of K is a root of a separable polynomial with coefficients over F . A field extension $K \supset F$ which is not separable is called **inseparable**.

Proposition

28.14

Any algebraic extension of a perfect field is separable.

29. Finite Fields

Lemma

29.1

Let F be a finite field containing a subfield E with $|E| = q$. Then, $|F| = q^{[F:E]}$.

Proof. Let $m := [F : E]$. If $b_1, \dots, b_m$ is a basis for F over E , then each element of F has a unique expression $x_1 b_1 + \dots + x_m b_m$ with x_i in E . Since each x_i can take on q distinct values, we see that $|F| = q^m$. $\square$

Corollary

29.2

If F is a finite field, then $|F| = p^n$, where $p = \text{char } F$ and $n = [F : \mathbb{F}_p]$.

Lemma

29.3

Suppose F is a finite field with q elements, and $E \subset F$ is a subfield. Then,

- (1) Every element $a \in F$ satisfies $a^q = a$.
- (2) The polynomial $x^q - x \in E[x]$ factors in $F[x]$ as

$$x^q - x = \prod_{a \in F} (x - a)$$

Proof. For (1), if $a = 0$, then $a^q = 0 = a$. If $a \neq 0$, then $a \in F^* = F \setminus \{0\}$. The multiplicative group $(F^*, \cdot)$ has $q - 1$ elements, hence $a^{q-1} = 1$, so $a^q = a$.

For (2), we know that every element of F is a root of $x^q - x$ by (1). Moreover, $x^q - x$ can have at most q roots, hence it has exactly q roots (all the elements of F). $\square$

Existence and Uniqueness of Finite Fields (E. Galois)

29.4

For every prime p and $n \geq 1$, there exists a finite field with p^n . Moreover, any finite field with $q = p^n$ elements is isomorphic to the splitting field of $x^q - x$ over $\mathbb{F}_p$.

Proof. Consider the splitting field F of $f(x) = x^q - x$ over $\mathbb{F}_p$. Note that f has q distinct roots in F , since $f'(x) = -1$, so $\gcd(f, f') = 1$. Let $\mathcal{S} := \{a \in F \mid a^q - a = 0\}$. Note that $0, 1 \in \mathcal{S}$, and if $a, b \in \mathcal{S}$, then

$$(a + b)^q - (a + b) = a^q + b^q - a - b = 0$$

so $a + b \in \mathcal{S}$, and for $b \neq 0$, $ab^{-1} \in \mathcal{S}$ as well. Hence, $\mathcal{S}$ is a field. Moreover, $f(x)$ splits completely over $\mathcal{S}$ by definition, hence $F = \mathcal{S}$, and $|F| = |\mathcal{S}| = q$.

For uniqueness: Let F be any finite field of order $q = p^n$. Then, $F \supset \mathbb{F}_p$, and by Lemma 29.3, F is the splitting field of $x^q - x$ over $\mathbb{F}_p$. The result follows from the uniqueness of splitting fields. $\square$

Theorem

29.5

Let F be any field, and $H \leq (F^\times, \cdot)$ be a finite subgroup of $F^\times$. Then, H is cyclic, and consists of roots of unity in F .

In particular, if F is finite, then $F^\times$ is cyclic.

Proof. Suppose $|H| = n$. Then, $\alpha^n = 1$ for all $\alpha \in H$, hence all elements of H are root of $x^n - 1$, which is a polynomial of at most n roots. Hence, H consists exactly of the n -th roots of unity in F . By the classification of finite abelian groups,

$$H \cong \mathbb{Z}/d_1\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/d_k\mathbb{Z}$$

as an additive group, where $d_1 \mid d_2 \mid \cdots \mid d_k$, and $n = d_1 \cdots d_k$. So, each element in H has order dividing d_k , so is a root of $x^{d_k} - 1$. This polynomial has at most d_k roots in F , so we conclude that $d_k = n$, $k = 1$, and $H \cong \mathbb{Z}/n\mathbb{Z}$ is cyclic. $\square$

Corollary

29.6

- (1) Any finite field $\mathbb{F}_{q^m}$ is a simple extension of $\mathbb{F}_q$ (so generated by a single element).
- (2) For all $m \geq 1$, there exists an irreducible polynomial in $\mathbb{F}_q[x]$ of degree m .

Proof.

- (1) If $(\mathbb{F}_{q^m}^\times, \cdot)$ is generated by $\zeta \in \mathbb{F}_{q^m}^\times$, then clearly $\mathbb{F}_{q^m} = \mathbb{F}_q(\zeta)$.
- (2) Since $[\mathbb{F}_q(\zeta) : \mathbb{F}_q] = m$, the minimal polynomial of ζ over $\mathbb{F}_q$ has degree m .

$\square$

30. Galois Theory

Definition

30.1

For any field K , the set of automorphisms of K (isomorphisms $K \rightarrow K$) forms a group under composition, denoted **Aut** K .

The set **Gal**(K/F) of all F -automorphisms of K (i.e. elements in $\text{Aut } K$ which are the identity on F) is called the **Galois group of K over F** .

Proposition

30.2

Let $K \supset F$ be a field extension, and suppose $\alpha \in K$ is algebraic over F . Then, for any $\sigma \in \text{Gal}(K/F)$, $\sigma(\alpha)$ is a root of the minimal polynomial of α over F . So, the elements of $\text{Gal}(K/F)$ permute the roots of polynomials in $F[x]$.

Proof. If $f(x) = c_0 + c_1x + \cdots + c_nx^n \in F[x]$ is the minimal polynomial of α over F , then

$$\begin{aligned} f(\alpha) = 0 &\implies \sigma(f(\alpha)) = 0 \implies 0 = \sigma(c_0 + c_1\alpha + \cdots + c_n\alpha^n) \\ &= \sigma(c_0) + \sigma(c_1)\sigma(\alpha) + \cdots + \sigma(c_n)\sigma(\alpha)^n \\ &= c_0 + c_1\sigma(\alpha) + \cdots + \sigma(c_n)\sigma(\alpha)^n. \end{aligned}$$

$\square$

Example 30.3. 1. $\text{Gal}(F/F) = \{1\}$

2. $\text{Gal}(\mathbb{C}/\mathbb{R}) = \{1, z \mapsto \bar{z}\} \cong \mathbb{Z}/2\mathbb{Z}$. [Indeed, since $\mathbb{C} = \mathbb{R} \oplus \mathbb{R}i$, $\sigma \in \text{Gal}(\mathbb{C}/\mathbb{R})$ is determined by $\sigma(i)$. But, as $x^2 + 1$ is the minimal polynomial of i over $\mathbb{R}$, $\sigma(i) = \pm i$.] More generally, if $K \supset F$ is a quadratic extension, and $\text{char } F \neq 2$, then $K = F(\sqrt{\delta})$, so the same work shows $\text{Gal}(K/F) \cong \mathbb{Z}/2\mathbb{Z}$.
3. $\text{Gal}(\mathbb{Q}(\sqrt[3]{7})/\mathbb{Q}) \cong \{1\}$ [If $\sigma \in \text{Gal}(\mathbb{Q}(\sqrt[3]{7})/\mathbb{Q})$, then σ is determined by $\sigma(\sqrt[3]{7})$. Now, $\sigma(\sqrt[3]{7}) \in \{\sqrt[3]{7}, \omega\sqrt[3]{7}, \omega^2\sqrt[3]{7}\}$ where ω is a third root of unity. However $\mathbb{Q}(\sqrt[3]{7}) \subset \mathbb{R}$ and the other two options for $\sigma(\sqrt[3]{7})$ are not in $\mathbb{R}$. Hence, $\sigma = 1$.]

Now, given any field extension $F \subset K$, we've associated to it some group $\text{Gal}(K/F)$. Let's go the other way.

Definition

30.4

Given $G \leq \text{Aut } K$, we say that the **fixed field** of G is the subfield

$$K^G := \{x \in K \mid \sigma(x) = x \forall \sigma \in G\},$$

which one checks is indeed a field.

Now, suppose $K \supset F$ is a field extension, and $G = \text{Gal}(K/F)$. Then, we have maps:

$$\begin{array}{ccc}
 K & & 1 \\
 \uparrow & & \downarrow \\
 E & \rightsquigarrow & \text{Gal}(K/E) \\
 \uparrow & & \downarrow \\
 F & & G
 \end{array}
 \qquad
 \begin{array}{ccc}
 1 & & K \\
 \downarrow & & \uparrow \\
 H & \rightsquigarrow & K^H \\
 \downarrow & & \uparrow \\
 G & & F
 \end{array}$$

It's easy to see that $K \rightsquigarrow 1, F \rightsquigarrow G, 1 \rightsquigarrow K$ at the three corners. But, at the fourth corner, $G \rightsquigarrow F$ is not always true, i.e. $K^{\text{Gal}(K/F)}$ may be different than F (example 3 in the above example). This motivates the following definition:

Definition

30.5

Let $K \supset F$ be a field extension. K is said to be **Galois over F** and $K \supset F$ is said to be a **Galois extension** if K is algebraic over F , and F is the fixed field of $\text{Gal}(K/F)$, i.e. $F = K^{\text{Gal}(K/F)}$. [We only consider finite Galois extensions, so algebraic is automatic]

Theorem

30.6

Let $K \supset F$ be a finite field extension. Then, $|\text{Gal}(K/F)| \leq [K : F]$. Furthermore, TFAE:

- (i) $K \supset F$ is Galois.
- (ii) $|\text{Gal}(K/F)| = [K : F]$
- (iii) $K \supset F$ is normal and separable.
- (iv) K is the splitting field of a separable polynomial $f \in F[x]$.

Remark

30.7

- (1) Part (iv) tells us how to construct Galois extensions.
- (2) We've already shown (iii) $\iff$ (iv) without the separable condition.

Fundamental Theorem of Galois Theory

30.8

Let $K \supset F$ be a Galois extension. Then there is a bijection between subfields of K containing F , and subgroups of $\text{Gal}(K/F)$ given by

$$E \longmapsto \text{Gal}(K/E)$$

$$K^H \longleftarrow H$$

which are inverses to each other. Moreover:

- (1) These correspondences are inclusion reversing, i.e. $H_1 \leq H_2 \iff K^{H_1} \supset K^{H_2}$
- (2) $[K : E] = |H|$ and $[E : F] = |\text{Gal}(K/F) : H|$ when $E \leftrightarrow H$.
- (3) $K \supset E$ is always Galois, with Galois group $H = \text{Gal}(K/E)$.
- (4) E is Galois over $F \iff H \trianglelefteq \text{Gal}(K/F)$. In this case, $\text{Gal}(E/F) \cong \text{Gal}(K/F)/H$

Remark

30.9

Given a finite extension $K \supset F$, and $f \in F[x]$, the Galois group $\text{Gal}(K/F)$ permutes the roots of $f(x)$. If K is the splitting field of f , then $K \cong F[\alpha_1, \dots, \alpha_n]$, where the α_i are the roots of $f(x)$. If $X := \{\alpha_1, \dots, \alpha_n\}$, we get a permutation representation $\varphi : \text{Gal}(K/F) \rightarrow S(X)$, where $\varphi(g)$ is the permutation of X induced by g .

Moreover, the representation φ is faithful, i.e. φ is injective [an automorphism $g \in \text{Gal}(K/F)$ acting trivially on X must be the identity, since X generates K over F]. So, $\text{Gal}(K/F)$ is realized as the set of permutations of X which come from F -automorphisms of K .

Remark

30.10

Since $\text{Gal}(K/F)$ is finite, it has a finite number of subgroups. Hence, if K/F is Galois, the fundamental theorem implies that there are only finitely many subfields E with $F \subset E \subset K$.

Proposition

30.11

Let $K \supset F$ be a finite extension of fields. Then, there exists an element α of K such that $K = F(\alpha)$ (so K/F is simple) $\iff$ there exist only a finite number of fields E with $F \subset E \subset K$.

Proof. If K is a finite field, then we know $K = F(\alpha)$ for some $\alpha \in K$, since $(K^\times, \cdot)$ is cyclic. Also, clearly there are only finitely many subfields of K . Now, we may assume that F is an infinite field (otherwise K would be finite since the extension is finite).

Suppose there are only finitely many fields E with $F \subset E \subsetneq K$, say $E_1, \dots, E_n$. Then, $\bigcup_{i=1}^n E_i \neq K$ (from HW #1 of last semester, where we showed a vector space over an infinite field can't be written as the union of finitely many proper subspaces). Now, choose $\alpha \in K \setminus \bigcup_{i=1}^n E_i$. Then,

clearly $F(\alpha) = K$, since $F(\alpha) \neq E_i$ for all $i = 1, \dots, n$.

Conversely, suppose $K = F(\alpha)$, and let $f \in F[x]$ be the minimal polynomial of α over F . Given $F \subset E \subset K$, let $g_E(x) \in E[x]$ be the minimal polynomial of α over E . Then, g_E divides f . So, there can only be finitely many such polynomials g_E . We claim that the map $E \mapsto g_E$ is injective, which will conclude the proof.

Indeed, let $E_0 \subset E$ be the subfield generated over F by the coefficients of g_E . Then, g_E has coefficients in E_0 , and is irreducible over E_0 since it's irreducible over E . Hence, the degree of α over E_0 is the same as the degree of α over E . Therefore, $E_0 = E$, since $K = F(\alpha) \supset E \supset E_0$ and $F(\alpha) = E_0(\alpha) = E(\alpha)$. This finishes the proof. $\square$

Theorem of the Primitive Element

30.12

Given $K \supset F$ a finite Galois extension, there exists $\gamma \in K$ such that $K = F(\gamma)$.

Remark

30.13

More generally, this is true for any finite separable extension, since any such is contained in a Galois extension.

Proof. If $K = F(\alpha_1, \dots, \alpha_n)$, f_i is the minimal polynomial of α_i , and L is the splitting field of $f_1 \cdots f_n$, then $F \subset K \subset L$, and L/F is Galois by Theorem 30.6. $\square$

Example 30.14. Any finite extension K of $\mathbb{Q}$ can be written as $\mathbb{Q}(\gamma)$.

31. Examples of Galois Correspondence

Consider the following four examples.

Example 31.1. Suppose $\text{char } F \neq 2$, and $K \supset F$ is a quadratic extension. Then, we know $K = F(\sqrt{a})$ for some $a \in F$, and then $\text{Gal}(K/F) = \{1, \sigma\}$, where $\sigma(x + y\sqrt{a}) = x - y\sqrt{a}$. This corresponds to the diagram:

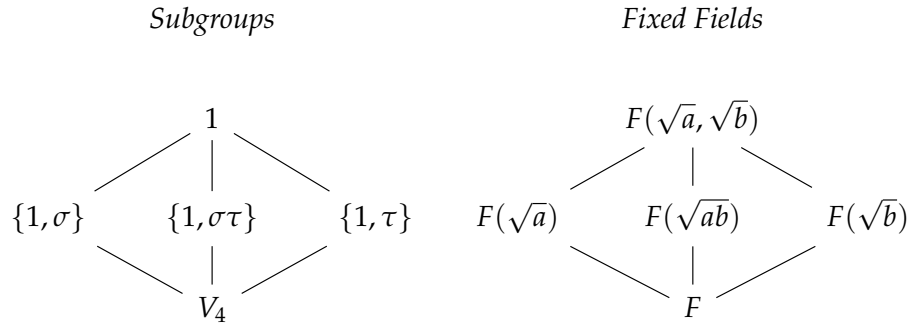
$$\begin{array}{ccc} 1 & & F(\sqrt{a}) \\ | & & | \\ G = \{1, \sigma\} & & F \end{array}$$

Example 31.2. An extension $K \supset F$ is **biquadratic** if $[K : F] = 4$, and K is generated by the roots of two quadratic polynomials in $F[x]$. Assume $\text{char } F \neq 2$. Then, we have $K = F(\sqrt{a}, \sqrt{b})$ for some $a, b \in F$ such that $\sqrt{a}, \sqrt{b} \notin F$. Then, the condition $[K : F] = 4$ is equivalent to $\sqrt{ab} \notin F$. [Indeed: if $K = F(\sqrt{a})(\sqrt{b})$ has degree 4 over F , then $\sqrt{b} \notin F[\sqrt{a}]$, so there does not exist $x, y \in F$ with $\sqrt{b} = x + y\sqrt{a} \iff x = \sqrt{b} - y\sqrt{a}$. By squaring, we see this equality holds if and only if $\sqrt{ab} \in F$.] Now, $K = F(\sqrt{a}, \sqrt{b})$ is the splitting field of $(x^2 - a)(x^2 - b)$, a separable polynomial over F . Hence, $K \supset F$ is a Galois extension. By our first example, there exists a unique $F(\sqrt{a})$ -automorphism σ of K with $\sigma(\sqrt{b}) = -\sqrt{b}$. Similarly, we have a $F(\sqrt{b})$ -automorphism τ with $\tau(\sqrt{a}) = -\sqrt{a}$. We thus also have their composition $\sigma\tau$ as a F -automorphism, which maps $\sqrt{a} \mapsto -\sqrt{a}$ and $\sqrt{b} \mapsto -\sqrt{b}$. Since

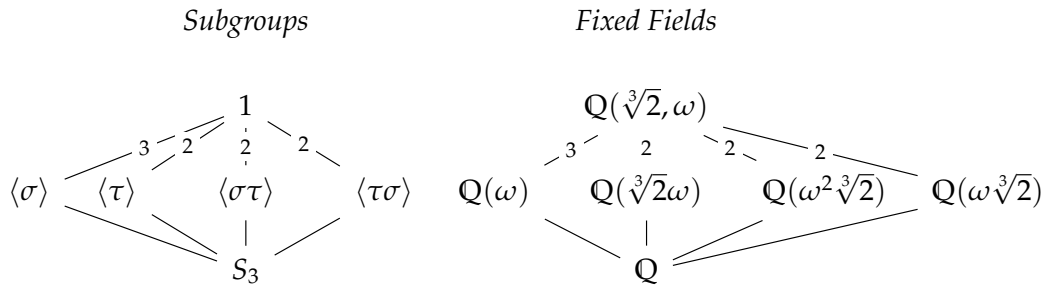
$|\text{Gal}(K/F)| = [K : F] = 4$, these must be all the automorphisms, i.e.

$$\text{Gal}(K/F) = \{1, \sigma, \tau, \sigma\tau\} \cong V_4 \cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}.$$

This has the correspondence diagram



Example 31.3. Consider the splitting field $K = \mathbb{Q}(\sqrt[3]{2}, \omega)$ of $x^3 - 2$ over $\mathbb{Q}$, where ω is a primitive third root of unity. Note that $[\mathbb{Q}(\omega) : \mathbb{Q}] = 2$ since $\omega^2 + \omega + 1 = 0$. Moreover, $[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3$, so $2, 3 \mid [K : \mathbb{Q}] \leq 3!$, hence $[K : \mathbb{Q}] = 6$ and $|\text{Gal}(K/\mathbb{Q})| = 6$. Since $x^3 - 2$ has three roots, $\text{Gal}(K/\mathbb{Q})$ is a subgroup of $S_3 = S(\{\text{roots}\})$, so by size considerations, $\text{Gal}(K/\mathbb{Q}) \cong S_3$. By similar work to example 2, $\text{Gal}(K/\mathbb{Q}) = \langle \sigma, \tau \rangle$, where $\sigma(\sqrt[3]{2}) = \omega\sqrt[3]{2}$ is a $\mathbb{Q}(\omega)$ -automorphism, and $\tau(\omega) = \omega^2$ is a $\mathbb{Q}(\sqrt[3]{2})$ -automorphism. This has the correspondence diagram



Notice how $\mathbb{Q}(\omega)/\mathbb{Q}$ is Galois, corresponding to the normal subgroup $\langle \sigma \rangle$ of S_3 . The other three extensions are not Galois over $\mathbb{Q}$, corresponding to the other three subgroups which are not normal.

Example 31.4 (The generic Galois extension). Let F be any field, $x_1, \dots, x_n$ indeterminants, and consider the field $K := F(x_1, \dots, x_n)$ of rational functions in $x_1, \dots, x_n$. The symmetric group S_n acts on K by permuting the variables: $\sigma f(x_1, \dots, x_n) = f(x_{\sigma(1)}, \dots, x_{\sigma(n)})$. Consider the fixed field $S := K^{S_n}$, consisting of rational functions which are **symmetric** in $x_1, \dots, x_n$. In S , we have the elementary symmetric polynomials

$$s_1 := x_1 + \cdots + x_n \quad s_j := \sum_{i_1 < \cdots < i_j} x_{i_1} \cdots x_{i_j} \quad s_n := x_1 \cdots x_n.$$

We claim $S = F(s_1, \dots, s_n)$, $[K : S] = n!$, and $K \supset S$ is Galois with Galois group S_n .

Indeed, note $S_n \subset \text{Gal}(K/S)$ by definition, so $|\text{Gal}(K/S)| \geq n!$, hence $|\text{Gal}(K/S)| = n!$. Consider a variable t and the polynomial

$$f(t) := \prod_{i=1}^n (t - x_i) = \sum_{i=0}^n (-1)^i s_i t^{n-i} = t^n - s_1 t^{n-1} + \cdots + (-1)^n s_n$$

in $S[t]$. K is clearly the splitting field of $f(t)$ over $F(s_1, \dots, s_n)$. Also $K \supset S \supset F(s_1, \dots, s_n)$, so $[K : S] \leq [K : F(s_1, \dots, s_n)] \leq n!$. On the other hand by [Theorem 30.6](#), we have $n! \leq |\text{Gal}(K/S)| \leq [K : S] \leq n!$. This proves the claim.

Remark

31.5

In the fourth example, the roots of $f(t)$ are independent variables, and each permutation of these roots extends to an automorphism of $F(x_1, \dots, x_n) = K$

Corollary

31.6

Any finite group G is the Galois group of some field extension

Proof. We know that G is a subgroup of S_n for some n . Let $K \supset F$ be a Galois extension with $\text{Gal}(K/F) = S_n$. Then, by the Galois correspondence ([Theorem 30.8](#)), $K \supset K^G \supset F = K^{S_n}$ and $K \supset K^G$ is Galois with $\text{Gal}(K/K^G) \cong G$. $\square$

Open Question (Inverse Galois Problem)

Is every finite group G a Galois group over $\mathbb{Q}$? That is, does there exist a Galois extension $K \supset \mathbb{Q}$ with $\text{Gal}(K/\mathbb{Q}) = G$.

Definition

31.7

If $f \in F[x]$ is a separable polynomial, the **Galois group of f** is $\text{Gal}(K/F)$, where K is a splitting field of f over F .

Theorem

31.8

Let $f \in \mathbb{Q}[x]$ be irreducible with $\deg f = p$ a prime. If f has precisely 2 non-real roots in $\mathbb{C}$, then the Galois group of f over $\mathbb{Q}$ is S_p .

Proof. Consider the following lemma:

Lemma

31.9

If H is a subgroup of S_p and H contains both a p -cycle and a transposition (ij) , then $H = S_p$.

Proof. Exercise. $\square$

Now, let K be the splitting field of f over $\mathbb{Q}$, and α a root of f . Then, $[\mathbb{Q}(\alpha) : \mathbb{Q}] = \deg f = p$ since f is irreducible. Now, $K \supset \mathbb{Q}(\alpha) \supset \mathbb{Q}$, hence

$$p = [\mathbb{Q}(\alpha) : \mathbb{Q}] [K : \mathbb{Q}(\alpha)] = |\text{Gal}(K/\mathbb{Q})|.$$

Also, we know $G := \text{Gal}(K/\mathbb{Q})$ is a subgroup of S_p , since its elements permute the roots of $f(x)$. By Cauchy's theorem, G has an element σ of order p , and σ maps to a p -cycle in S_p .

Let $X = \{z, \bar{z}, \alpha_3, \dots, \alpha_p\}$ be the set of all roots of f , where $z, \bar{z}$ are the two complex roots, and all other $\alpha_j \in \mathbb{R}$. Complex conjugation $\tau(x) = \bar{x}$ maps $z \mapsto \bar{z}$, $\bar{z} \mapsto z$, and fixes the α_j , so restricts to an element of $\text{Gal}(K/\mathbb{Q})$, which is a transposition (ij) when mapped to S_p . Thus, $\text{Gal}(K/\mathbb{Q}) = S_p$.

by the lemma. $\square$

Example 31.10. When $p = 5$, let $f(x) = 2x^5 - 10x + 5$, which is irreducible over $\mathbb{Q}$. Using calculus, one can show that f has three real roots, hence two complex roots, so its Galois group over $\mathbb{Q}$ is S_5 .

Remark

31.11

One can show that S_n is a Galois group over $\mathbb{Q}$ for all $n \geq 1$.

Theorem

31.12

$x^n - x + 1$ has Galois group S_n over $\mathbb{Q}$ (might be $x^n - x - 1$).

Theorem

31.13

Let $q = p^n$ for some prime p and $G = \text{Gal}(\mathbb{F}_{q^m}/\mathbb{F}_q)$. Then, the elements of G are the automorphisms $\sigma_0 = 1$ and $\sigma_1, \dots, \sigma_{m-1}$ with $\sigma_j(x) := x^{q^j}$. Hence, $\mathbb{F}_{q^m}/\mathbb{F}_q$ is a Galois extension with cyclic Galois group of order m .

Proof. Observe that each σ_j is an injective endomorphism of $\mathbb{F}_{q^m}$, hence an automorphism because $\mathbb{F}_{q^m}$ is finite. Since $\mathbb{F}_q$ consists of the roots of $x^q - x$, we see that $\sigma_j|_{\mathbb{F}_q} = \text{id}_{\mathbb{F}_q}$, so $\sigma_j \in G$ for each j . Note that $\sigma_0, \dots, \sigma_{m-1}$ are distinct since they take distinct values on a primitive element ζ of $\mathbb{F}_{q^m}$. Therefore, $m = [\mathbb{F}_{q^m} : \mathbb{F}_q] \geq |G| \geq m$, so $G = \{\sigma_i : 0 \leq i \leq m-1\}$ and $G \cong \mathbb{Z}/m\mathbb{Z}$ is cyclic. $\square$

32. Proofs of Theorems

We'll now attempt to prove [Theorem 30.6](#) and [Theorem 30.8](#). We begin with the following fact.

Proposition

32.1

Let G be a finite group of automorphisms of a field K , and let $F := K^G$ be its fixed field (so K/F is a Galois extension). Choose $\beta := \beta_1 \in K$, and let $\{\beta_1, \dots, \beta_r\}$ be its orbit under the action of G . Then, β is algebraic over F of degree r , and its minimal polynomial over F is $g(x) := (x - \beta_1) \cdots (x - \beta_r)$.

Proof. The coefficients of $g(x)$ are elementary symmetric polynomials in $\beta_1, \dots, \beta_r$, hence they are fixed under the action of any $\sigma \in G$. Hence, $g(x) \in K^G[x] = F[x]$. Let $f(x)$ be the minimal polynomial of β over F . Since $g(\beta) = 0$, we have $f(x) \mid g(x)$. But, since G permutes the roots of irreducible polynomials, each β_i must also be a root of $f(x)$. Hence, $g(x) \mid f(x)$. $\square$

Example 32.2. Let $K = \mathbb{Q}(\sqrt{2}, \sqrt{3})$, $F = \mathbb{Q}$, and $G = \langle \sigma, \tau \rangle \cong V_4$ from an earlier example. Then, the minimal polynomial of $\beta := 7\sqrt{2} + 5\sqrt{3}$ over $\mathbb{Q}$ is

$$g(x) = (x - 7\sqrt{2} + 5\sqrt{3})(x - 7\sqrt{2} - 5\sqrt{3})(x + 7\sqrt{2} + 5\sqrt{3})(x + 7\sqrt{2} - 5\sqrt{3}) \in \mathbb{Q}[x]$$

We can now prove [Theorem 30.6](#). Here it is again:

Restating Theorem 30.6

Let $K \supset F$ be a finite field extension. Then, $|\text{Gal}(K/F)| \leq [K : F]$. Furthermore, TFAE:

- (i) $K \supset F$ is Galois.
- (ii) $|\text{Gal}(K/F)| = [K : F]$
- (iii) $K \supset F$ is normal and separable.
- (iv) K is the splitting field of a separable polynomial $f \in F[x]$.

Proof.

Proposition**32.3**

If K is a field, and $\sigma_1, \dots, \sigma_n \in \text{Aut } K$ are distinct, then there do not exist elements $a_1, \dots, a_n \in K$ (not all zero) such that $\sum a_i \sigma_i(u) = 0$ for all $u \in K$.

Proof. Assume contrarily that we have a relation of this type with as few terms as possible. Upon renumbering, we may assume the minimal relation is $a_1 \sigma_1(u) + \dots + a_m \sigma_m(u) = 0$ with $a_i \neq 0$ for all $1 \leq i \leq m$. Clearly, $m > 1$, since $m = 1 \implies \sigma_1 = 0$ or $a_1 = 0$. So, there exists $c \in K$ with $\sigma_1(c) \neq \sigma_m(c)$. Now,

$$\begin{aligned} \sum_{i=1}^m a_i \sigma_i(cu) = 0 &\implies \sum_{i=1}^m a_i \sigma_i(c) \sigma_i(u) = 0 \\ \sigma_1(c) \sum_{i=1}^m a_i \sigma_i(u) = 0 &\implies \sum_{i=1}^m a_i \sigma_1(c) \sigma_i(u) = 0, \end{aligned}$$

so subtracting gives

$$\sum_{i=1}^m a_i (\sigma_i(c) - \sigma_1(c)) \sigma_i(u) = 0,$$

so we obtain a contradiction with $b_i = a_i (\sigma_i(c) - \sigma_1(c))$ for $2 \leq i \leq m$. □

Now, let $G := \{\sigma_1 = 1, \sigma_2, \dots, \sigma_n\}$ be a subgroup of $\text{Aut } K$, and let $F := K^G$. We claim $[K : F] = [K : K^G] = n = |G|$.

Proof. Indeed, suppose $n > [K : F]$. Let $w_1, \dots, w_m$ be a basis of K over F (so $m < n$). The system

$$\begin{cases} \sigma_1(w_1)x_1 + \dots + \sigma_n(w_1)x_n = 0 \\ \vdots \\ \sigma_1(w_m)x_1 + \dots + \sigma_n(w_m)x_n = 0 \end{cases} \quad (7)$$

of m equations in n unknowns has a non-zero solution $\beta_1, \dots, \beta_n$. Given any $u \in K$, write $u = \sum_{i=1}^m a_i w_i$ with $a_i \in F$. Multiplying the i 'th equation in (7) by a_i , and noticing that $\sigma_j(a_i) = a_i$ for all i, j , adding the resulting equation gives

$$\sigma_1(u)\beta_1 + \dots + \sigma_n(u)\beta_n = 0$$

for all $u \in K$. This contradicts [Proposition 32.3](#). On the other hand, if $n < [K : F]$, then there are

more than n F -linearly independent elements of K , say $\alpha_1, \dots, \alpha_{n+1}$. Then, they system

$$\begin{cases} \sigma_1(\alpha_1)x_1 + \dots + \sigma_1(\alpha_{n+1})x_{n+1} = 0 \\ \vdots \\ \sigma_n(\alpha_1)x_1 + \dots + \sigma_n(\alpha_{n+1})x_{n+1} = 0 \end{cases} \quad (8)$$

has a solution $\beta_1, \dots, \beta_{n+1}$ in K , where not all β_i are zero. Moreover, at least one β_i is not in F , as otherwise the $\{\alpha_i\}$ would be linearly dependent over F by the first equation in (8), since $\sigma_1 = 1$. Now, choose a solution $\beta_1, \dots, \beta_r$ with the minimal number of non-zero β 's. Dividing by β_r , we may assume $\beta_r = 1$. After reordering the system now reads

$$\sigma_i(\alpha_1)\beta_1 + \dots + \sigma_i(\alpha_{r-1})\beta_{r-1} + \sigma_i(\alpha_r) = 0$$

for $1 \leq i \leq n$. Assume $\beta_1 \notin F$. Then, there exists σ_k such that $\sigma_k(\beta_1) \neq \beta_1$. Applying σ_k to the equation above yields

$$\sigma_k\sigma_i(\alpha_1)\sigma_k(\beta_1) + \dots + \sigma_k\sigma_i(\alpha_r) = 0$$

for $1 \leq i \leq n$. But, since G is a group, we have $\{\sigma_k\sigma_1, \dots, \sigma_k\sigma_n\} = \{\sigma_1, \dots, \sigma_n\}$, so by reindexing, we write

$$\sigma_j(\alpha_1)\sigma_k(\beta_1) + \dots + \sigma_j(\alpha_{r-1})\sigma_k(\beta_{r-1}) + \sigma_j(\alpha_r) = 0.$$

Then subtraction this system from the first of the three system, we obtain

$$\sigma_j(\alpha_1)(\beta_1 - \sigma_k(\beta_1)) + \dots + \sigma_j(\alpha_{r-1})(\beta_{r-1} - \sigma_k(\beta_{r-1})) = 0$$

for $1 \leq j \leq n$. This is a solution to (8), and $\beta_1 - \sigma_k(\beta_1) \neq 0$, contradicting minimality. $\square$

Now, back to [Theorem 30.6](#). If $K \supset F$ is any field extension, then for $G = \text{Gal}(K/F)$, we have $K \supset K^G \supset F$. Hence, $|G| = [K : K^G] \leq [K : F]$, with equality if and only if $F = K^G$. This proves (i) $\iff$ (ii).

(i) $\implies$ (ii): Let $G \supset F$ be Galois, and take $\alpha \in K \setminus F$. [Proposition 32.1](#) implies that the minimal polynomial of α over F is $\prod_{i=1}^r (x - \sigma_i(\alpha))$, where $\sigma_1(\alpha), \dots, \sigma_r(\alpha)$ are the distinct elements of the G -orbit $G \cdot \alpha := \{\sigma(\alpha) : \sigma \in G\}$. Hence, K/F is normal and separable.

(iii) $\implies$ (iv): We've shown that $K \supset F$ being normal implies K is the splitting field of a polynomial $f \in F[x]$. If $K \supset F$ is separable, we can take f to be separable.

(iv) $\implies$ (i),(ii): Suppose K is the splitting field of a separable polynomial $f \in F[x]$. We use induction on $[K : F]$, so assume $[K : F] > 1$. Let $p(x)$ be an irreducible factor of $f(x)$ with $\deg p(x) > 1$. Let α be a root of $p(x)$ and $E := F(\alpha)$, so $F \subsetneq E \subset K$. Then, K is the splitting field of f over E , so by induction, K/E is Galois. Let $H := \text{Gal}(K/E) \leq G = \text{Gal}(K/F)$. Let $\alpha_1, \dots, \alpha_r$ be the distinct roots of $p(x)$, so $[E : F] = r$. By the isomorphism extension theorem, there exists $\tau_i \in G$ with $\tau_i(\alpha) = \alpha_i$ for $1 \leq i \leq r$. That is τ_i extends $F(\alpha) \cong F(\alpha_i)$ to K .

We claim that the cosets $\tau_i H$ are distinct for $1 \leq i \leq r$. Indeed, if $i \neq j$, and $\tau_i \tau_j^{-1} \in H$, then $\tau_i^{-1} \tau_j(\alpha) = \alpha$, hence $\alpha_i = \tau_i(\alpha) = \tau_j(\alpha) = \alpha_j$, a contradiction. So, we conclude that

$$G = |G : H| \cdot |H| \geq r \cdot |H| = [E : F] \cdot [K : E] = [K : F].$$

But, we always have $|G| \leq [K : F]$, hence $|G| = [K : F]$, proving (ii). $\square$

Next, recall the statement of [Theorem 30.8](#).

Restating Theorem 30.8

Let $K \supset F$ be a Galois extension. Then there is a bijection between subfields of K containing F , and subgroups of $\text{Gal}(K/F)$ given by

$$E \longmapsto \text{Gal}(K/E)$$

$$K^H \longleftarrow H$$

which are inverses to each other. Moreover:

- (1) These correspondences are inclusion reversing.
- (2) $[K : E] = |H|$ and $[E : F] = |\text{Gal}(K/F) : H|$ when $E \leftrightarrow H$.
- (3) $K \supset E$ is always Galois, with Galois group $H = \text{Gal}(K/E)$.
- (4) E is Galois over $F \iff H \trianglelefteq \text{Gal}(K/F)$. In this case, $\text{Gal}(E/F) \cong \text{Gal}(K/F)/H$

Proof. Let φ be the map $H \mapsto K^H$, and ψ the map $E \mapsto \text{Gal}(K/E)$. The definitions essentially imply that they are inclusion reversing.

φ is injective: Let $E_i := K^{H_i}$ for $i = 1, 2$, and $E_1 = E_2 = E$. Then, each H_i is a subgroup of $\text{Gal}(K/E)$, while $|\text{Gal}(K/E)| = |\text{Gal}(K/K^{H_i})| = |H_i|$ for both $i = 1, 2$, hence $H_1 = H_2$.

φ is surjective: As K/F is Galois, K is the splitting field of a separable $f \in F[x]$ over F , so K/E is a splitting field of f over E , hence K/E is Galois. So, $E = K^{\text{Gal}(K/E)}$, so $\varphi(\text{Gal}(K/E)) = E$.

Now, if $\varphi(H) = E$, then $|H| = |\text{Gal}(K/E)| = [K : E]$. Also $|G| = [K : F]$. So, dividing gives $[G : H] = |G|/|H| = [E : F]$. We've therefore proved statements 1, 2, and 3.

Now for the last part. We first claim that $E \supset F$ is Galois $\iff$ for all $\sigma \in G$, $\sigma(E) \subset E$. First, note that $E \supset F$ is separable because $K \supset F$ is separable. By the primitive element theorem, $E = F(\alpha)$ for some $\alpha \in E$.

($\implies$) If $E \supset F$ is Galois, it is normal, so the minimal polynomial $p(x)$ of α over F has all its roots in E . But, for any $\sigma \in G$, $\sigma(\alpha)$ is also a root of $p(x)$, so lies in E . Hence, $\sigma(E) \subset E$, since $E = F(\alpha)$.

($\impliedby$) Suppose $\sigma(\alpha) \in E$ for all $\sigma \in \text{Gal}(K/F)$. Then, E is the splitting field of $p(x) = \prod_{\sigma \in G} (x - \sigma(\alpha)) \in F[x]$, and hence is a normal extension of F . Hence, $E \supset F$ is Galois.

Now, let $H := \text{Gal}(K/E)$. Then, $E \supset F$ is normal $\iff$ for all $\sigma \in G, \tau \in H, \alpha \in E$, we have $\tau\sigma(\alpha) = \sigma(\alpha) \iff (\sigma^{-1}\tau\sigma)(\alpha) = \alpha \iff \sigma^{-1}\tau\sigma \in H$, since $E = F(\alpha)$. Hence, E/F is normal $\iff H$ is a normal subgroup of G . Also, we have a homomorphism $\psi : \text{Gal}(K/F) \rightarrow \text{Gal}(E/F)$ mapping $\sigma \mapsto \sigma|_E$. Clearly, $\ker \psi = \text{Gal}(K/E) = H$. Also, ψ is surjective, since

$$|\text{Im } \psi| = |G/H| = [G : H] = [E : F] = |\text{Gal}(E/F)|.$$

□

33. Vieta's Substitution on Cubic's

We want to solve $x^3 + ax^2 + bx + c = 0$. Set $x := y - \frac{a}{3}$ to get $y^3 + py + q = 0$. Next, set $y := z - \frac{p}{3z}$ to get $z^3 + q - \frac{p^3}{27z^3} = 0$. Then $z^6 + qz^3 - \frac{p^3}{27} = 0$, which is a quadratic equation in z^3 , with solutions $W = -\frac{q}{2} \pm \sqrt{\frac{p^3}{27} + \frac{q^2}{4}}$. If w_1, w_2, w_3 are the three cube roots of W , then the roots of the original equation are $w_1 - \frac{p}{3w_1}, w_2 - \frac{p}{3w_2}$, and $w_3 - \frac{p}{3w_3}$.

34. Solvability by Radicals

We want to make precise the notion of expressing a root by “radicals”.

One way to begin is by considering the extension $\mathbb{Q}(\sqrt[4]{2}) \supset \mathbb{Q}$, but the issue is that this extension is not Galois (the roots of the minimal polynomial $x^4 - 2$ are not all in $\mathbb{Q}(\sqrt[4]{2})$).

We restrict our work to characteristic zero.

Definition

34.1

A **simple radical extension** of a field F is an extension of the form $F(\sqrt[n]{a})$ for some $n \geq 1$ and $a \in F$. A finite extension $K \supset F$ is **cyclic** if K/F is Galois with a cyclic Galois group.

Theorem

34.2

Let F be a field which contain the n -th roots of unity.

- (1) If $a \in F$, then the extension $F(\sqrt[n]{a}) \supset F$ is cyclic of degree dividing n .
- (2) Conversely, if $K \supset F$ is cyclic of degree n , then $K = F(\sqrt[n]{a})$ for some $a \in F$.

Proof. For (1), since F contains the n -th roots of unity $1, \zeta, \dots, \zeta^{n-1}$, and the roots of $x^n - a$ are

$$\sqrt[n]{a}, \zeta \sqrt[n]{a}, \dots, \zeta^{n-1} \sqrt[n]{a},$$

we deduce that $F(\sqrt[n]{a})$ is the splitting field of $x^n - a$ over F , hence $F(\sqrt[n]{a}) \supset F$ is Galois. Moreover, given $\sigma \in \text{Gal}(F(\sqrt[n]{a})/F)$, $\sigma(\sqrt[n]{a})$ is again a root of $x^n - a$, so $\sigma(\sqrt[n]{a}) = \zeta_\sigma \sqrt[n]{a}$ for some $\zeta_\sigma \in \mu_n := \{x \in F \mid x^n = 1\}$. Moreover, one checks that $\zeta_{\sigma\tau} = \zeta_\sigma \zeta_\tau$, so the map $\varphi : \text{Gal}(F(\sqrt[n]{a})/F) \rightarrow \mu_n \cong \mathbb{Z}/n\mathbb{Z}$ sending $\sigma \mapsto \zeta_\sigma$ is an injective homomorphism. Therefore, $\text{Gal}(F(\sqrt[n]{a})/F)$ is a subgroup of $\mathbb{Z}/n\mathbb{Z}$.

For (2), suppose $K \supset F$ is a cyclic with $G := \text{Gal}(K/F) =: \langle \sigma \rangle$. Given $\alpha \in K$, and ζ any n -th root of unity in F , we define

$$(\alpha, \zeta) := \sum_{k=0}^{n-1} \zeta^k \sigma^k(\alpha) \in K$$

Since $\sigma(\zeta) = \zeta$, we have

$$\sigma(\alpha, \zeta) = \sum_{k=0}^{n-1} \zeta^k \sigma^{k+1}(\alpha) = \zeta^{-1} \sum_{k=0}^{n-1} \zeta^{k+1} \sigma^{k+1}(\alpha) = \zeta^{-1} \cdot (\alpha, \zeta).$$

Therefore, $\sigma((\alpha, \zeta)^n) = \zeta^{-n} (\alpha, \zeta)^n = (\alpha, \zeta)^n$, hence $(\alpha, \zeta)^n$ is fixed by all $\sigma \in G$, so as the extension is Galois, we have $(\alpha, \zeta)^n \in F$.

Now, let ζ be a primitive n -th root of unity in F , so $\mu_n = \langle \zeta \rangle$. Since the automorphisms $1, \sigma, \dots, \sigma^{n-1}$ are linearly independent, there exists $\alpha \in K$ with $(\alpha, \zeta) \neq 0$. Moreover, iterating $\sigma(\alpha, \zeta) = \zeta^{-1} \cdot (\alpha, \zeta)$ yields

$$\sigma^j(\alpha, \zeta) = \zeta^{-j}(\alpha, \zeta)$$

for all $j \in [0, n-1]$. It follows that (α, ζ) cannot lie in any proper subfield of K , and so $K = F((\alpha, \zeta))$. Finally, since $a := (\alpha, \zeta)^n \in F$, we get $K = F(\sqrt[n]{a})$. $\square$

Definition**34.3**

An element α which is algebraic over F can be **expressed by radicals** if $\alpha \in K$, where K satisfies

$$F =: K_0 \subset K_1 \subset \dots \subset K_{s-1} \subset K_s := K \quad (9)$$

such that $K_{i+1} = K(\sqrt[n_i]{a_i})$ with $a_i \in K_i$ for all $i \in [0, s-1]$. Such a field is called a **root extension of F** . A polynomial $f(x) \in F[x]$ is said to be **solved by radicals** if the splitting field of f is contained in a root extension of F .

Lemma**34.4**

If K is a root extension of F as in (9), then K is contained in a root extension which is Galois over F , and such that each intermediate extension K_{i+1}/K_i is cyclic.

Proof. There exists a finite extension $L \supset K \supset F$ such that $L \supset F$ is Galois, since $K \supset F$ is separable. Let $G = \text{Gal}(L/F)$, and consider the composite field

$$E := \prod_{\sigma \in G} \sigma(K).$$

Since E is stable under the action of G (i.e. $\tau E \subset E$ for all $\tau \in G$), we know $E \supset F$ is Galois (E is called the **Galois closure** of K). Also, one checks that $E \supset F$ is a root extension, since each $\sigma(K) \supset F$ is a root extension, and the composite KK' of two root extensions $K, K' \supset F$ is again a root extension. Hence, $E \supset F$ is a Galois root extension of F . Next, we adjoin to F the n_i -th roots of unity for all the roots $\sqrt[n_i]{a_i}$ which appear in (9) for $K \supset F$, to obtain a field F' . So, $F' \supset F$ is Galois and a root extension, with cyclic intermediate extensions. Moreover, we have composites

$$F \subset F' = F'K_0 \subset F'K_1 \subset \dots \subset F'K_s = F'K.$$

Using splitting fields, one sees that the composite of two Galois extensions is itself a Galois extension, so $F'K \supset F$ is Galois, and each $F'K_{i+1} \supset F'K_i$ is a simple radical extension. Moreover, since F' contains the appropriate roots of unity, we deduce from [Theorem 34.2](#) that all intermediate extensions of $K'F \supset F$ are cyclic. $\square$

Solvable Groups (Recall)**34.5**

A group G is **solvable** $\iff$ there exists subgroups $G := G_0 \supseteq G_1 \supseteq \dots \supseteq G_{s-1} \supseteq G_s = 1$ such that G_i/G_{i+1} is cyclic for all i .

Galois' Theorem**34.6**

The polynomial $f(x) \in F[x]$ can be solved by radicals if and only if the Galois group of $f(x)$ is a solvable group.

Proof.

($\implies$) Let E be the splitting field of $f(x)$ over F . By [Lemma 34.4](#), $E \subset K$, where K/F is Galois and each K_{i+1}/K_i is cyclic. Let $G := \text{Gal}(K/F)$, and $G_i \leq G$ be the subgroup associated to the intermediate fields K_i . Since $\text{Gal}(K_{i+1}/K_i) = G_i/G_{i+1}$ is cyclic, we see that $\text{Gal}(K/F)$ is solvable. Now, $K \supset E \supset F$, and $\text{Gal}(E/F)$ is a quotient group of $\text{Gal}(K/F)$, hence $\text{Gal}(E/F)$ is also solvable (a result from last semester).

($\impliedby$) Let K be the splitting field of $f \in F[x]$, and assume $G := \text{Gal}(K/F)$ is a solvable group. Then, there is a chain of subgroups $G = G_0 \supseteq G_1 \supseteq \cdots \supseteq G_s = 1$ with G_{i-1}/G_i cyclic. Taking the fixed fields of the G_i produces a chain of subfields of K , namely

$$F =: K_0 \subset K_1 \subset \cdots \subset K_s := K$$

with K_{i+1}/K_i a cyclic extension with degree n_i . Let F' be the field obtained from F by adjoining all n_i -th roots of unity for $0 \leq i \leq s-1$, and we form the composite fields

$$F \subset F' = F'K_0 \subset F'K_1 \subset \cdots \subset F'K_s = F'K.$$

Claim: The extension $F'K_{i+1}/F'K_i$ is cyclic of degree dividing n_i .

Note that this claim proves the theorem: since we have all appropriate roots of unity in the base field, [Theorem 34.2](#) implies each of these cyclic extensions is a simple radical extension. All the roots of $f(x)$ are thus contained in the root extension $F'K$, so $f(x)$ can be solved by radicals. $\square$

The claim is itself proved by the following lemma:

Lemma**34.7**

Let $K \supset F$ be a Galois extension and $F' \supset F$ by any finite extension. Then, $KF' \supset F$ is Galois, and

$$\text{Gal}(KF'/F') = \text{Gal}(K/K \cap F'),$$

so $\text{Gal}(KF'/F)$ is a subgroup of $\text{Gal}(K/F)$.

Proof. K is a splitting field of a (separable) polynomial $f \in F[x]$. Hence, KF' is a splitting field of $f(x)$ over F' , so $KF' \supset F'$ is Galois. If $\alpha_1, \dots, \alpha_n$ are the roots of $f(x)$, then $K = F(\alpha_1, \dots, \alpha_n)$, so $KF' = F'(\alpha_1, \dots, \alpha_n)$. If $\sigma \in \text{Gal}(KF'/F')$, then $\sigma(K) \subset K$, since σ fixes F and permutes the roots of $\alpha_1, \dots, \alpha_n$ of f . We therefore have a well defined map $\varphi : \text{Gal}(KF'/F') \rightarrow \text{Gal}(K/F)$ sending $\sigma \mapsto \sigma|_K$. Now, we compute

$$\ker \varphi = \{\sigma \in \text{Gal}(KF'/F') : \sigma|_K = \text{id}_K\} = 1,$$

since any $\sigma \in \text{Gal}(KF'/F')$ satisfies $\sigma|_{F'} = \text{id}_{F'}$. Therefore, φ is an injection, so $\text{Gal}(KF'/F')$ is realized as a subgroup of $\text{Gal}(K/F)$. Finally, we need to show $\text{Img } \varphi = \text{Gal}(K/K \cap F')$.

If $\sigma \in \text{Gal}(KF'/F')$, then $\varphi(\sigma)|_{K \cap F'} = \text{id}_{K \cap F'}$ since $\sigma|_{F'} = \text{id}_{F'}$. Therefore, $K \cap F' \subset K^{\text{Im} \varphi}$ (the fixed field). Conversely, if $x \in K \setminus F'$, then there exists $\rho \in \text{Gal}(KF'/F')$ such that $\rho(x) \neq x$. Therefore, $\varphi(\rho)(x) \neq x$ as well, so $x \notin K^{\text{Im} \varphi}$. Hence, $K^{\text{Im} \varphi} \subset K \cap F'$. So, we've shown $K^{\text{Im} \varphi} = K \cap F'$, hence the Galois correspondence implies $\text{Im} \varphi \cong \text{Gal}(K/K \cap F')$. $\square$

Corollary of Lemma 34.7

34.8

In the above situation,

$$[KF' : F] = \frac{[K : F][F' : F]}{[K \cap F' : F]}$$

Proof. Exercise. $\square$

Corollary of Galois' Theorem

34.9

The general equation of degree n cannot be solved by radicals for $n \geq 5$.

Proof. The symmetric group S_n is not solvable for $n \geq 5$, since its composition series is $S_n \supseteq A_n \supseteq 1$ with A_n simple. $\square$

Example 34.10 (Osada, J. Number Theory 25 (1987), 230-238). $x^n - x - 1$ has Galois group S_n over $\mathbb{Q}$.

Representations

35. Group Representations

We will work with representations of finite groups. These are very similar to (continuous) representations of compact groups like $S^1 := \{z \in \mathbb{C} : |z| = 1\}$, $S^1 \times \cdots \times S^1$, $\mathcal{O}(n)$, and $U(n)$.

Definition

35.1

Let G be a finite group. A **matrix representation** of G is a group homomorphism $\rho : G \rightarrow GL_n(\mathbb{C})$ for some n . The **degree** of ρ is n . If V is a complex vector space, a **(linear) representation of G on V** is a homomorphism $\rho : G \rightarrow \text{Aut}(V) \equiv GL(V)$. The representation is **faithful** if ρ is injective, so G is realized as a subgroup of $GL(V)$. When ρ is given, we say that V is a **representation** of G (by abusive language). Each $g \in G$ gives an element $\rho_g := \rho(g) \in GL(V)$, which we may denote by $g : V \rightarrow V$. Another way to view this is that V is a representation of $G \iff G$ acts on V by linear transformations.

We may use a field other than $\mathbb{C}$, but there's many reasons we like $\mathbb{C}$. Specifically, it needs to be an algebraically closed field of characteristic 0.

Remark

35.2

A linear representation of G becomes a matrix representation once a basis of V is chosen. More on this later.

Definition

35.3

A map of two representations V and W of G is a linear map $T : V \rightarrow W$ such that the diagram

$$\begin{array}{ccc} V & \xrightarrow{T} & W \\ \downarrow g & & \downarrow g \\ V & \xrightarrow{T} & W \end{array}$$

commutes for all $g \in G$. Precisely, $T(gv) = gT(v)$ for all $g \in G$. We say that T is **G -linear** or **G -equivariant**. This allows us to define **isomorphisms** of two representations in the obvious way.

Example 35.4. (1) A representation of degree 1 of G is a group homomorphism $\rho : G \rightarrow GL_1(\mathbb{C}) = \mathbb{C}^*$. Since G is a finite group, each element of G has finite order, so $\rho(g)$ must be a root of unity for every $g \in G$. If we take $\rho(g) = 1$ for all g , we get the **trivial representation** of G .

(2) The cyclic and dihedral groups C_n and D_n act in the usual way on $\mathbb{R}^2$ by rotations and reflections of a regular n -gon centered around the origin. These are representations in $GL_2(\mathbb{R})$, which extend to representations in $GL_2(\mathbb{C})$.

- (3) Let $\mathbb{C}G := \bigoplus_{g \in G} \mathbb{C}e_g$ be the free $\mathbb{C}$ -module (vector space) with basis indexed by the elements of G . For each $h \in G$, let $\rho_h : \mathbb{C}G \rightarrow \mathbb{C}G$ be the linear map sending $e_g \mapsto e_{hg}$. This defines a G -representation on $\mathbb{C}G$ called the **regular representation** of G .
- (4) More generally than (3), suppose X is a finite set on which G acts, inducing a permutation representation $G \rightarrow S(X)$. Let $V := \bigoplus_{x \in X} \mathbb{C}e_x$ be the $\mathbb{C}$ -vector space with basis $\{e_x \mid x \in X\}$. Then, the resulting linear representation of G on V is also called a **permutation representation** of G .
- (5) Recall that $\mathbb{C}G$ is a non-commutative $\mathbb{C}$ -algebra under

$$\left(\sum a_h e_h\right)\left(\sum b_k e_k\right) = \sum_{g \in G} \left(\sum_{hk=g} a_h b_k\right) e_g.$$

So, a complex vector space V is a (left) $\mathbb{C}G$ -module in the usual sense, such that the scalars in $\mathbb{C}G$ ($c \cdot 1_G$) act on the usual way on V [i.e. $\lambda e_g(v) = \lambda v$ when $g = 1$]. So, a $\mathbb{C}G$ -module by definition is a complex vector space V , together with a compatible system of actions $\rho_g \in GL(V)$, i.e. a homomorphism $\rho : G \rightarrow GL(V)$. That is, the concepts of a linear representation of G and a $\mathbb{C}G$ -module are equivalent. We may transfer concepts from module theory to representation theory using this idea. For example, a homomorphism of representations is a just a homomorphism of $\mathbb{C}G$ -modules. Note that this is somewhat similar to our discussion of $\mathbb{C}[x]$ -modules, which were defined with respect to some linear map.

Definition

35.5

Let $\rho : G \rightarrow GL(V)$ be a representation of G . Suppose $W \leq V$ is a subspace. W is said to be **stable** under the action of G , or **G -invariant**, if $\rho(g)W \subset W$ for all $g \in G$. In this case, $\rho(g)$ restricts to an isomorphism $W \rightarrow W$ for all $g \in G$, and $\rho|_W : G \rightarrow GL(W)$ is a representation of G on W . We say that W is a **subrepresentation** of V . This is equivalent to W being a $\mathbb{C}G$ -submodule of V .

Example 35.6. Let $V := \mathbb{C}G$ be the regular representation of G , let $x := \sum_{g \in G} e_g$, and let $W := \langle x \rangle$, a 1-dimensional subspace. Then, we have $\rho_g(x) = x$ for all $g \in G$, so W is a subrepresentation spanned by x , isomorphic to the trivial representation of G .

Definition

35.7

We can define the **direct sum** of representations V, W of G as the vector space $V \oplus W$ with the obvious action $\rho := \rho_V \oplus \rho_W$. In matrix form, this looks like

$$\begin{pmatrix} [\rho_V(g)] & [0] \\ [0] & [\rho_W(g)] \end{pmatrix}.$$

Definition

35.8

A G -representation V is **irreducible** if it has no G -invariant subspaces, besides 0 and V .

As usual, we want to decompose a representation as a direct sum of irreducible subrepresentations. The key to doing this is the following theorem. Note that this theorem fails in characteristic p .

Maschke's Theorem

35.9

Let G be a finite group, and V be (finite dimensional) $\mathbb{C}G$ -module. If W is any $\mathbb{C}G$ -submodule of V , there exists a $\mathbb{C}G$ -submodule U of V such that $V = U \oplus W$.

Proof. Let $\langle \cdot, \cdot \rangle_1$ be any hermitian inner product on V (this always exists, since $V \cong \mathbb{C}^n$, so we may identify any inner product on $\mathbb{C}^n$ with an inner product on V). We define a **G -invariant** inner product $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{C}$ by

$$\langle v_1, v_2 \rangle := \sum_{g \in G} \langle gv_1, gv_2 \rangle_1.$$

It's easy to check that this is indeed a hermitian inner product on V , and moreover $\langle gx, gy \rangle = \langle x, y \rangle$ for all $x, y \in V$. Our claim is that $W^\perp$, defined under this inner product, is a G -invariant subspace.

Indeed, if $y \in W^\perp$ and $g \in G$, we just need to show $gy \in W^\perp$. Well,

$$\langle gy, w \rangle = \langle gy, g(g^{-1}w) \rangle = \langle y, g^{-1}w \rangle,$$

and $g^{-1}w \in W$ because W is G -invariant. Hence, $\langle y, g^{-1}w \rangle = 0$. This concludes the claim, and the theorem with $U := W^\perp$. $\square$

Corollary

35.10

Every finite dimensional representation of G is a direct sum of irreducible representations.

Proof. This follows immediately by induction on $\dim V$ using Maschke's theorem. $\square$

Question

Suppose $V = W_1 \oplus \cdots \oplus W_k$ is a decomposition of V into a direct sum of irreducible representations. Is this decomposition unique?

No. For example, $\rho : G \rightarrow GL_n(\mathbb{C})$ given by $\rho(g) = I_n$ for all $g \in G$ shows that this is not true in general [every subspace is invariant, so any decomposition of $\mathbb{C}^n$ into a direct sum of lines is direct sum of irreducible representations]. We will see that the number of W_i which are isomorphic to a given irreducible representation does not change. Also, the *isotypical summands* are uniquely determined.

Given a representation V of G via $\rho : G \rightarrow GL(V)$, we get a matrix representation of G by choosing a basis $\mathcal{B}$ of V , and then $\rho_{\mathcal{B}}(g) := (\rho(g))_{\mathcal{B}}$. We use the shorthand $(g)_{\mathcal{B}}$.

Proposition

35.11

The $\mathbb{C}G$ -modules V and W are isomorphic $\iff$ there are bases $\mathcal{B}$ of V and $\mathcal{C}$ of W such that $(g)_{\mathcal{B}} = (g)_{\mathcal{C}}$ for all $g \in G$. In other words, the matrix representations induced by $\mathcal{B}$ and $\mathcal{C}$ coincide.

Proof. Let $T : V \rightarrow W$ be a G -linear isomorphism, let $\mathcal{B} := \{v_1, \dots, v_n\}$ be any basis of V , and let $\mathcal{C} := (T(v_1), \dots, T(v_n))$. From $T(gv_i) = g(T(v_i))$, it follows that $(g)_{\mathcal{B}} = (g)_{\mathcal{C}}$ for all $g \in G$. Conversely, suppose $\mathcal{B} = \{v_i\}$ and $\mathcal{C} = \{w_i\}$ are ordered bases for V and W such that $(g)_{\mathcal{B}} = (g)_{\mathcal{C}}$

for all $g \in G$. Then, let $T : V \rightarrow W$ by the isomorphism sending v_i to w_i . Since $T(gv_i) = gT(v_i) = gw_i$, we see that T is G -linear and an isomorphism. $\square$

Corollary**35.12**

Two matrix representations $\rho_1, \rho_2 : G \rightarrow GL_n(\mathbb{C})$ are **equivalent** [i.e. they induce isomorphism representations of $\mathbb{C}^n$] $\iff$ there exists a matrix $P \in GL_n(\mathbb{C})$ such that $\rho_2(g) = P\rho_1(g)P^{-1}$ for all $g \in G$. This P is the change of basis matrix given by T above.

36. Schur's Lemma**Schur's Lemma****36.1**

If V, W are irreducible representations of G , and $\varphi : V \rightarrow W$ is a G -linear map, then

- (1) Either φ is an isomorphism, or $\varphi = 0$.
- (2) If $V = W$, then $\varphi = \lambda I$ for some $\lambda \in \mathbb{C}$, where I is the identity.

Proof. For (1), observe that $\ker \varphi$ and $\text{Im } \varphi$ are G -invariant subspaces of V and W , respectively. But, V and W are irreducible, so $\ker \varphi = 0$ and $\text{Im } \varphi = W$, or vice-versa. For (2), since $\mathbb{C}$ is algebraically closed, φ must have an eigenvalue $\lambda \in \mathbb{C}$. This is equivalent to $\varphi - \lambda I$ having a non-zero kernel, but $\varphi - \lambda I$ is G -invariant, hence $\varphi - \lambda I = 0$ by the same reasoning. $\square$

Theorem**36.2**

For any representation V of (a finite group) G , there is a decomposition

$$V = V_1^{\oplus n_1} \oplus \cdots \oplus V_k^{\oplus n_k},$$

where the V_i are distinct irreducible representations. Moreover, the decomposition of V into a direct sums of the k factors is unique (they are called the **isotypical components of V**), as are the V_i that occur, and their multiplicities n_i .

Proof. First, suppose W is another representation of G with decomposition $W = \bigoplus W_j^{\oplus m_j}$. Any G -linear map $\varphi : V \rightarrow W$ by its action on each summand V_i . If $\pi_j : W \rightarrow W_j$ is the projection onto any summand, the composition $\pi_j \circ \varphi|_{V_i}$ is a G -linear map $V_i \rightarrow W_j$, so it must be zero unless $W_j \cong V_i$ by Schur's Lemma. It follows that φ must map the factor $V_i^{\oplus n_i}$ into that factor $W_j^{\oplus m_j}$ for which $V_i \cong W_j$. When we apply this to the identity map $\text{id} : V \rightarrow V$, the uniqueness statement follows. $\square$

Questions

1. Describe all irreducible representations of G .
2. Determine the direct sum decomposition for a representation of G (more importantly, find the multiplicities of any irreducible summand n_i).

Example 36.3 (Representation Theory of Finite Abelian Groups). Suppose G is an abelian group, and V a representation of G . Generally speaking, for $g \in G$, the map $g : V \rightarrow V$ won't be G -linear. But, when

G is abelian, it is, because $g \cdot (x \cdot v) = x \cdot (g \cdot v)$. If V is irreducible, Schur's lemma says that for all $v \in V$, $g \cdot v = \lambda_g \cdot v$ for some $\lambda_g \in \mathbb{C}$. It follows that any subspace of V is G -invariant, therefore $\dim V = 1$. Now, let $G \cong C_{n_1} \times \cdots \times C_{n_r}$, a product of cyclic groups. Let $C_{n_i} = \langle c_i \rangle$ and $g_i := (1, \dots, 1, c_i, 1, \dots, 1)$ with c_i in the i 'th slot, so

$$G = \langle g_1, \dots, g_r \mid g_i^{n_i} = g_i g_j g_i^{-1} g_j^{-1} = 1 \rangle.$$

Any irreducible representation V of G is 1-dimensional, and each $g \in G$ acts by multiplication by a scalar. Therefore, $g_i v = \lambda_i v$, where $\lambda_i \in \mathbb{C}$ must be an n_i -th root of unity. Moreover, the values $\lambda_1, \dots, \lambda_r$ determine the representation completely, since for each $g \in G$,

$$g v = (g_1^{k_1} \cdots g_r^{k_r}) v = (\lambda_1^{k_1} \cdots \lambda_r^{k_r}) v.$$

Conversely, given any n_i -th root of unity $\lambda_1, \dots, \lambda_r$,

$$g_1^{k_1} \cdots g_r^{k_r} \mapsto \lambda_1^{k_1} \cdots \lambda_r^{k_r}$$

defines a map $G \rightarrow \text{GL}_1(\mathbb{C}) = \mathbb{C}^\times$, which is a 1-dimensional representation of G .

Theorem

36.4

Let $G = C_{n_1} \times \cdots \times C_{n_r}$ be an abelian group. The representations $\rho_{\lambda_1, \dots, \lambda_r}$ constructed above are irreducible, and have degree 1. There are $|G|$ such representations, and every irreducible representation is equivalent to one of them. A general representation V of G will be decomposed as a direct sum of such 1-dimensional representations.

Here's some motivation for what's coming: given an abelian group G , the action $g : V \rightarrow V$ is determined as soon as we know the eigenvalues $\lambda_1, \dots, \lambda_r$ for the action. The important thing is, we actually only need to know $\sum_i \lambda_i$ for all $g \in G$, because, then we'd know $\sum_i \lambda_i^k$ for all $k \geq 1$, using $g^k : V \rightarrow V$, and this is enough to know all the λ_i . But, $\sum \lambda_i = \text{Tr}(g : V \rightarrow V) =: \chi_V(g)$. We'll see in general that $\chi_V : G \rightarrow \mathbb{C}$ determines V (not just for abelian groups).

Example 36.5 (More Important Examples of New Representations from Old Ones). 1. Given

V, W two representations of G , the tensor product $V \otimes W$ becomes a G -representation in a natural way: $\rho_1 \otimes \rho_2 : G \rightarrow \text{GL}(V \otimes W)$ satisfies $g \cdot (v \otimes w) = gv \otimes gw$.

2. $\text{Hom}(V, W)$ is also a representation of G : For any $\varphi \in \text{Hom}(V, W)$ and $g \in G$, we define $g\varphi : V \rightarrow W$ so that the diagram

$$\begin{array}{ccc} V & \xrightarrow{\varphi} & W \\ g \downarrow & & \downarrow g \\ V & \xrightarrow{g\varphi} & W \end{array}$$

commutes. That is, $(g\varphi)(g \cdot v) = g \cdot \varphi(v)$, which is equivalent (replace v with $g^{-1}v$) to $(g\varphi)(v) := v \cdot \varphi(g^{-1}v)$

3. If $W = \mathbb{C}$ and $\rho_W : G \rightarrow \mathbb{C}^*$ is the trivial representation, then we get the definition of $V^* := \text{Hom}(V, \mathbb{C})$ as a $\mathbb{C}G$ -module. So, for any $f \in V^*$, $(gf)(v) = f(g^{-1}v)$. In matrix notation, viewing things as column vectors,

$$f(g^{-1}v) = f^t \cdot ((g^{-1})^t)^t \cdot v = ((g^{-1})^t f)^t \cdot v \implies gf = (g^{-1})^t \cdot f.$$

4. For V a representation of G , set

$$V^G := \{v \in V \mid gv = v, \forall g \in G\} \leq V.$$

This is the sum of all trivial factors in the decomposition of V into irreducible G -representations.

Given any two representations V, W of G , let $\mathbf{Hom}_G(V, W)$ be the space of all G -linear maps $V \rightarrow W$. We claim $\mathbf{Hom}_G(V, W) = \mathbf{Hom}(V, W)^G$.

Indeed, $\mathbf{Hom}(V, W)^G = \{\varphi : V \rightarrow W \mid (g\varphi)(v) = \varphi(v), \forall v \in V, \forall g \in G\} = \{\varphi : V \rightarrow W \mid g\varphi(g^{-1}v) = \varphi(v), \forall v \in V, \forall g \in G\} = \mathbf{Hom}_G(V, W)$.

Exercise

Show examples (1)-(3) are compatible with the isomorphism $V \otimes W \cong \mathbf{Hom}(V^*, W)$.

37. Characters

Definition

37.1

Let $\rho : G \rightarrow \mathbf{GL}(V)$ be a representation of G . The **character** of the representation is the function $\chi_V : G \rightarrow \mathbb{C}$ with $\chi_V(g) := \mathrm{Tr}(\rho(g))$ for all $g \in G$. The **degree** is the degree ρ (i.e. $\dim V$). χ_V is said to be **irreducible** if ρ is irreducible.

The definitions in this section are somewhat scattered, so we state or restate them here:

Definition

37.2

Given V are representation of G , we set $V^G = \{v \in V \mid gv = v, \forall g \in G\}$. A **class function** from $G \rightarrow \mathbb{C}$ is a function which is fixed under the conjugacy classes of G . Given two class functions $f_1, f_2 : G \rightarrow \mathbb{C}$, we define their **inner product** to be

$$\langle f_1, f_2 \rangle = \frac{1}{|G|} \sum_{g \in G} f_1(g) \overline{f_2(g)}$$

which turns the $\mathbb{C}$ -vector space of all class functions on G into a hermitian inner product space, with dimension equal to the number of conjugacy classes of G .

Proposition

37.3

Suppose χ is a character of a degree n representation. Then, for all $g, h \in G$

- (i) $\chi(1) = n$
- (ii) $\chi(g^{-1}) = \overline{\chi(g)}$
- (iii) $\chi(hgh^{-1}) = \chi(g)$

Proof. For (i), $\rho(1) = I_n \in \mathbf{GL}(V)$, and $\mathrm{Tr}(I_n) = n$. For (ii), note that the eigenvalues $\lambda_1, \dots, \lambda_n$ of $\rho(g)$ are roots of unity, so satisfy $|\lambda_i|^2 = 1 \iff \overline{\lambda_i} = \lambda_i^{-1}$ for all i . Hence,

$$\overline{\chi(g)} = \sum_{i=1}^n \overline{\lambda_i} = \sum_{i=1}^n \lambda_i^{-1} = \mathrm{Tr}(\rho(g)^{-1}) = \mathrm{Tr}(\rho(g^{-1})) = \chi(g^{-1}).$$

Finally, (iii) follows for the property $\text{Tr}(PAP^{-1}) = \text{Tr}(A)$. $\square$

Remark

37.4

Part (iii) from the above proposition says χ is a class function.

Proposition

37.5

Suppose $\rho_1 : G \rightarrow \text{GL}(V)$ and $\rho_2 : G \rightarrow \text{GL}(W)$ are two representations of G . Then,

- (i) $\chi_{V \oplus W} = \chi_V + \chi_W$
- (ii) $\chi_{V^*} = \overline{\chi_V}$
- (iii) $\chi_{V \otimes W} = \chi_V \chi_W$.

Proof. For (i), given ρ_1, ρ_2 in matrix form, we have $\rho_1 \oplus \rho_2(g) = \begin{pmatrix} \rho_1(g) & 0 \\ 0 & \rho_2(g) \end{pmatrix}$. For (ii),

$$\chi_{V^*}(g) = \text{Tr}(\rho_1^*(g)) = \text{Tr}(\rho_1(g^{-1})^t) = \chi_V(g^{-1}) = \overline{\chi_V(g)}.$$

Similarly, for (iii), $\text{Tr}(\rho_1(g) \otimes \rho_2(g)) = \text{Tr}(\rho_1(g)) \text{Tr}(\rho_2(g))$ $\square$

Question

Given V , find V^G explicitly.

Well, for any $g \in G$, the map $g : V \rightarrow V$ is generally not G -linear. However, if we take the **average** $\varphi = \frac{1}{|G|} \sum_{g \in G} g : V \rightarrow V$, then φ is G -linear, since

$$\sum_{g \in G} g = \sum_{g \in G} hgh^{-1}, \forall h \in G \implies \left(\sum_{g \in G} g \right) h = h \left(\sum_{g \in G} g \right).$$

Proposition

37.6

φ is a projection of V onto V^G [so $\varphi : V \twoheadrightarrow V^G$, $\varphi^2 = \varphi$].

Proof. If $v = \varphi(w) = \frac{1}{|G|} \sum_{g \in G} gw$, then for all $h \in G$,

$$hv = \frac{1}{|G|} \sum_{g \in G} hgw = \frac{1}{|G|} \sum_{g \in G} gw = v.$$

Thus, $\text{Im } \varphi \subset V^G$. Conversely, if $v \in V^G$, then $\varphi(v) = \frac{1}{|G|} \sum_{g \in G} gv = v$, so $V^G \subset \text{Im } \varphi$. Also $\varphi \circ \varphi = \varphi$, so φ is a projection. $\square$

Question

How do we find $\dim V^G$?

This will be the **trace** of the projection φ . So,

$$\dim(V^G) = \text{Tr}(\varphi) = \frac{1}{|G|} \sum_{g \in G} \text{Tr}(g) = \frac{1}{|G|} \sum_{g \in G} \chi_V(g). \quad (10)$$

If V is irreducible and not the trivial representation, then by definition $V^G = 0$. Therefore, we de-

duce that $\sum_{g \in G} \chi_V(g) = 0$. Now, let V, W be any two representations of G . Recall $\text{Hom}(V, W)^G = \text{Hom}_G(V, W)$, so if V is irreducible, by Schur's lemma $\dim \text{Hom}(V, W)^G = \text{multiplicity of } V \text{ in } W$. Similarly, if W is irreducible, then $\dim \text{Hom}(V, W)^G$ is the multiplicity of W in V . Hence, if both V and W are irreducible, then $\dim \text{Hom}_G(V, W) = \begin{cases} 1 & V \cong W \\ 0 & V \not\cong W \end{cases}$. Moreover, the character $\chi_{\text{Hom}(V, W)}(g) = \chi_{V^* \otimes W}(g) = \overline{\chi_V(g)} \cdot \chi_W(g)$. So, applying (10), we obtain

$$\frac{1}{|G|} \sum_{g \in G} \overline{\chi_V(g)} \chi_W(g) = \begin{cases} 1 & V \cong W \\ 0 & V \not\cong W \end{cases}.$$

Let $\mathcal{C}$ be the $\mathbb{C}$ -vector space of class functions on G . Define a hermitian inner product on $\mathcal{C}$ by

$$\langle f_1, f_2 \rangle := \frac{1}{|G|} \sum_{g \in G} f_1(g) \overline{f_2(g)} \quad \text{“=”} \quad \int_G f_1(g) \overline{f_2(g)} dg$$

Theorem

37.7

The characters of the irreducible representations of G are orthonormal.

Remark

37.8

1. The number of irreducible representations is $\leq$ the number of conjugacy classes in G (we'll see later that they're in fact equal).
2. Suppose $\dim \mathcal{C} = k$, and $g_1, \dots, g_k$ are representatives of the conjugacy classes of G . If C_{g_i} is the centralizer of g_i , then

$$\langle \chi, \psi \rangle = \frac{1}{|G|} \sum_{i=1}^k |C_{g_i}| \chi(g_i) \overline{\psi(g_i)} = \sum_{i=1}^k \frac{\chi(g_i) \overline{\psi(g_i)}}{|C_G(g_i)|}$$

Theorem

37.9

Let V be a representation of G with character χ_V , and suppose $V = V_1^{\oplus n_1} \oplus \dots \oplus V_k^{\oplus n_k}$ is the decomposition of V into a direct sum of distinct irreducible representations. If W is an irreducible representation of G with character χ_W , then the number n_j of V_j 's isomorphic to W equals $\langle \chi_V, \chi_W \rangle$.

Proof. We have $\chi_V = \sum_{i=1}^k n_i \chi_{V_i}$, hence

$$\langle \chi_V, \chi_W \rangle = \sum_{i=1}^k n_i \langle \chi_{V_i}, \chi_W \rangle = n_j,$$

where $W \cong V_j$. □

Corollary

37.10

Two representations with the same character are isomorphic.

Corollary**37.11**

A representation V is irreducible if and only if $\langle \chi_V, \chi_V \rangle = 1$.

Proof. If $V \cong V_1^{\oplus n_1} \oplus \cdots \oplus V_k^{\oplus n_k}$ as usual, then $\langle \chi_V, \chi_V \rangle = \sum_{i=1}^k n_i^2$, which is 1 if and only if V is irreducible (i.e. we have $n_i = 1$ once and the rest of the n_i are zero). $\square$

38. Decomposition of the Regular Representation CG

From now on, the irreducible characters of G will be denoted $\chi_1, \dots, \chi_k$, and their degrees are $d_1, \dots, d_k$. This means $d_i = \chi_i(1)$. Recall the regular representation

$$R := \mathbb{C}G = \bigoplus_{g \in G} \mathbb{C}e_g.$$

Let χ_R be its character.

If $g \in G, g \neq 1$, then $g \cdot e_g = e_{gh} \neq e_h$. So, the diagonal entries in the matrix $(g)_E$ representing the action of $g : R \rightarrow R$ are all zero, hence $\text{Tr}(g) = 0$. Thus,

$$\chi_R(g) = \begin{cases} |G| & g = 1 \\ 0 & g \neq 1 \end{cases}.$$

We therefore see that R is not irreducible by [Corollary 37.11](#) when $G \neq \{1\}$. In fact, if $R = \bigoplus_{i=1}^k V_i^{\oplus n_i}$, then

$$n_i = \langle \chi_R, \chi_i \rangle = \frac{1}{|G|} (\chi_{V_i}(1) \cdot |G|) = \chi_{V_i}(1) = d_i.$$

This produces the following result:

Theorem**38.1**

The decomposition of $\mathbb{C}G$ into irreducible representations looks like

$$\mathbb{C}G \cong \bigoplus_{i=1}^k V_i^{\oplus \dim V_i}.$$

Moreover,

- (i) Every irreducible representation V of G appears in the regular representation $\dim V$ times.
- (ii) The degrees $d_1, \dots, d_k$ of the corresponding irreducible characters satisfy $\sum_{i=1}^k d_i^2 = |G|$
- (iii) $\chi_R(g) = \sum_{i=1}^k d_i \chi_{V_i}(g) = 0$ for all $g \neq 1$

Example 38.2 (Characters of S_3). Some of the work in this example is skipped, motivated by facts we will soon prove connecting irreducible representations, characters, and conjugacy classes ([Theorem 38.4](#)).

S_3 has two degree 1 representations: χ_U the trivial representation, χ_U the sgn representation. Moreover, S_3 acts on $\mathbb{C}^3$ by permuting e_1, e_2, e_3 . This has an invariant subspace $U = \langle (1, 1, 1) \rangle \subset \mathbb{C}^3$, which is a copy of the trivial representation. The complement V of U is

$$V = U^\perp = \{(z_1, z_2, z_3) \in \mathbb{C}^3 \mid z_1 + z_2 + z_3 = 0\}.$$

Now, $\chi_{\mathbb{C}^3} = (3, 1, 0)$, so $\chi_V = \chi_{\mathbb{C}^3} - \chi_U = (3, 1, 0) - (1, 1, 1) = (2, 0, -1)$. V is sometimes called the standard representation of S_3 . This has the following character table (definition below):

S_3	1	(12)	(123)
U	1	1	1
U'	1	-1	1
V	2	0	-1

Question

Which linear combinations of the endomorphisms $g : V \rightarrow V$ are G -linear?

Proposition

38.3

Let $f : G \rightarrow \mathbb{C}$ be any function on the group G . For V a representation of V , set

$$\varphi_{f,V} := \sum_{g \in G} f(g)g \in \text{Hom}(V, V).$$

Then, $\varphi_{f,V}$ is G -linear for all representations V if and only if f is a class function

Proof. Suppose f is a class function. Then, for $h \in G, v \in V$,

$$\varphi(hv) = \sum_{g \in G} f(g)ghv = \sum_{g \in G} f(hgh^{-1})(hgh^{-1}) \cdot hv = h \sum_{g \in G} f(g)g \cdot v = h\varphi(v),$$

so φ is G -linear. The other direction is left as an exercise [show that if f is not a class function, then $\varphi_{f,R}$ fails to be G -linear, where R is the regular representation of G]. $\square$

Theorem

38.4

The number of irreducible representations of G is equal to the number of conjugacy classes of G . Equivalently, their characters $\{\chi_V\}$ form an orthonormal basis for the space of class functions $\mathcal{C}$.

Proof. Suppose $f : G \rightarrow \mathbb{C}$ is a class function and $\langle f, \chi_V \rangle = 0$ for all irreducible representation V of G . We must show that $f = 0$ (this implies that the χ_V span $\mathcal{C}$)

Let V be any irreducible representation and consider

$$\varphi_{f,V} := \sum_{g \in G} f(g) \cdot g : V \rightarrow V,$$

which by the previous proposition is G -linear. So, by Schur's Lemma, $\varphi = \lambda \text{id}_V$, where

$$\lambda = \frac{1}{\dim V} \text{Tr}(\varphi_{f,V}) = \frac{1}{\dim V} \sum_{g \in G} f(g)\chi_V(g) = \frac{1}{\dim V} \langle f, \chi_{V^*} \rangle = 0.$$

Thus, $\varphi_{f,V} = 0$ for every irreducible representation V . Therefore, $\varphi_{f,V} = 0$ for every representation of V . So, take $V = R$ to be the regular representation of G . In $R = \mathbb{C}G$, the elements $\{g \mid g \in G\} \subset \text{Hom}(R, R)$ are linearly independent since $\{g(1) = e_g\}_{g \in G}$ are linearly independent. Since $0 = \varphi_{f,R} = \sum_{g \in G} f(g)g$, we deduce that $f = 0$. $\square$

Theorem**38.5**

If χ is an irreducible character of G , then $\chi(1) \mid |G|$

Proof. Let $C_1, \dots, C_k$ be the conjugacy classes of G , $C_i = [g_i]$. Then, by the orthogonality relations,

$$\frac{1}{|G|} \sum_{i=1}^k |C_i| \chi(g_i) \overline{\chi(g_i)} = \langle \chi, \chi \rangle = 1 \implies \frac{|G|}{\chi(1)} = \sum_{i=1}^k \frac{|C_i|}{\chi(1)} \chi(g_i) \overline{\chi(g_i)}.$$

We claim $\frac{|C_i|}{\chi(1)} \chi(g_i)$ is an algebraic integer. Note that this implies the theorem, since $\overline{\chi(g_i)}$ is also an algebraic integer for all i , hence $\frac{|G|}{\chi(1)}$ is both rational and an algebraic integer, thus an integer.

Let $C = [g]$ be a conjugacy class of G . Consider

$$\bar{C} := \sum_{x \in C} e_x \in \mathbb{C}G.$$

Observe that $\bar{C}$ lies in the center of $\mathbb{C}G$, since for all $g \in G$,

$$\bar{C} = \sum_{x \in C} e_{gxg^{-1}} \implies g \cdot \bar{C} = \bar{C} \cdot g.$$

Now, suppose V is an irreducible $\mathbb{C}G$ -submodule of $\mathbb{C}G$ (i.e. an irreducible representation of G) which character χ . Then, for all $g \in G, v \in V$, we have $\bar{C} \cdot (gv) = g(\bar{C} \cdot v)$, hence the map $v \mapsto \bar{C} \cdot v$ is g -linear. By Schur's Lemma, we must therefore have $\bar{C}v = \lambda v$ for all $v \in V$. Thus, taking traces, we get

$$\sum_{x \in C} x \cdot v = \lambda v \implies |C| \chi(g) = \sum_{x \in C} \chi(x) = \lambda \chi(1) \implies \lambda = \frac{\bar{C}}{\chi(1)} \chi(g)$$

where $C = [g]$. Let $g_1, \dots, g_n$ be the elements of G , so $\mathcal{B} := \{e_{g_i}\}$ is a basis of $\mathbb{C}G$. If $A := (\bar{C})_{\mathcal{B}}$, then A is a matrix with integer entries. So, the statement that $\bar{C} \cdot v = \lambda v$ with $v \neq 0$ implies λ is an eigenvalue of A , thus λ is an algebraic integer (it's the root of the characteristic polynomial which has integer coefficients), and we're done. $\square$

39. Character Tables and Orthogonality Relations

Definition**39.1**

Let $\chi_1, \dots, \chi_k$ be the irreducible characters of G , and $[g_i] = C_i$ be the distinct conjugacy classes of G . The $k \times k$ matrix $\{\chi_i(g_j)\}_{1 \leq i, j \leq k}$ is called the **character table** of G . We usually have $\chi_1 = 1_G$ and $g_1 = 1$. This is usually presented as follows:

$ C_i $	n_1	n_2	$\cdots$	n_k
g_i	$1 = g_1$	g_2	$\cdots$	g_k
$1_G = \chi_1$	$d_1 = 1$	1	$\cdots$	1
χ_2	d_2	$\cdots$		
$\vdots$				
χ_k	d_k	$\cdots$		

Orthogonality Relations**39.2**

Let δ_{rs} be 1 when $r = s$ and 0 otherwise. Then, we have the following relations:

Rows:

$$\sum_{i=1}^k \frac{\chi_r(g_i) \overline{\chi_s(g_i)}}{|C_G(g_i)|} = \langle \chi_r, \chi_s \rangle = \delta_{rs}$$

Columns:

$$\sum_{i=1}^k \chi_i(g_r) \overline{\chi_i(g_s)} = \delta_{rs} |C_G(g_r)|$$

Proof. We'll prove the column relation: For $1 \leq s \leq k$, let ψ_s be the class function which satisfies $\psi_s(g_r) = \delta_{rs}$. Since $\{\chi_i\}_{i=1}^k$ is an orthonormal basis of $\mathcal{C}$, we have $\psi_s = \sum_{i=1}^k \langle \psi_s, \chi_i \rangle \chi_i$. Now,

$$\langle \psi_s, \chi_i \rangle = \frac{1}{|G|} \sum_{g \in G} \psi_s(g) \overline{\chi_i(g)} = \frac{|C_s|}{|G|} \overline{\chi_i(g_s)} = \frac{\overline{\chi_i(g_s)}}{|C_g(g_s)|}.$$

Hence,

$$\delta_{rs} = \psi_s(g_r) = \sum_{i=1}^k \langle \psi_s, \chi_i \rangle \cdot \chi_i(g_r) = \sum_{i=1}^k \frac{\chi_i(g_r) \overline{\chi_i(g_s)}}{|C_G(g_s)|},$$

as desired. □

Example 39.3. Suppose we are given the following part of a character table of a group G :

$ C_i $	1	3	4	4
g_i	$1 = g_1$	g_2	g_3	g_4
χ_1	1	1	1	1
χ_2	1	1	ω	ω^2
χ_3	1	1	ω^2	ω
χ_4	?	?	?	?

where $\omega = e^{2\pi i/3}$. Fill in the missing entries.

Well, $|G| = 1 + 3 + 4 + 4 = 12 = 1^2 + 1^2 + 1^2 + d_4^2$, so $d_4 = 3$. Now, from orthogonality between columns 1 and 2,

$$1 \cdot 1 + 1 \cdot 1 + 1 \cdot 1 + 3 \cdot \overline{\chi_4(g_2)} = 0 \implies \chi_4(g_2) = -1.$$

Similarly, $\chi_4(g_3) = \chi_4(g_4) = 0$. So, we have

$ C_i $	1	3	4	4
g_i	$1 = g_1$	g_2	g_3	g_4
χ_1	1	1	1	1
χ_2	1	1	ω	ω^2
χ_3	1	1	ω^2	ω
χ_4	3	-1	0	0

This is the character table for A_4 (not obvious)

Example 39.4. For the cyclic group C_3 , we have the table

$ C_i $	1	1	1
g_i	1	a	a_2
χ_1	1	1	1
χ_2	1	ω	ω^2
χ_3	1	ω^2	ω

40. Burnside's Theorem

Burnside's Theorem

40.1

Any group G of order $p^a q^b$, with p, q prime, is solvable.

Lemma

40.2

Suppose $\zeta_1, \dots, \zeta_n \in \mathbb{C}$ are roots of unity such that $\frac{1}{n}(\zeta_1 + \dots + \zeta_n)$ is an algebraic integer. Then, either $\zeta_1 = \dots = \zeta_n$, or $\zeta_1 + \dots + \zeta_n = 0$.

Proof. Set $\alpha = \frac{1}{n}(\zeta_1 + \dots + \zeta_n)$ and assume not all ζ_i are equal. Then, the triangle inequality implies $|\zeta_1 + \dots + \zeta_n| < n$, or equivalently $|\alpha| < 1$. Let $p(x) \in \mathbb{Z}[x]$ be the minimal polynomial of α over $\mathbb{Q}$, and let K be the splitting field of p , with $G := \text{Gal}(K/\mathbb{Q})$. For any automorphism $\sigma \in G$, $\sigma(\alpha)$ is the average of a set of n roots of unity, so $|\sigma(\alpha)| \leq 1$. Therefore,

$$\left| \prod_{\sigma \in G} \sigma(\alpha) \right| < 1.$$

However, we have (exercise)

$$\prod_{\sigma \in G} (x - \sigma(\alpha)) = p(x)^m$$

where $m = [K : \mathbb{Q}(\alpha)]$. We therefore deduce that the constant term of $\prod (x - \sigma(\alpha))$ is an integer, but it's $\prod \sigma(\alpha)$ which has absolute value < 1 . Hence, $\prod \sigma(\alpha) = 0$, i.e. $\alpha = 0$. $\square$

Lemma

40.3

For any finite group G and representation $\rho : G \rightarrow \text{GL}_n(\mathbb{C})$, the matrix $\rho(g)$ is diagonalizable.

Proof. If $m = |G|$, then $\rho(g)^m = I$ in $\text{GL}_n(\mathbb{C})$. Hence, the minimal polynomial $f(x)$ of $\rho(g)$ divides $x^m - 1$, which is a polynomial with m distinct roots. Hence, $f(x)$ has distinct roots, so $\rho(g)$ is diagonalizable. $\square$

Theorem

40.4

Let V be an irreducible representation of G , and suppose C is a conjugacy class in G with $\gcd(|C|, \dim V) = 1$. Then, for any $g \in C$, either $\chi_V(g) = 0$, or g acts as a scalar on V .

Proof. There exists $a, b \in \mathbb{Z}$ such that $a \cdot |C| + b \cdot n = 1$, where $n = \dim V$. Hence,

$$\frac{\chi_V(g)}{n} = a \frac{|C|}{n} \chi_V(g) + b \chi_V(g),$$

but $\chi_V(g) = \lambda_1 + \dots + \lambda_n$ is an algebraic integer, when λ_i are the eigenvalues of g . By the proof

in Theorem 38.5, we prove the first term is an algebraic integer as well, hence $\frac{\chi_V(g)}{n}$ is an algebraic integer. By Lemma 40.3, the eigenvalues $\lambda_1, \dots, \lambda_n$ are roots of unity, hence the result follows by Lemma 40.2. $\square$

Theorem**40.5**

Let G be a finite group, and let C be a conjugacy class in G with $|C| = p^k$, where p is prime and $k > 0$. Then, G is not simple.

Proof. Choose an element $g \in C$. We know $g \neq 1$ because $k > 0$, and by the orthogonality of the columns of the character table of G , we have

$$\sum_{V \in \text{Irr}(G)} \dim V \cdot \chi_V(g) = 0, \quad (11)$$

where $\text{Irr}(G)$ is the set of k irreducible representations of G , up to isomorphism. We may divide $\text{Irr } G$ into three parts:

- (i) The trivial representation.
- (ii) S , the set of irreducible representations whose dimensions are divisible by p .
- (iii) T , the set of non-trivial irreducible representations whose dimension is not divisible by p .

Now, we make the following claim:

Claim 40.6. *There exists a $V \in T$ such that $\chi_V(g) \neq 0$.*

Proof. Indeed, if $V \in S$, the number $\frac{\dim V}{p} \chi_V(g)$ is an algebraic integer, so $a := \sum_{V \in S} \frac{\dim V}{p} \chi_V(g)$ is an algebraic integer. Now, by (11),

$$0 = \chi_C(g) + \sum_{V \in S} \dim(V) \chi_V(g) + \sum_{V \in T} \dim(V) \chi_V(g) = 1 + pa + \sum_{V \in T} \dim(V) \chi_V(g),$$

hence the last term is non-zero (because otherwise $a = -1/p$ is not an algebraic integer). The claim follows. $\square$

Now, we have $g \in C$, where $|C| = p^k$, and $V \in T$ with $\chi_V(g) \neq 0$, and $p \nmid \dim V$. So, by Theorem 40.4, g acts by a non-zero scalar on V . Let H be the subgroup generated by all elements ab^{-1} , with $a, b \in C$:

$$H := \langle ab^{-1} \mid a, b \in C \rangle.$$

Note that for any $x \in G$ $xab^{-1}x^{-1} = (xax^{-1})(xbx^{-1})^{-1} \in H$, so $H \trianglelefteq G$. Clearly, $|H| > 1$ since $|C| > 1$. Moreover, the elements of H act trivially on V , hence $H \neq G$ because V is not the trivial representation of G . This proves G is not simple. $\square$

We are now ready to prove Burnside's Theorem.

Proof. Let $|G| = p^a q^b$. Assume for the sake of contradiction that G is not solvable. Then, there exists a non-abelian simple group of order $p^a q^b$: otherwise, any such G has a nontrivial normal subgroup N , and by induction on $|G|$ we may assume $N, G/N$ are both solvable. This implies G is solvable.

If G is simple, we may consider the class equation

$$p^a q^b = |G| = \sum_{|C_i|=1} |C_i| + \sum_{|C_i|>1} |C_i|,$$

where the second term is $\cong 0 \pmod{pq}$. We deduce that there are at least 2 conjugacy classes C_i with $|C_i| = 1$, i.e. $Z(G) \neq 1$. This contradicts the simplicity of G . $\square$

Theorem

40.7

Let G be a finite group of order n , and $g \in G$. Then, TFAE:

- (a) For every representation V of G , $\chi_V(g) \in \mathbb{Q}$.
- (b) For every representation V of G , $\chi_V(g) \in \mathbb{Z}$.
- (c) The element g is conjugate to g^k for all $k \in \mathbb{Z}$ with $\gcd(k, n) = 1$.

Proof. Let $K = \mathbb{Q}(\zeta)$, with $\zeta = e^{2\pi i/n}$, i.e. K is the splitting field of $x^n - 1$ over $\mathbb{Q}$. Then, it's known that $\text{Gal}(K/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^\times$, where $k \in (\mathbb{Z}/n\mathbb{Z})^\times \mapsto (\sigma_k : \zeta \mapsto \zeta^k)$. For any $g \in G$ and for any representation V of G , we know $\chi_V(g) \in K$, and $\chi_V(g^k) = \sigma_k(\chi_V(g))$ for all k coprime to n .

Now, we know $\chi_V(g)$ is an algebraic integer, so clearly (a) $\iff$ (b). If $\chi_V(g) \in \mathbb{Z}$, we deduce that $\chi_V(g) = \chi_V(g^k)$, because if g, g^k were in two different conjugacy classes, then there is a class function $f \in \mathcal{C}$ for which $f(g) \neq f(g^k)$, hence there exists a character χ_V with $\chi_V(g) \neq \chi_V(g^k)$. (???)

If (c) holds, then $\sigma_k(\chi_V(g)) = \chi_V(g)$ for all k coprime to n , so $\chi_V(g)$ lies in $K^G = \mathbb{Q}$, hence (a) holds. $\square$

Exercise

Check that (iii) holds for all g in the symmetric group S_n . So, all character values for permutations in S_n are integers. What can be said about A_n ?

41. More on Group Characters

Question

Does the character table of G determine G ?

Answer

No.

For example, the groups D_4 (the dihedral group of order 8) and Q_8 (the quaternion group) have the same character table: Indeed, both groups have the same commutator quotient $G/[G, G] \cong V_4$, the Klein 4-group. It follows that they both have exactly four characters of degree 1, which are easily computed, and then the orthogonality relations can be used to find the fifth irreducible character, which has dimension 2. So the complete character table for Q_8 is which is identical to that of D_4 .

Sizes	1	1	2	2	2
Q_8	1	-1	i	j	k
χ_1	1	1	1	1	1
χ_2	1	1	-1	1	-1
χ_3	1	1	1	-1	-1
χ_4	1	1	-1	-1	1
χ_5	2	-2	0	0	0

Question

What information about a group can be deduced by its character table?

Lemma

41.1

Let $N \trianglelefteq G$ and W be a $\mathbb{C}(G/N)$ -module. Then, W admits a canonical $\mathbb{C}G$ -module structure, with a subspace of W being a $\mathbb{C}G$ -submodule $\iff$ it's a $\mathbb{C}(G/N)$ -submodule. If ψ is the character of the $\mathbb{C}(G/N)$ -module W , then the character of the $\mathbb{C}G$ -module W is $\psi \circ \pi$, where $\pi : G \rightarrow G/N$ is the projection map.

Proof. Given $g \in G, w \in W$, we define $g \cdot w := (gN) \cdot w$. □

Definition

41.2

Let $\rho : G \rightarrow \text{GL}(V)$ be a representation with character χ . Define $K_\chi := \{g \in G \mid \chi(g) = \chi(1)\}$, called the **kernel of χ** . We have $K_\chi = \ker \rho \trianglelefteq G$ [indeed, $\chi(g)$ is a sum of $\dim V$ eigenvalues, each of which is a root of unity. If $\chi(g) = \chi(1)$, then all the eigenvalues must be 1, so $\rho(g) = \text{id}$.] We write K_i instead of K_{χ_i} for $\chi_1, \dots, \chi_k$ the irreducible characters of G .

Proposition

41.3

All the normal subgroups of G are exactly the sets $\bigcap_{i \in I} K_i$ for all subsets $I \subset \{1, \dots, k\}$.

Proof. Since $K_i \trianglelefteq G$ for $i \in \{1, \dots, k\}$, then we know $\bigcap_{i \in I} K_i$ are normal subgroups of G .

Conversely, let $N \trianglelefteq G$ be a normal subgroup, let $W := \mathbb{C}(G/N)$, and let ψ be the character of W as a $\mathbb{C}(G/N)$ -module. Let χ be the character of W as a $\mathbb{C}G$ -module. As ψ is the regular character of G/N , we have $\chi(g) = \chi(1) \iff g \in N$. Therefore, $N = K_\chi$. Then, we may write $\chi = \sum_{i=1}^k a_i \chi_i$, and

$$|\chi(g)| \leq \sum_{i=1}^k |\chi_i(g)| \leq \sum_{i=1}^k a_i \chi_i(1) = \chi(1).$$

So, since $|\chi_U(g)| \leq \chi_U(1)$ for any $\mathbb{C}G$ -module U , we see that $g \in K_\chi \iff g \in K_i$ for all i for which $a_i > 0$. Therefore, $N = \bigcap_{i \in I} K_i$, where $I = \{1 \leq i \leq k \mid a_i > 0\}$. □

Corollary

41.4

G is simple if and only if the only irreducible character χ_i for which $\chi_i(g) = \chi_i(1)$ for some $g \neq 1$ is the principal character χ_1 .

Corollary

41.5

The character table of G can be used to determine whether or not G is solvable.

Proof. The character table allows us to determine the normal subgroups of G , and all the inclusion relations between them, so we can determine whether or not G has a normal series whose successive quotients are p -groups. $\square$

42. The Group Algebra

Definition

42.1

Recall that a complex vector space A is a **$\mathbb{C}$ -algebra** if there is a bilinear multiplication $A \times A \rightarrow A$ which is associative and has a multiplicative unit.

Example 42.2. 1. $M_n(\mathbb{C})$ is a $\mathbb{C}$ -algebra. Generally, for V a complex vector space, $\text{End}(V) := \text{Hom}(V, V)$ is a $\mathbb{C}$ -algebra.

2. $\mathbb{C}G$, where G is a finite group, is a $\mathbb{C}$ -algebra.

3. If R_1 and R_2 are $\mathbb{C}$ -algebras, then $R_1 \times R_2$ is a $\mathbb{C}$ -algebra.

Definition

42.3

An algebra A is **semisimple** if, given any two finite dimensional A -modules $N \subset M$, there exists an A -submodule $N' \subset M$ such that $M = N \oplus N'$.

Example 42.4. $\mathbb{C}G$ is a semisimple $\mathbb{C}$ -algebra.

We've seen that if $V_1, \dots, V_k$ are the irreducible representations of G , then $\mathbb{C}G \cong \bigoplus_{i=1}^k V_i^{\oplus \dim V_i}$. Now, we can refine this statement in terms of the group algebra.

Theorem

42.5

There exists an isomorphism of $\mathbb{C}$ -algebras

$$\mathbb{C}G \cong \bigoplus_{i=1}^k \text{End}(V_i) \cong \bigoplus_{i=1}^k (V_i^* \otimes V_i)$$

Proof. For any representation V of G , the group homomorphism $G \rightarrow \text{GL}(V)$ extends by linearity to a map of $\mathbb{C}$ -algebras $\mathbb{C}G \rightarrow \text{End}(V)$, since $\mathbb{C}G = \bigoplus \mathbb{C}g_i$. If we apply this to each of the k irreducible representations V_i of G , we get a canonical map

$$\mathcal{F} : \mathbb{C}G \rightarrow \bigoplus_{i=1}^k \text{End}(V_i) \cong \prod_{i=1}^k M_{d_i}(\mathbb{C})$$

where $d_i = \dim(V_i)$. We claim $\mathcal{F}$ is an isomorphism.

Indeed, if $x = \sum_g f(g) \cdot g \in \mathbb{C}G$ for some function $f : G \rightarrow \mathbb{C}$ and $\mathcal{F}(x) = 0$, then $\mathcal{F}(x) = 0$ as an operator on each V_i . It follows (since $\mathbb{C}G = \bigoplus V_i^{\oplus d_i}$) that $x = 0$ on $\mathbb{C}G$, hence $x = 0$, because the representation of G on $\mathbb{C}G$ is faithful. Therefore, $\mathcal{F}$ is injective, and since the dimensions are equivalent, we see that $\mathcal{F}$ is an isomorphism. $\square$

Homework 11 (B1) interprets $\mathcal{F}$ as a Fourier transform, and discusses its inverse. The converse of the theorem holds, and the precise result is a part of Wedderburn Theory:

Theorem

42.6

1. Let F be any field, and A a finite dimensional F -algebra with a unit. Then, A is semisimple if and only if A is isomorphic to a direct sum of matrix algebras $M_r(D)$ over division algebras D (fields that don't need commutativity).
2. If F is algebraically closed, then any finite dimensional semisimple F -algebra is isomorphic to a direct sum of matrix algebras over F .

43. Compact Groups

Definition

43.1

A **topological group** G is a group with a topology such that multiplication by elements of G is continuous. A **compact group** is a topological group that is compact. A **left-invariant measure** dg on a group G is such that

- (i) For all $f : G \rightarrow \mathbb{C}$ continuous, $\int_G f(g)dg = \int_G f(hg)dg$ for all $h \in G$
- (ii) $\text{Vol } G := \int_G dg = 1$

Remark

43.2

The properties of dg imply that its also invariant under right translation.

Example 43.3. 1. Suppose $|G| = n$ is finite. Then, dg assigns to each $x \in G$ the measure $1/n$.

2. Let $G = S^1 := \{z \in \mathbb{C} : |z| = 1\} = \{e^{i\theta} \mid \theta \in \mathbb{R}\}$. Then, $dg = d\theta$, normalized so that $\int_{S^1} d\theta = 1$. In other words, if $f(e^{i\theta})$ is a function on S^1 viewed as a 2π -periodic function $\tilde{f} : \mathbb{R} \rightarrow \mathbb{R}$, then $\int_{\theta \in S^1} f(e^{i\theta})d\theta = \frac{1}{2\pi} \int_0^{2\pi} \tilde{f}(x)dx$.

Definition

43.4

A **linear representation** of G on a complex vector space V is a homomorphism $\rho : G \rightarrow \text{GL}(V)$ which is continuous, where $\text{GL}(V) \cong \text{GL}_n(\mathbb{C})$ is viewed with the natural topology. This is equivalent to saying that the multiplication map $G \times V \rightarrow V$ sending $(g, x) \mapsto \rho(g)(x)$ is a continuous function of the two variables $g \in G, x \in V$. **Irreducible representations, characters**, etc. are defined in the same manner as for finite groups.

Definition

43.5

If $\varphi, \psi : G \rightarrow \mathbb{C}$ are any two functions (say continuous), their **scalar product** is defined by

$$\langle \varphi, \psi \rangle := \int_G \varphi(g) \overline{\psi(g)} dg$$

Most of the properties from the representation theory of finite groups carry over to complex groups replacing $\frac{1}{|G|} \sum_{g \in G} f(g)$ by $\int_G f(g) dg$. More precisely:

Remark

43.6

The same:

1. Maschke's Theorem remains true, so every representation of a compact group is a sum of irreducible representations.
2. Schur's Lemma holds
3. The irreducible characters of G are orthonormal elements of

$$C(G) := \{\text{continuous functions } f : G \rightarrow \mathbb{C}\}.$$

Different:

1. There may be infinitely many irreducible representations
2. There may be infinitely many irreducible characters

The “naive” analogue of $\mathbb{C}G$, namely $C(G)$, is no longer a finite dimensional $\mathbb{C}$ -vector space. However, it is an inner product space, along with the space $\mathcal{C}$ of all class functions on G . But, importantly, $C(G)$ is not complete with respect to the norm $\|f\|_2 = \sqrt{\int_G |f(g)|^2 dg}$ coming from $\langle \cdot, \cdot \rangle$ [but it is complete with respect to the supremum norm $\|\cdot\|_\infty$]. The completion of $(C(G), \|\cdot\|_2)$ is the space $L^2(G)$ of **square integrable measurable functions** G : the functions f such that $\int_G |f(g)|^2 dg < \infty$.

Peter-Weyl Theorem

43.7

The characters of irreducible representations span a dense subspace of the space of continuous class functions on G . That is, if f is a continuous class function, then for all $\varepsilon > 0$, there exists $\chi_1, \dots, \chi_n$ irreducible characters $\lambda_i \in \mathbb{C}$ such that

$$\left\| f - \sum_{i=1}^n \lambda_i \chi_i \right\|_2 < \varepsilon.$$

Moreover, there exists a **Hilbert space decomposition**

$$L^2(G) = \bigoplus_{i=1}^{\infty} V_i^{\oplus \dim V_i}$$

where the V_i are the irreducible representations of G , and this sum is an **orthogonal direct sum**.

$C(G)$ is still a “representation” of G with the group action $(gf)(t) = f(g^{-1}t)$ for all $f \in C(G), g \in$

G. We're putting representation in quotes because it's infinite dimensional, so we can't speak of its character, but the Hilbert space decomposition makes sense of this.

Example 43.8. Consider the compact group S^1 . Since S^1 is abelian, its characters are 1-dimensional. If $f : S^1 \rightarrow \mathbb{C}^*$ is a continuous group homomorphism, then since S^1 is compact, $\text{Im} f \subset S^1 \subset \mathbb{C}^*$. Exercise: Show that all characters are of the form $\chi_n : z \mapsto z^n$, $n \in \mathbb{Z}$. Then,

$$L^2(S^1) = \bigoplus_{n \in \mathbb{Z}} \mathbb{C} \cdot \chi_n.$$

So, for any square integrable $f : S^1 \rightarrow \mathbb{C}$, $f \in L^2(S^1)$,

$$f = \sum_{n \in \mathbb{Z}} \langle f, \chi_n \rangle \chi_n = \sum_{n \in \mathbb{Z}} a_n \chi_n$$

where

$$a_n = \frac{1}{2\pi} \int_0^{2\pi} f(x) e^{-inx} dx$$

is the n -th Fourier coefficient of f .

44. Representation Theory of the Symmetric Group

Recall that the conjugacy classes in S_n are indexed by partitions of n . For example, S_4 has the identity, two-cycles, three-cycles, four-cycles, and products of disjoint two-cycles.

Definition

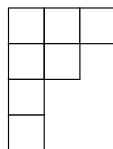
44.1

A **partition** of n is a decreasing sequence of integers $\lambda = (\lambda_1 > \lambda_2 > \cdots > \lambda_k > 0)$ such that

$$|\lambda| := \sum_{i=1}^k \lambda_i = n.$$

We write $\lambda \vdash n$ if λ is a partition of n . The number k of (non-zero) partitions λ_i is the **length** of λ , denoted $l(\lambda)$. A partition λ is identified with its **Young diagram** of boxes, with λ_i boxes in each row. By a **numbering** of $\lambda \vdash n$ we mean a numbering of the boxes in the diagram of λ using $1, \dots, n$. There are $n!$ possible numbering of λ .

Example 44.2. The partition $(3, 2, 1, 1)$ has the Young diagram



Consider $R := \mathbb{C}[x_1, \dots, x_n]$. S_n acts linearly on R by permuting the variables. The irreducible representations of S_n will be constructed as S_n submodules of R , one for each $\lambda \vdash n$.

Given λ and a fixed numbering T of λ , define the polynomial

$$f_T := \prod_{i \text{ north of } j} (x_i - x_j)$$

where the product is over all pairs (i, j) such that i lies due north of j . Then, f_T is the product of all Vandermonde determinants corresponding to the columns of T .

For example, consider the partition, $\lambda = (3, 2, 1, 1)$ with the following numbering:

7	1	6
3	5	
4		
2		

Then,

$$f_T = (x_7 - x_3)(x_7 - x_4)(x_7 - x_2)(x_3 - x_4)(x_3 - x_2)(x_4 - x_2)(x_1 - x_5).$$

Definition

44.3

Define $V_\lambda \subset \mathbb{C}[x_1, \dots, x_n]$ to be the $\mathbb{C}$ -vector space spanned by all $n!$ polynomials f_T , for all numberings of the Young diagram of λ . S_n acts on V_λ by permuting the variables. Hence, $V_\lambda = \mathbb{C}[S_n] \cdot f_T$ is the $\mathbb{C}[S_n]$ -submodule of $\mathbb{C}[x_1, \dots, x_n]$ spanned by f_T , for any numbering T of λ . V_λ is sometimes called the **Specht module**.

Theorem

44.4

For $\lambda \vdash n$, the V_λ forms a complex set of the irreducible representations of S_n

Example 44.5.

(1) Let $n = 2$. We have partitions (2) and $(1, 1)$. Then,

$$V_2 = \langle 1 \rangle \subset \mathbb{C}[x_1, x_2] \text{ is the trivial representation}$$

$$V_{1,1} = \langle x_1 - x_2, x_2 - x_1 \rangle = \langle x_1 - x_2 \rangle \text{ is the alternating representation of } S_2$$

(2) Let $n = 3$. Then, we have the partitions (3) , $(1, 1, 1)$, $(2, 1)$. Then,

$$V_3 = \langle 1 \rangle$$

$$V_{1,1,1} = \langle (x_1 - x_2)(x_1 - x_3)(x_2 - x_3) \rangle \text{ is the alternating representation}$$

$$V_{2,1} = \langle x_i - x_j \mid 1 \leq i < j \leq 3 \rangle = \langle x_1 - x_2, x_1 - x_3 \rangle \text{ is the standard representation}$$

(3) For any n , V_n is the trivial representation, and $V_{1,\dots,1}$ is the alternating representation.

(4) Let $n = 4$. Besides the above 2, we have

$$V_{3,1} = \langle x_1 - x_i \mid i = 2, 3, 4 \rangle \text{ is three dimensional}$$

$$V_{2,1,1} = \langle (x_1 - x_i)(x_1 - x_j)(x_i - x_j) \mid 1 < i < j \leq 4 \rangle \text{ is three dimensional}$$

$$V_{2,2} = \langle (x_1 - x_2)(x_3 - x_4), (x_1 - x_3)(x_2 - x_4) \rangle \text{ is two dimensional.}$$

Note that our presentation for $V_{2,2}$ is not obvious. Namely, we have the numbering

1	2
4	3

which induces the polynomial $(x_1 - x_4)(x_2 - x_3)$, but this polynomial in fact equals

$$(x_1 - x_3)(x_2 - x_4) - (x_1 - x_2)(x_3 - x_4).$$

This motivates us to think about whether or not the numbers are increasing along the rows.

Definition

44.6

Any numbering T of λ is called a **standard tableau** if the numbers are increasing along each row and down each column.

Theorem

44.7

The polynomials f_T where T is a standard tableau of shape λ form a basis for V_λ . Hence, $f^\lambda := \dim_{\mathbb{C}} V_\lambda$ equals the number of standard tableau shapes of λ .

Proof. We'll prove that the f_T are linearly independent. The proof that they span is left as an exercise.

Consider the monomials $x^\alpha := x_1^{\alpha_1} \cdots x_n^{\alpha_n}$, where $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{Z}^n$ with $\alpha_i \geq 0$. We order them as follows: $x^\alpha < x^\beta \iff \alpha < \beta$ is the lexicographic order on $\mathbb{Z}_{\geq 0}^n$. For example,

$$(0, 1, 2) < (0, 2, 1) < (1, 0, 2) < (1, 2, 0) < (2, 0, 1) < (2, 1, 0) \\ \implies x_2x_3^2 < x_2^2x_3 < x_1x_3^2 < x_1x_2^2 < x_1^2x_3 < x_1^2x_2.$$

For T a standard tableau of λ , let m_T be the monomial $x_1^{k_1} \cdots x_n^{k_n}$ where $k_i = r - 1$ if i lies in the r 'th row of T . For example:

$$\begin{array}{|c|c|} \hline 1 & 3 \\ \hline 2 & 5 \\ \hline 4 & \\ \hline \end{array}, \quad m_T = x_2x_4^2x_5, \quad f_T = (x_1 - x_2)(x_1 - x_4)(x_2 - x_4)(x_3 - x_5)$$

It's left as an exercise to show that if T is standard, then $f_T = \pm m_T +$ a later monomial.

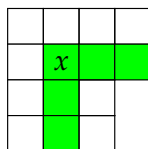
Let $T_1, \dots, T_s$ be the standard tableau of shape λ , so $s = f^\lambda$. The monomials $m_{T_1}, \dots, m_{T_s}$ are all distinct, and we may assume that $m_{T_1} < \cdots < m_{T_s}$. We claim $f_{T_1}, \dots, f_{T_s}$ are linearly independent. Indeed, if $\sum_{i=1}^s c_i f_{T_i} = 0$, the coefficient of m_T on the LHS is $\pm c_1$. Hence $c_1 = 0$. Repeating the argument shows that $c_2 = 0$, and so on. It is "much harder" but "completely trivial" to show that f_T for T standard span V_λ . $\square$

Definition

44.8

Given a box x in λ , the **hook** H_x at x is the together with all boxes due east and south of x . The **hook length** h_x is the number of boxes in H_x .

Example 44.9. The box labeled x has the following hook:

**Frame-Robinson-Thrall Theorem**

44.10

$$f^\lambda = \frac{n!}{\prod_{x \in \lambda} h_x}.$$

Example 44.11.

$$f^{(2,2)} = \frac{4!}{3 \cdot 2 \cdot 2 \cdot 1} = 2.$$

As a matter of fact,

$$f^{(n,n)} = \frac{1}{n+1} \binom{2n}{n} = C_n,$$

the n 'th Catalan number.

Corollary

44.12

$\prod_{x \in \lambda} h_x$ divides $n!$ for $\lambda \vdash n$. Moreover,

$$\sum_{\lambda \vdash n} (f^\lambda)^2 = n!$$

45. The Character χ_λ of V_λ

We want to compute $\chi_\lambda(\mu)$ for $\lambda, \mu \vdash n$, where λ corresponds to the irreducible representations of V_λ , and μ corresponds to the conjugacy classes of S_n . Let $k = l(\lambda)$, and introduce the variables $x_1, \dots, x_k$. Moreover, let

$$\Delta(x) := \prod_{1 \leq i < j \leq k} (x_i - x_j).$$

Suppose $C = C[m]$ is a conjugacy class of S_n consisting of permutations with m_1 1-cycles, m_2 2-cycles, $\dots$, m_d d -cycles, so $\mu = (1^{m_1} 2^{m_2} \dots d^{m_d})$.

Frobenius Formula

45.1

$\chi_\lambda(C[m])$ is the coefficient of $x_1^{h_{11}} \dots x_k^{h_{k1}}$ in $\Delta(x) \cdot \prod_{j=1}^d p_j(x)^{m_j}$, where $p_j(x) = x_1^j + \dots + x_k^j$. Here, $h_{1j} = \lambda_j + k - j$ are the hook lengths of λ in the 1st column.

Definition

45.2

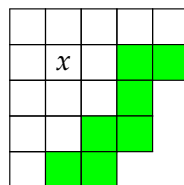
The **Murnaghan-Nakayama Rule** is a combinatorial algorithm for computing character values, described below.

Definition

45.3

If $v \subset \lambda$ is a subset of the diagram of λ , write $\lambda \setminus v$ for the set-theoretic difference. For x a box in λ , the **rim hook** α of λ corresponding to x is obtained by projecting the hook H_x along diagonals to the southeast boundary of λ . The **height** $\text{ht}(\alpha)$ of α is one less than the number of rows it occupies.

Example 45.4. The rim hook of x is filled in green.



Theorem

45.5

If $\lambda \vdash n$ and μ is another partition of n corresponding to a conjugacy class in S_n , then

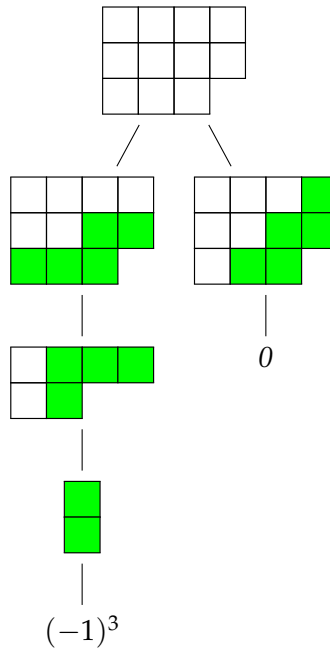
$$\chi_\lambda(\mu) = \sum_{|\alpha|=\mu_1} (-1)^{\text{ht}(\alpha)} \chi_{\lambda \setminus \alpha}(\mu \setminus \mu_1)$$

summed over all rim hooks α such that $|\alpha| = \mu_1$

Example 45.6. We may now compute the character recursively. For example,

$$\begin{aligned} \chi_{(4,4,3)}((5,4,2)) &= -\chi_{(4,2)}(4,2) + \chi_{3,2,1}(4,2) \\ &= (-1)(-1)\chi_{(1,1)}(2) + 0 = -(-(-(\chi_0(0)))) = (-1)^3 = -1. \end{aligned}$$

This corresponds to the following recursive tree of rim hooks:



Example 45.7. We'll compute the character values for the standard two dimensional representation $V_{2,1}$ of S_3 . We have three conjugacy classes: $(1,1,1)$, $(2,1)$, (3) .

$$\chi((1,1,1)) = 1 + 1 = 2$$

$$\chi((2,1)) = 0 \text{ since no rim hook of length 2 can be subtracted}$$

$$\chi(3) = -1$$

46. Polynomial Representations of $GL_n(\mathbb{C})$

In this section, we use $G := GL_n(\mathbb{C})$

Definition

46.1

A matrix representation $\rho : G \rightarrow GL_N(\mathbb{C})$ is called **polynomial** if for any matrix $A \in GL_n(\mathbb{C})$, the entries of $\rho(A)$ are polynomials in the entries of A . If V is a complex vector space with

$\dim_{\mathbb{C}} V = N$, the isomorphism $V \cong \mathbb{C}^N$ may be used to define polynomial representations $\rho : G \rightarrow \mathrm{GL}(V)$.

- Example 46.2.**
1. $V = \mathbb{C}^n$ with the usual action of G on V is the **standard representation** of G .
 2. For any $k \geq 1$, $T^k V, S^k V, \wedge^k V$ are all polynomial representations of G (for example, if $g \in G$, then $g(v_1 \otimes \cdots \otimes v_k) = gv_1 \otimes \cdots \otimes gv_k$). Moreover, $S^k V$ and $\wedge^k V$ are irreducible, but $T^k V$ is not for $k \geq 2$.
 3. $\det V = \wedge^n V$ is a degree 1 representation of G . That is, $\rho : G \rightarrow \mathbb{C}^*$ given by $\rho(A) = \det A$.
 4. G has finite dimensional non-polynomial representations, which are instead rational or meromorphic. For example, $\rho(A) = (\det A)^{-1}$.
 5. $U(n) \subset \mathrm{GL}_n(\mathbb{C})$ is a maximal compact subgroup of $\mathrm{GL}_n(\mathbb{C})$. Weyl's "unitary trick" ensures that the polynomial representations of $U(n)$ and $\mathrm{GL}_n(\mathbb{C})$ actually coincide. Since $U(n)$ is compact, it has more manageable representation theory.

47. Schur Modules

Definition

47.1

For any partition λ and vector space V , we define the **Schur module** $s_\lambda(V)$ as follows:

Number the boxes in the Young diagram of λ with the integers $1, 2, \dots, p = |\lambda|$ in order going from left to right and top to bottom. This gives a standard tableau T on λ . Let R (resp. C) denote the subgroup of S_p consisting of the elements which permute the entries in each row (resp. column) of T with themselves. Now, consider the elements

$$a_\lambda := \sum_{u \in C} \mathrm{sgn}(u)u, \quad b_\lambda := \sum_{v \in R} v$$

in $\mathbb{C}[S_p]$. Define the **Young symmetrizer** $c_\lambda := a_\lambda b_\lambda$. Now, given any complex vector space V , let $M = T^p V = V \otimes \cdots \otimes V$. Then, the group S_p acts on the right on M by permuting the factors:

$$(v_1 \otimes \cdots \otimes v_p)\sigma := v_{\sigma(1)} \otimes \cdots \otimes v_{\sigma(p)}.$$

Moreover, the group $G := \mathrm{GL}(V)$ acts on the left on M by

$$g(v_1 \otimes \cdots \otimes v_p) = (gv_1) \otimes \cdots \otimes (gv_p).$$

One can see that the two actions commute with each other. Finally, the **Schur module** $s_\lambda(V)$ is defined at the image of the map $M \rightarrow M$ which is right multiplication by c_λ . This construction is **functorial**: a linear map $f : V \rightarrow W$ of vector spaces induces a linear map $s_\lambda(f) : s_\lambda(V) \rightarrow s_\lambda(W)$ with $s_\lambda(f_1 \circ f_2) = s_\lambda(f_1) \circ s_\lambda(f_2)$ and $s_\lambda(\mathrm{id}_V) = \mathrm{id}_{s_\lambda(V)}$.

Example 47.2. If $\lambda = (4, 3, 2)$, then T is

1	2	3	4
5	6	7	
8	9		

Example 47.3. 1. Let $p = 2$, and write $S_2 = \{1, \sigma\}$. The two partitions of 2 are (2) and $(1, 1)$, with

Young symmetrizers $c_2 = 1 + \sigma$ and $c_{1,1} = 1 - \sigma$. We have $s_2(V) = S^2V$ and $s_{1,1} = \wedge^2 V$, sitting inside T^2V . In fact, $V \otimes V = S^2V \oplus \wedge^2 V$.

2. Generalizing (1), for any $p \geq 1$, we have $c_p = \sum_{\sigma \in S_p} \sigma$, and $c_{1^p} = \sum_{\sigma \in S_p} \text{sgn}(\sigma)\sigma$, so $s_p(V) = S^pV$ and $s_{1^p}(V) = \wedge^p V$.
3. Given any matrix $A = \{a_{ij}\}$ and partition λ , the **Schur matrix** is $s_\lambda(A)$, generalizing $S^p A$ and $\wedge^p A$. No general formula is known.

Theorem

47.4

Let $V = \mathbb{C}^n$ be the standard representation of $G := \text{GL}_n(\mathbb{C})$, and let λ be any partition of length $l(\lambda) \leq n$. Then, $s_\lambda(V)$ is an irreducible polynomial representation of G . Moreover, $\{s_\lambda(V) \mid l(\lambda) \leq n\}$ is a complete set of irreducible polynomial CG-modules.

Question

What is $\dim s_\lambda(V)$?

Definition

47.5

A **semistandard tableau** T on λ is a filling of the boxes in λ with positive integers so that the entries are weakly increasing along each row, and strictly increasing down each column.

Example 47.6.

1	1	3	3	4
2	2	4	4	
4	5	5		

Proposition

47.7

The dimension $\dim s_\lambda(V)$ is the number of semistandard tableaux of shape λ with entries $\leq n$.

Example 47.8. The module $\wedge^p \mathbb{C}^n$ is the Schur module for 1^p , so its dimension is the number of ways to fill in the vertical line of boxes so that it's semistandard. This is $\binom{n}{p}$. The dimension $S^p \mathbb{C}^n$ corresponds to the semistandard fillings of the horizontal line of boxes, which is $\binom{n+p-1}{p}$.

If A is a diagonal matrix with eigenvalues $x_1, \dots, x_n$, then $\text{Tr}(s_\lambda(A) : s_\lambda(V) \rightarrow s_\lambda(V))$ is the **Schur polynomial** $s_\lambda(x_1, \dots, x_n)$. This is the character of $s_\lambda(V)$ in the following sense:

The diagonalizable matrices form a dense open subset of $\text{GL}_n(\mathbb{C})$. So, any polynomial character $\chi : \text{GL}_n(\mathbb{C}) \rightarrow \mathbb{C}$ is determined by its values on diagonalizable elements.

Example 47.9. $s_1 = x_1 + \dots + x_n$. $s_{1^p} = e_p(x_1, \dots, x_n)$, the p -th elementary symmetric polynomial.

$$s_p = \sum_{|\alpha|=p} x^\alpha,$$

the sum of all monomials of degree p .

D. Littlewood's Theorem (not that Littlewood)

47.10

In general,

$$s_\lambda(x_1, \dots, x_n) = \sum_T x^{c(T)},$$

where the sum runs over all semistandard tableaux T on λ with entries $\leq n$, and $c(T) = (c_1, \dots, c_n)$ is the **content vector** of T , where $c_i :=$ the number of entries equal to i in T .

Example 47.11. Let $\lambda = (2, 1)$. A semistandard Young tableau of shape λ is of the form $\begin{array}{|c|c|} \hline a & b \\ \hline c & \\ \hline \end{array}$ where $1 \leq a \leq b$ and $a < c$. So, if $x = (x_1, \dots, x_n)$, then

$$s_{2,1}(x) = \sum_{\substack{a \leq b \\ a < c}} x_a x_b x_c = \sum_{a \neq b} x_a^2 x_b + 2 \sum_{a < b < c} x_a x_b x_c$$

Remark

47.12

$s_\lambda(x_1, \dots, x_n)$ is a symmetric, homogeneous polynomial in $x_1, \dots, x_n$ of degree $|\lambda|$.

Much else is known about Schur polynomials.

Proposition

47.13

$$s_\lambda(x_1, \dots, x_n) = \frac{\det(x_i^{\lambda_i + n - j})_{1 \leq i, j \leq n}}{\det(x_i^{n - j})_{1 \leq i, j \leq n}}$$

This was Cauchy's original definition of Schur polynomials.

Consider the ring of symmetric polynomials $\mathbb{C}[e_1, \dots, e_n]$. As it turns out, this is

$$\bigoplus_{\lambda, l(\lambda) \leq n} \mathbb{C} s_\lambda(x) \cong \text{Polynomial representation ring of } \text{GL}_n(\mathbb{C})$$

and also

$$\mathbb{C}[s_1, \dots, s_n].$$

The products correspond to

$$s_\lambda(V) \otimes s_\mu(V) = \bigoplus_{l(v) \leq n} s_v(V)$$

and

$$s_\lambda(x) s_\mu(x) = \sum_{l(v) \leq n} c_{\lambda\mu}^v s_v(x).$$

Definition

47.14

If $|\lambda| = m$, we have an irreducible character χ_λ of S_m . For any partition $\mu := (\mu_1, \dots, \mu_r)$, let $p_\mu(x) := p_{\mu_1}(x) \cdots p_{\mu_r}(x)$ be the μ -th **power sum**, where $p_k(x) = x_1^k + \cdots + x_n^k$, where $n \geq l(\lambda)$.

Frobenius' Theorem

47.15

The sets $\{s_\lambda(x)\}_{|\lambda|=m}$ and $\{p_\lambda(x)\}_{|\lambda|=m}$ are two $\mathbb{C}$ -bases of the vector space $\mathbb{C}[x_1, \dots, x_n]_{(m)}^{S_n}$ of symmetric polynomials in $x_1, \dots, x_n$ which are homogeneous of degree m . These are related by the following change of basis formula: for any $\mu \vdash m$,

$$p_\mu(x) = \sum_{\lambda \vdash m} \chi_\lambda(\mu) \cdot s_\lambda(x),$$

where χ_λ is the irreducible character of S_m indexed by λ , and μ in $x_\lambda(\mu)$ corresponds to a conjugacy class in S_m . We also have

$$s_\lambda(x) = \frac{1}{m!} \sum_{\sigma \in S_m} \chi_\lambda(\sigma) p_\sigma(x)$$